Best practices for your confirmatory factor analysis

A JASP and lavaan tutorial

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Abstract

Confirmatory factor analysis (CFA) is a fundamental method for evaluating the internal structural validity of measurement instruments. In most CFA applications, the measurement model serves as a means to an end rather than an end in itself. To select the appropriate model, prior validity evidence is crucial, and items are typically assessed on an ordinal scale, which has been used in the applied social sciences. However, textbooks on structural equation modeling (SEM) often overlook this specific case, focusing on applications estimable using maximum likelihood (ML) instead. Unfortunately, several popular commercial SEM software packages lack suitable solutions for handling this 'typical CFA', leading to confusion and suboptimal decision-making when conducting CFA in this context. This article conceptually contributes to this ongoing discussion by presenting a set of guidelines for conducting a typical CFA, drawing from recent empirical research. We provide a practical contribution by introducing and developing a tutorial example within the JASP and lavaan software platforms. Supplementary materials such as videos, files, and scripts are freely available.

0.1 Introduction

Confirmatory factor analysis (CFA) is a fundamental method for evaluating the internal structural validity of measurement instruments. This tutorial demonstrates best practices for conducting CFA using the WHOQOL-BREF instrument, a widely used quality of life assessment tool.

All analyses in this tutorial use the DWLS estimator (called WLSMV in lavaan), which is appropriate for ordinal data. This is the "typical CFA" scenario in applied social sciences, where items are measured on Likert scales rather than continuous variables.

> **!** Tutorial Objective
>
> This tutorial is designed to be **fully self-contained**. All code, explanations, and results are presented in this single document, making it easy to follow the entire CFA workflow from data loading through model comparison and power analysis.

The WHOQOL-BREF dataset (N = 1,047) was obtained from Mendeley Data (Rogers, 2021) and supports the methodological framework described in Rogers (2022). The instrument contains 24 items measuring quality of life across four domains: Psychological (6 items), Physical (7 items), Social (3 items), and Environment (8 items).

0.2 Setup

First, we load all required R packages for the analysis:

```r
library(lavaan)     # CFA estimation and model fitting
```

```
Warning: pacote 'lavaan' foi compilado no R versão 4.4.3

This is lavaan 0.6-21
lavaan is FREE software! Please report any bugs.
```

```r
library(semTools)   # Reliability and additional SEM tools
```

```
Warning: pacote 'semTools' foi compilado no R versão 4.4.3


###############################################################################

This is semTools 0.5-7

All users of R (or SEM) are invited to submit functions or ideas for functions.

###############################################################################
```

```r
library(semPlot)    # Path diagrams
```

```
Warning: pacote 'semPlot' foi compilado no R versão 4.4.3
```

```r
library(psych)      # Descriptive statistics and omega coefficients
```

```
Warning: pacote 'psych' foi compilado no R versão 4.4.3

Anexando pacote: 'psych'

Os seguintes objetos são mascarados por 'package:semTools':

    reliability, skew

O seguinte objeto é mascarado por 'package:lavaan':

    cor2cov
```

```r
library(simsem)     # Power analysis via simulation
```

```
Warning: pacote 'simsem' foi compilado no R versão 4.4.3


#################################################################

This is simsem 0.5-17

simsem is BETA software! Please report any bugs.

simsem was first developed at the University of Kansas Center for

Research Methods and Data Analysis, under NSF Grant 1053160.

#################################################################

Anexando pacote: 'simsem'
```

```
O seguinte objeto é mascarado por 'package:psych':

    sim

O seguinte objeto é mascarado por 'package:lavaan':

    inspect
```

```
library(MASS)       # General statistical functions
library(dynamic)    # Dynamic Fit Index computation
```

```
Beta version. Please report bugs: https://github.com/melissagwolf/dynamic/issues.
```

Source: Article Notebook

Set global options and seed for reproducibility:

```
options(max.print = 1000000)  # Allow printing large outputs
set.seed(123456)              # Ensure reproducible results for simulations
```

Source: Article Notebook

0.3 Step 1: ETL — Extract, Transform, Load

The WHOQOL-BREF data are loaded directly from Mendeley Data (Rogers, 2021). Items Q3, Q4, and Q26 have already been reverse-coded in the source data.

0.3.1 Loading the Data

```
# Define the database URL
url_data <- "https://data.mendeley.com/datasets/rdky78bk8r/2/files/b58a7054-4978-4239-81db-00f9ef

# Read data without variable names
df0 <- read.table(url_data)

# Define the URL of variable names
url_labels <- "https://data.mendeley.com/datasets/rdky78bk8r/2/files/ae2a5c01-2fc2-43e5-b31c-8db8

# Read variable names from the first column
column_labels <- read.table(url_labels)
column_labels <- column_labels$V1

# Assign variable names to the dataset
names(df0) <- column_labels
```

Source: Article Notebook

0.3.2 Exploring the Data

```
knitr::kable(head(df0))
```

Table 1: First rows of the WHOQOL-BREF dataset

Q3_F	Q4_F	Q10_F	Q15_F	Q16_F	Q17_F	Q18_F	Q5_P	Q6_P	Q7_P	Q11_P	Q19_P	Q26_P	Q20_S	Q21_S	Q22_S	Q8_A	Q9_A	Q12_A	Q13_A	Q14_A	Q23_A	Q24_A	Q25_A
5	4	2	5	3	4	3	3	3	2	2	2	4	4	2	2	4	3	3	4	3	2	2	2
5	5	3	3	3	3	2	3	2	2	3	3	4	2	3	2	3	2	2	2	3	2	4	2
5	4	3	4	5	2	2	4	4	2	4	3	2	4	5	5	3	2	3	4	3	5	5	5
5	5	5	5	5	4	5	4	5	5	5	4	5	4	5	5	4	4	5	5	4	5	3	5
3	4	3	4	4	4	4	3	5	3	2	4	4	4	5	4	2	2	3	3	2	5	3	3
2	4	4	4	4	4	4	3	4	3	4	4	4	4	2	4	3	3	2	4	3	4	2	4

Source: Article Notebook

```r
# Display all variable names
names(df0)
```

```
 [1] "Q3_F"  "Q4_F"  "Q10_F" "Q15_F" "Q16_F" "Q17_F" "Q18_F" "Q5_P"  "Q6_P" 
[10] "Q7_P"  "Q11_P" "Q19_P" "Q26_P" "Q20_S" "Q21_S" "Q22_S" "Q8_A"  "Q9_A" 
[19] "Q12_A" "Q13_A" "Q14_A" "Q23_A" "Q24_A" "Q25_A"
```

Source: Article Notebook

```r
# Display data structure
str(df0)
```

```
'data.frame':	1047 obs. of  24 variables:
 $ Q3_F : int  5 5 5 5 3 2 5 5 5 4 ...
 $ Q4_F : int  4 5 4 5 4 4 4 5 5 5 ...
 $ Q10_F: int  2 3 3 5 3 4 4 3 4 4 ...
 $ Q15_F: int  5 3 4 5 4 4 5 3 5 5 ...
 $ Q16_F: int  3 3 5 5 4 4 2 4 3 3 ...
 $ Q17_F: int  4 3 2 4 4 4 4 3 2 4 ...
 $ Q18_F: int  3 2 2 5 4 4 4 3 2 3 ...
 $ Q5_P : int  3 3 4 4 3 3 4 4 3 3 ...
 $ Q6_P : int  3 2 4 5 5 4 5 4 3 4 ...
 $ Q7_P : int  2 2 2 5 3 3 4 3 4 4 ...
 $ Q11_P: int  2 3 4 5 2 4 4 3 4 3 ...
 $ Q19_P: int  2 3 3 4 4 4 4 3 2 4 ...
 $ Q26_P: int  4 4 2 5 4 4 5 4 4 4 ...
 $ Q20_S: int  4 2 4 4 4 4 5 5 4 3 ...
 $ Q21_S: int  2 3 5 5 5 2 2 3 2 2 ...
 $ Q22_S: int  2 2 5 5 4 4 5 5 3 3 ...
 $ Q8_A : int  4 3 3 4 2 3 2 3 3 3 ...
 $ Q9_A : int  3 2 2 4 2 3 1 4 4 4 ...
 $ Q12_A: int  3 2 3 5 3 2 4 3 3 4 ...
 $ Q13_A: int  4 2 4 5 3 4 4 3 5 5 ...
 $ Q14_A: int  3 3 3 4 2 3 5 2 3 3 ...
 $ Q23_A: int  2 2 5 5 5 4 5 4 4 3 ...
 $ Q24_A: int  2 4 5 3 3 2 5 5 3 3 ...
 $ Q25_A: int  2 2 5 5 3 4 5 2 5 3 ...
```

Source: Article Notebook

```r
knitr::kable(describe(df0), digits = 2)
```

Table 2: Descriptive statistics for all WHOQOL-BREF items

	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se
Q3_F	1	1047	4.17	0.93	4	4.31	1.48	1	5	4	-0.99	0.27	0.03
Q4_F	2	1047	4.10	1.01	4	4.26	1.48	1	5	4	-1.02	0.31	0.03
Q10_F	3	1047	3.61	0.91	4	3.65	1.48	1	5	4	-0.18	-0.43	0.03
Q15_F	4	1047	4.42	0.79	5	4.57	0.00	1	5	4	-1.52	2.41	0.02
Q16_F	5	1047	3.50	1.03	4	3.49	1.48	2	5	3	-0.13	-1.16	0.03
Q17_F	6	1047	3.72	0.92	4	3.78	0.00	2	5	3	-0.51	-0.53	0.03
Q18_F	7	1047	3.75	0.96	4	3.81	1.48	2	5	3	-0.50	-0.64	0.03
Q5_P	8	1047	3.29	0.82	3	3.35	1.48	1	5	4	-0.44	-0.09	0.03
Q6_P	9	1047	3.90	0.99	4	4.02	1.48	1	5	4	-0.88	0.48	0.03
Q7_P	10	1047	3.61	0.86	4	3.66	1.48	1	5	4	-0.53	0.12	0.03
Q11_P	11	1047	3.82	0.99	4	3.92	1.48	1	5	4	-0.62	-0.09	0.03
Q19_P	12	1047	3.72	0.95	4	3.77	1.48	2	5	3	-0.42	-0.70	0.03
Q26_P	13	1047	3.91	0.92	4	4.03	0.00	1	5	4	-1.19	1.68	0.03
Q20_S	14	1047	3.64	0.89	4	3.67	1.48	2	5	3	-0.25	-0.66	0.03
Q21_S	15	1047	3.46	1.00	4	3.45	1.48	2	5	3	-0.05	-1.09	0.03
Q22_S	16	1047	3.58	0.88	4	3.60	1.48	2	5	3	-0.16	-0.69	0.03
Q8_A	17	1047	2.94	0.98	3	2.97	1.48	1	5	4	-0.11	-0.43	0.03
Q9_A	18	1047	3.47	0.91	4	3.52	1.48	1	5	4	-0.64	0.26	0.03
Q12_A	19	1047	3.45	1.10	3	3.50	1.48	1	5	4	-0.25	-0.54	0.03
Q13_A	20	1047	3.93	0.82	4	3.98	1.48	1	5	4	-0.57	0.15	0.03
Q14_A	21	1047	3.34	0.96	3	3.34	1.48	1	5	4	-0.29	-0.26	0.03

Table 2: Descriptive statistics for all WHOQOL-BREF items

vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se
Q23_A22	1047	3.98	0.97	4	4.09	1.48	2	5	3	-0.70	-0.45	0.03
Q24_A23	1047	3.73	0.97	4	3.79	1.48	2	5	3	-0.40	-0.79	0.03
Q25_A24	1047	3.83	0.98	4	3.91	1.48	2	5	3	-0.50	-0.72	0.03

Source: Article Notebook

Data Overview

The dataset contains N = 1047 observations on 24 variables. All items are measured on a 5-point Likert scale (1-5), representing ordinal data. Items Q3, Q4, and Q26 have been reverse-coded so that higher values consistently indicate better quality of life.

0.4 Population Model

For the calculation of the Dynamic Fit Index (DFI) and *a priori* power analysis, we use a population model based on the meta-analysis by Lin & Yao (2022). In their study, Lin and Yao validated the factor structure of the WHOQOL-BREF using a meta-analysis of exploratory factor analyses combined with social network analysis.

Important Caveat

The example population model (*popModel*) should be understood as a **teaching illustration**. Ideally, one should look for a robust national survey that has used the WHOQOL-BREF with CFA for ordinal data, providing more appropriate starting values. The loadings from Lin & Yao (2022) come primarily from exploratory factor analysis studies using principal components with varimax rotation—a method that has known limitations (Rogers, 2022).

0.4.1 Defining the Population Model

Given the ordinal nature of the variables, we assume equidistant thresholds following a linearity assumption, with proportions of 12%, 23%, 31%, and 12%. For a five-point Likert scale (1 to 5), typical equidistant thresholds would be -1.2, -0.4, 0.4, and 1.2. While it is unlikely that thresholds will be available in previous studies, if response frequencies for each category are provided, thresholds can be estimated using the inverse normal distribution.

```
pop_model <- "
 # Define factors and set loadings according to Lin and Yao (2022)
psycho =~ 0.92*Q5_P + 0.81*Q6_P + 0.94*Q7_P + 0.73*Q11_P + 0.75*Q19_P + 0.63*Q26_P
physical =~ 0.84*Q3_F + 0.83*Q4_F + 0.63*Q10_F + 0.71*Q15_F + 0.68*Q16_F
```

```
            + 0.89*Q17_F + 0.61*Q18_F
social =~ 0.94*Q20_S + 0.89*Q21_S + 0.78*Q22_S
environment =~ 0.37*Q8_A + 0.49*Q9_A + 0.79*Q12_A + 0.81*Q13_A + 0.77*Q14_A
               + 0.75*Q23_A + 0.64*Q24_A + 0.73*Q25_A
 # Set the maximum acceptable correlation between Q4 and Q3
Q4_F ~~ 0.3*Q3_F
 # Set factor variances to 1 and correlations to 0.3
 # According to Lin and Yao (2022), correlations between factors ranged from 0.08 to 0.3
psycho ~~ 1*psycho + 0.3*physical + 0.3*social + 0.3*environment
physical ~~ 1*physical + 0.3*social + 0.3*environment
social ~~ 1*social + 0.3*environment
environment ~~ 1*environment
 # Set all item residual variances according to defined loadings (1-L^2)
 # Except Q3 and Q4 due to the predicted correlation between the two
Q5_P ~~ 0.15*Q5_P
Q6_P ~~ 0.34*Q6_P
Q7_P ~~ 0.12*Q7_P
Q11_P ~~ 0.47*Q11_P
Q19_P ~~ 0.44*Q19_P
Q26_P ~~ 0.6*Q26_P
Q10_F ~~ 0.6*Q10_F
Q15_F ~~ 0.5*Q15_F
Q16_F ~~ 0.54*Q16_F
Q17_F ~~ 0.21*Q17_F
Q18_F ~~ 0.63*Q18_F
Q20_S ~~ 0.12*Q20_S
Q21_S ~~ 0.21*Q21_S
Q22_S ~~ 0.39*Q22_S
Q8_A ~~ 0.86*Q8_A
Q9_A ~~ 0.76*Q9_A
Q12_A ~~ 0.38*Q12_A
Q13_A ~~ 0.34*Q13_A
Q14_A ~~ 0.41*Q14_A
Q23_A ~~ 0.44*Q23_A
Q24_A ~~ 0.59*Q24_A
Q25_A ~~ 0.47*Q25_A
 # Set equidistant thresholds for all items
Q5_P | -1.2*t1
Q5_P | -0.4*t2
Q5_P | 0.4*t3
Q5_P | 1.2*t4
Q6_P | -1.2*t1
Q6_P | -0.4*t2
Q6_P | 0.4*t3
Q6_P | 1.2*t4
Q7_P | -1.2*t1
Q7_P | -0.4*t2
Q7_P | 0.4*t3
Q7_P | 1.2*t4
```

```
Q11_P | -1.2*t1
Q11_P | -0.4*t2
Q11_P | 0.4*t3
Q11_P | 1.2*t4
Q19_P | -1.2*t1
Q19_P | -0.4*t2
Q19_P | 0.4*t3
Q19_P | 1.2*t4
Q26_P | -1.2*t1
Q26_P | -0.4*t2
Q26_P | 0.4*t3
Q26_P | 1.2*t4
Q3_F | -1.2*t1
Q3_F | -0.4*t2
Q3_F | 0.4*t3
Q3_F | 1.2*t4
Q4_F | -1.2*t1
Q4_F | -0.4*t2
Q4_F | 0.4*t3
Q4_F | 1.2*t4
Q10_F | -1.2*t1
Q10_F | -0.4*t2
Q10_F | 0.4*t3
Q10_F | 1.2*t4
Q15_F | -1.2*t1
Q15_F | -0.4*t2
Q15_F | 0.4*t3
Q15_F | 1.2*t4
Q16_F | -1.2*t1
Q16_F | -0.4*t2
Q16_F | 0.4*t3
Q16_F | 1.2*t4
Q17_F | -1.2*t1
Q17_F | -0.4*t2
Q17_F | 0.4*t3
Q17_F | 1.2*t4
Q18_F | -1.2*t1
Q18_F | -0.4*t2
Q18_F | 0.4*t3
Q18_F | 1.2*t4
Q20_S | -1.2*t1
Q20_S | -0.4*t2
Q20_S | 0.4*t3
Q20_S | 1.2*t4
Q21_S | -1.2*t1
Q21_S | -0.4*t2
Q21_S | 0.4*t3
Q21_S | 1.2*t4
Q22_S | -1.2*t1
```

```
Q22_S | -0.4*t2
Q22_S | 0.4*t3
Q22_S | 1.2*t4
Q8_A | -1.2*t1
Q8_A | -0.4*t2
Q8_A | 0.4*t3
Q8_A | 1.2*t4
Q9_A | -1.2*t1
Q9_A | -0.4*t2
Q9_A | 0.4*t3
Q9_A | 1.2*t4
Q12_A | -1.2*t1
Q12_A | -0.4*t2
Q12_A | 0.4*t3
Q12_A | 1.2*t4
Q13_A | -1.2*t1
Q13_A | -0.4*t2
Q13_A | 0.4*t3
Q13_A | 1.2*t4
Q14_A | -1.2*t1
Q14_A | -0.4*t2
Q14_A | 0.4*t3
Q14_A | 1.2*t4
Q23_A | -1.2*t1
Q23_A | -0.4*t2
Q23_A | 0.4*t3
Q23_A | 1.2*t4
Q24_A | -1.2*t1
Q24_A | -0.4*t2
Q24_A | 0.4*t3
Q24_A | 1.2*t4
Q25_A | -1.2*t1
Q25_A | -0.4*t2
Q25_A | 0.4*t3
Q25_A | 1.2*t4
"
```

Source: Article Notebook

0.4.2 Dynamic Fit Index (DFI)

The Dynamic Fit Index provides simulation-based cutoff values tailored to a specific model and sample size, as an alternative to generic rules of thumb for fit indices.

> **Warning**
>
> The DFI computation is commented out because it is very time-consuming due to extensive simulations. The code below shows how to compute it if needed. These cutoffs would be valid for all subsequent analyses under the assumption

that *popModel* represents the correct population model.

```
# Patient! This takes a long time due to simulations
# dynamic::catHB(pop_model, manual = TRUE, n = 1047, plot = TRUE)
```

Source: Article Notebook

We could calculate a DFI for each of the estimated models, assuming that the model in question would be the correct population model. However, this would lose comparability, as DFI methods for bifactor models are not yet available, and DFI for hierarchical models are still in development.

0.5 4-Factor CFA Model

The main model specifies four correlated latent factors (Psychological, Physical, Social, and Environment) with all items treated as ordinal. Based on preliminary analysis and theoretical considerations, items Q4 and Q3 are allowed to correlate.

0.5.1 Model Specification

```
# Endogenous (latent) variables are correlated by default in lavaan
# Items Q3, Q4, and Q26 have already been reverse-coded
# We allow Q4 and Q3 to have correlated residuals
spe_4fa <- "
psycho =~ Q5_P + Q6_P + Q7_P + Q11_P + Q19_P + Q26_P
physical =~ Q3_F + Q4_F + Q10_F + Q15_F + Q16_F + Q17_F + Q18_F
social =~ Q20_S + Q21_S + Q22_S
environment =~ Q8_A + Q9_A + Q12_A + Q13_A + Q14_A + Q23_A + Q24_A + Q25_A
Q4_F ~~ Q3_F
"
```

Source: Article Notebook

0.5.2 Model Estimation

```
est_4fa <- cfa(
  spe_4fa,
  data = df0,
  ordered = TRUE,       # All items are ordinal
  std.lv = TRUE,        # Standardize latent variables (variance = 1)
  estimator = "WLSMV"  # DWLS estimator for ordinal data
)
```

Source: Article Notebook

0.5.3 Model Evaluation

> **Evaluating CFA Models**
>
> When evaluating CFA models, watch for:
>
> 1. **Heywood cases**: Standardized loadings > 1 or negative variances
> 2. **Overall fit**: SRMR < .08, RMSEA < .06, CFI/TLI > .95 (scaled versions for WLSMV)
> 3. **Local fit**: Standardized estimates, residual correlations, and modification indices (MI)
> 4. **Reliability**: Cronbach's alpha and omega > .70 for adequate internal consistency

0.5.3.1 Overall Fit and Parameter Estimates

```r
summary(est_4fa,
  fit.measures = TRUE,    # Include fit indices
  standardized = TRUE,    # Include standardized estimates
  rsq = TRUE              # Include R-squared for items
)
```

```
lavaan 0.6-21 ended normally after 34 iterations

  Estimator                                       DWLS
  Optimization method                           NLMINB
  Number of model parameters                       117

  Number of observations                          1047

Model Test User Model:
                                              Standard      Scaled
  Test Statistic                              1339.798    1732.820
  Degrees of freedom                               245         245
  P-value (Unknown)                                 NA       0.000
  Scaling correction factor                                  0.807
  Shift parameter                                           73.409
    simple second-order correction

Model Test Baseline Model:

  Test statistic                             71098.143   25699.871
  Degrees of freedom                               276         276
  P-value                                           NA       0.000
  Scaling correction factor                                  2.786

User Model versus Baseline Model:

  Comparative Fit Index (CFI)                    0.985       0.941
  Tucker-Lewis Index (TLI)                       0.983       0.934
```

```
  Robust Comparative Fit Index (CFI)                         0.861
  Robust Tucker-Lewis Index (TLI)                            0.844

Root Mean Square Error of Approximation:

  RMSEA                                          0.065       0.076
  90 Percent confidence interval - lower         0.062       0.073
  90 Percent confidence interval - upper         0.069       0.080
  P-value H_0: RMSEA <= 0.050                    0.000       0.000
  P-value H_0: RMSEA >= 0.080                    0.000       0.033

  Robust RMSEA                                               0.083
  90 Percent confidence interval - lower                     0.079
  90 Percent confidence interval - upper                     0.087
  P-value H_0: Robust RMSEA <= 0.050                         0.000
  P-value H_0: Robust RMSEA >= 0.080                         0.899

Standardized Root Mean Square Residual:

  SRMR                                           0.057       0.057

Parameter Estimates:

  Parameterization                               Delta
  Standard errors                           Robust.sem
  Information                                 Expected
  Information saturated (h1) model        Unstructured

Latent Variables:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
  psycho =~
    Q5_P              0.740    0.017   44.811    0.000    0.740    0.740
    Q6_P              0.670    0.018   36.744    0.000    0.670    0.670
    Q7_P              0.664    0.018   36.636    0.000    0.664    0.664
    Q11_P             0.659    0.020   33.747    0.000    0.659    0.659
    Q19_P             0.860    0.012   71.456    0.000    0.860    0.860
    Q26_P             0.640    0.020   32.579    0.000    0.640    0.640
  physical =~
    Q3_F              0.508    0.026   19.363    0.000    0.508    0.508
    Q4_F              0.423    0.029   14.698    0.000    0.423    0.423
    Q10_F             0.817    0.012   66.850    0.000    0.817    0.817
    Q15_F             0.639    0.025   25.441    0.000    0.639    0.639
    Q16_F             0.593    0.023   25.768    0.000    0.593    0.593
    Q17_F             0.921    0.007  123.217    0.000    0.921    0.921
    Q18_F             0.886    0.009  102.178    0.000    0.886    0.886
  social =~
    Q20_S             0.818    0.018   44.690    0.000    0.818    0.818
    Q21_S             0.583    0.027   21.430    0.000    0.583    0.583
    Q22_S             0.751    0.020   38.204    0.000    0.751    0.751
```

```
  environment =~
    Q8_A              0.510    0.028   18.083    0.000    0.510    0.510
    Q9_A              0.624    0.024   25.625    0.000    0.624    0.624
    Q12_A             0.712    0.021   34.427    0.000    0.712    0.712
    Q13_A             0.649    0.023   28.015    0.000    0.649    0.649
    Q14_A             0.766    0.020   38.821    0.000    0.766    0.766
    Q23_A             0.571    0.027   21.322    0.000    0.571    0.571
    Q24_A             0.482    0.028   17.030    0.000    0.482    0.482
    Q25_A             0.529    0.028   19.203    0.000    0.529    0.529

Covariances:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
 .Q3_F ~~
   .Q4_F              0.388    0.025   15.711    0.000    0.388    0.497
  psycho ~~
    physical          0.909    0.010   90.825    0.000    0.909    0.909
    social            0.771    0.019   41.000    0.000    0.771    0.771
    environment       0.725    0.021   34.939    0.000    0.725    0.725
  physical ~~
    social            0.638    0.024   26.546    0.000    0.638    0.638
    environment       0.630    0.022   28.693    0.000    0.630    0.630
  social ~~
    environment       0.550    0.028   19.549    0.000    0.550    0.550

Thresholds:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
    Q5_P|t1          -2.053    0.089  -22.982    0.000   -2.053   -2.053
    Q5_P|t2          -0.989    0.046  -21.268    0.000   -0.989   -0.989
    Q5_P|t3           0.150    0.039    3.860    0.000    0.150    0.150
    Q5_P|t4           1.833    0.075   24.535    0.000    1.833    1.833
    Q6_P|t1          -1.931    0.081  -23.937    0.000   -1.931   -1.931
    Q6_P|t2          -1.319    0.054  -24.483    0.000   -1.319   -1.319
    Q6_P|t3          -0.598    0.041  -14.442    0.000   -0.598   -0.598
    Q6_P|t4           0.541    0.041   13.237    0.000    0.541    0.541
    Q7_P|t1          -2.244    0.106  -21.085    0.000   -2.244   -2.244
    Q7_P|t2          -1.238    0.052  -23.927    0.000   -1.238   -1.238
    Q7_P|t3          -0.290    0.039   -7.373    0.000   -0.290   -0.290
    Q7_P|t4           1.183    0.050   23.465    0.000    1.183    1.183
    Q11_P|t1         -2.033    0.088  -23.148    0.000   -2.033   -2.033
    Q11_P|t2         -1.296    0.053  -24.344    0.000   -1.296   -1.296
    Q11_P|t3         -0.415    0.040  -10.378    0.000   -0.415   -0.415
    Q11_P|t4          0.589    0.041   14.262    0.000    0.589    0.589
    Q19_P|t1         -1.062    0.048  -22.193    0.000   -1.062   -1.062
    Q19_P|t2         -0.405    0.040  -10.134    0.000   -0.405   -0.405
    Q19_P|t3          0.829    0.044   18.847    0.000    0.829    0.829
    Q26_P|t1         -1.846    0.075  -24.465    0.000   -1.846   -1.846
    Q26_P|t2         -1.378    0.056  -24.791    0.000   -1.378   -1.378
    Q26_P|t3         -0.796    0.044  -18.275    0.000   -0.796   -0.796
    Q26_P|t4          0.723    0.043   16.938    0.000    0.723    0.723
```

Q3_F|t1	-2.528	0.143	-17.684	0.000	-2.528	-2.528
Q3_F|t2	-1.471	0.059	-25.106	0.000	-1.471	-1.471
Q3_F|t3	-0.819	0.044	-18.676	0.000	-0.819	-0.819
Q3_F|t4	0.116	0.039	2.996	0.003	0.116	0.116
Q4_F|t1	-2.188	0.101	-21.680	0.000	-2.188	-2.188
Q4_F|t2	-1.302	0.053	-24.380	0.000	-1.302	-1.302
Q4_F|t3	-0.754	0.043	-17.522	0.000	-0.754	-0.754
Q4_F|t4	0.167	0.039	4.292	0.000	0.167	0.167
Q10_F|t1	-2.344	0.117	-19.953	0.000	-2.344	-2.344
Q10_F|t2	-1.285	0.053	-24.272	0.000	-1.285	-1.285
Q10_F|t3	-0.112	0.039	-2.872	0.004	-0.112	-0.112
Q10_F|t4	0.924	0.045	20.355	0.000	0.924	0.924
Q15_F|t1	-2.528	0.143	-17.684	0.000	-2.528	-2.528
Q15_F|t2	-1.820	0.074	-24.599	0.000	-1.820	-1.820
Q15_F|t3	-1.248	0.052	-24.007	0.000	-1.248	-1.248
Q15_F|t4	-0.167	0.039	-4.292	0.000	-0.167	-0.167
Q16_F|t1	-0.738	0.043	-17.231	0.000	-0.738	-0.738
Q16_F|t2	-0.116	0.039	-2.996	0.003	-0.116	-0.116
Q16_F|t3	0.917	0.045	20.245	0.000	0.917	0.917
Q17_F|t1	-1.070	0.048	-22.293	0.000	-1.070	-1.070
Q17_F|t2	-0.476	0.040	-11.781	0.000	-0.476	-0.476
Q17_F|t3	0.906	0.045	20.080	0.000	0.906	0.906
Q18_F|t1	-1.041	0.047	-21.941	0.000	-1.041	-1.041
Q18_F|t2	-0.479	0.040	-11.842	0.000	-0.479	-0.479
Q18_F|t3	0.793	0.044	18.217	0.000	0.793	0.793
Q20_S|t1	-1.178	0.050	-23.421	0.000	-1.178	-1.178
Q20_S|t2	-0.248	0.039	-6.326	0.000	-0.248	-0.248
Q20_S|t3	0.993	0.047	21.321	0.000	0.993	0.993
Q21_S|t1	-0.783	0.043	-18.044	0.000	-0.783	-0.783
Q21_S|t2	-0.035	0.039	-0.896	0.370	-0.035	-0.035
Q21_S|t3	0.985	0.046	21.215	0.000	0.985	0.985
Q22_S|t1	-1.150	0.050	-23.152	0.000	-1.150	-1.150
Q22_S|t2	-0.153	0.039	-3.922	0.000	-0.153	-0.153
Q22_S|t3	1.070	0.048	22.293	0.000	1.070	1.070
Q8_A|t1	-1.416	0.057	-24.945	0.000	-1.416	-1.416
Q8_A|t2	-0.492	0.041	-12.146	0.000	-0.492	-0.492
Q8_A|t3	0.558	0.041	13.600	0.000	0.558	0.558
Q8_A|t4	1.728	0.069	24.978	0.000	1.728	1.728
Q9_A|t1	-1.859	0.076	-24.391	0.000	-1.859	-1.859
Q9_A|t2	-1.096	0.049	-22.587	0.000	-1.096	-1.096
Q9_A|t3	-0.141	0.039	-3.613	0.000	-0.141	-0.141
Q9_A|t4	1.372	0.055	24.763	0.000	1.372	1.372
Q12_A|t1	-1.648	0.065	-25.169	0.000	-1.648	-1.648
Q12_A|t2	-0.947	0.046	-20.681	0.000	-0.947	-0.947
Q12_A|t3	0.080	0.039	2.069	0.039	0.080	0.080
Q12_A|t4	0.826	0.044	18.790	0.000	0.826	0.826
Q13_A|t1	-2.668	0.168	-15.900	0.000	-2.668	-2.668
Q13_A|t2	-1.603	0.064	-25.218	0.000	-1.603	-1.603
Q13_A|t3	-0.653	0.042	-15.578	0.000	-0.653	-0.653

```
    Q13_A|t4          0.692    0.042   16.349    0.000    0.692    0.692
    Q14_A|t1         -1.820    0.074  -24.599    0.000   -1.820   -1.820
    Q14_A|t2         -0.913    0.045  -20.190    0.000   -0.913   -0.913
    Q14_A|t3          0.109    0.039    2.810    0.005    0.109    0.109
    Q14_A|t4          1.291    0.053   24.309    0.000    1.291    1.291
    Q23_A|t1         -1.212    0.051  -23.722    0.000   -1.212   -1.212
    Q23_A|t2         -0.665    0.042  -15.816    0.000   -0.665   -0.665
    Q23_A|t3          0.410    0.040   10.256    0.000    0.410    0.410
    Q24_A|t1         -1.062    0.048  -22.193    0.000   -1.062   -1.062
    Q24_A|t2         -0.386    0.040   -9.705    0.000   -0.386   -0.386
    Q24_A|t3          0.748    0.043   17.406    0.000    0.748    0.748
    Q25_A|t1         -1.118    0.049  -22.826    0.000   -1.118   -1.118
    Q25_A|t2         -0.487    0.040  -12.025    0.000   -0.487   -0.487
    Q25_A|t3          0.606    0.041   14.622    0.000    0.606    0.606

Variances:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
   .Q5_P              0.452                               0.452    0.452
   .Q6_P              0.551                               0.551    0.551
   .Q7_P              0.559                               0.559    0.559
   .Q11_P             0.566                               0.566    0.566
   .Q19_P             0.260                               0.260    0.260
   .Q26_P             0.590                               0.590    0.590
   .Q3_F              0.742                               0.742    0.742
   .Q4_F              0.821                               0.821    0.821
   .Q10_F             0.333                               0.333    0.333
   .Q15_F             0.591                               0.591    0.591
   .Q16_F             0.649                               0.649    0.649
   .Q17_F             0.151                               0.151    0.151
   .Q18_F             0.214                               0.214    0.214
   .Q20_S             0.332                               0.332    0.332
   .Q21_S             0.660                               0.660    0.660
   .Q22_S             0.436                               0.436    0.436
   .Q8_A              0.740                               0.740    0.740
   .Q9_A              0.610                               0.610    0.610
   .Q12_A             0.492                               0.492    0.492
   .Q13_A             0.578                               0.578    0.578
   .Q14_A             0.414                               0.414    0.414
   .Q23_A             0.675                               0.675    0.675
   .Q24_A             0.768                               0.768    0.768
   .Q25_A             0.720                               0.720    0.720
    psycho            1.000                               1.000    1.000
    physical          1.000                               1.000    1.000
    social            1.000                               1.000    1.000
    environment       1.000                               1.000    1.000

R-Square:
                   Estimate
    Q5_P              0.548
```

```
Q6_P                 0.449
Q7_P                 0.441
Q11_P                0.434
Q19_P                0.740
Q26_P                0.410
Q3_F                 0.258
Q4_F                 0.179
Q10_F                0.667
Q15_F                0.409
Q16_F                0.351
Q17_F                0.849
Q18_F                0.786
Q20_S                0.668
Q21_S                0.340
Q22_S                0.564
Q8_A                 0.260
Q9_A                 0.390
Q12_A                0.508
Q13_A                0.422
Q14_A                0.586
Q23_A                0.325
Q24_A                0.232
Q25_A                0.280
```

Source: Article Notebook

0.5.3.2 Modification Indices

Modification indices (MI) suggest which parameters, if freed, would improve model fit the most. An MI > 3.84 is statistically significant at $= .05$.

```r
knitr::kable(modificationindices(est_4fa,
  sort. = TRUE,             # Sort by MI value
  minimum.value = 3.84,     # Only show significant MIs
  maximum.number = 20       # Limit to top 20
), digits = 3)
```

Table 3: Top modification indices for the 4-factor model

	lhs	op	rhs	mi	epc	sepc.lv	sepc.all	sepc.nox
289	Q5_P	~~	Q14_A	177.218	0.278	0.278	0.643	0.643
254	environment	=~	Q5_P	175.788	0.503	0.503	0.503	0.503
544	Q24_A	~~	Q25_A	124.552	0.282	0.282	0.379	0.379
216	physical	=~	Q5_P	119.025	-1.039	-1.039	-1.039	-1.039
467	Q17_F	~~	Q18_F	92.645	0.162	0.162	0.900	0.900
263	environment	=~	Q15_F	89.086	0.347	0.347	0.347	0.347
214	psycho	=~	Q24_A	80.033	-0.333	-0.333	-0.333	-0.333
231	physical	=~	Q24_A	78.731	-0.289	-0.289	-0.289	-0.289
200	psycho	=~	Q10_F	55.182	0.631	0.631	0.631	0.631
265	environment	=~	Q17_F	48.628	-0.251	-0.251	-0.251	-0.251

Table 3: Top modification indices for the 4-factor model

	lhs	op	rhs	mi	epc	sepc.lv	sepc.all	sepc.nox
266	environment	=~	Q18_F	44.622	-0.236	-0.236	-0.236	-0.236
280	Q5_P	~~	Q17_F	44.099	-0.193	-0.193	-0.739	-0.739
281	Q5_P	~~	Q18_F	41.613	-0.188	-0.188	-0.603	-0.603
258	environment	=~	Q19_P	37.766	-0.254	-0.254	-0.254	-0.254
428	Q10_F	~~	Q18_F	34.964	-0.123	-0.123	-0.462	-0.462
252	social	=~	Q24_A	33.765	-0.206	-0.206	-0.206	-0.206
203	psycho	=~	Q17_F	33.378	-0.549	-0.549	-0.549	-0.549
394	Q3_F	~~	Q15_F	32.487	0.181	0.181	0.273	0.273
212	psycho	=~	Q14_A	31.896	0.221	0.221	0.221	0.221
223	physical	=~	Q21_S	30.562	0.233	0.233	0.233	0.233

Source: Article Notebook

> ⚠ Interpreting Modification Indices
>
> High MI values often indicate local misfit. However, not all high MIs should be addressed—only those that make theoretical sense. In this model, we observe:
> - High MI for correlations between physical and psychological factors
> - Suggestions to allow Q5 to load on multiple factors (a red flag!)
> - Q5 appears in many high MIs, indicating it may be problematic

0.5.3.3 Residual Correlations

```r
# Correlation matrix of residuals
residuals(est_4fa, type = "cor")
```

```
$type
[1] "cor.bollen"

$cov
        Q5_P   Q6_P   Q7_P  Q11_P  Q19_P  Q26_P   Q3_F   Q4_F  Q10_F  Q15_F
Q5_P   0.000
Q6_P   0.087  0.000
Q7_P  -0.069  0.019  0.000
Q11_P -0.069 -0.002  0.028  0.000
Q19_P -0.072  0.020 -0.016 -0.012  0.000
Q26_P -0.020  0.064  0.010 -0.023  0.010  0.000
Q3_F  -0.063 -0.103 -0.024  0.065 -0.090  0.039  0.000
Q4_F  -0.025 -0.069 -0.008  0.078 -0.073  0.037  0.000  0.000
Q10_F -0.002  0.016  0.064  0.092  0.004  0.096  0.045  0.015  0.000
Q15_F -0.099 -0.064 -0.033  0.047 -0.094 -0.055  0.163  0.136 -0.031  0.000
Q16_F -0.010 -0.031 -0.011 -0.046  0.038  0.070 -0.031  0.002 -0.018 -0.088
Q17_F -0.166 -0.060  0.020 -0.037  0.034  0.002 -0.017 -0.018 -0.044 -0.086
Q18_F -0.163 -0.050  0.064 -0.042  0.047 -0.036 -0.028 -0.033 -0.086 -0.055
Q20_S  0.021  0.042 -0.027 -0.033  0.026 -0.029 -0.071 -0.090 -0.017 -0.055
```

```
Q21_S -0.007  0.012 -0.047  0.009  0.022  0.003  0.042 -0.015  0.062  0.004
Q22_S  0.041  0.026 -0.083 -0.011 -0.009 -0.061 -0.030 -0.050 -0.026 -0.041
Q8_A   0.081  0.002  0.074  0.017 -0.023  0.005  0.032  0.013  0.051  0.075
Q9_A   0.058 -0.011  0.036  0.025 -0.008 -0.007  0.031  0.044  0.080  0.088
Q12_A  0.090 -0.081 -0.030  0.033 -0.047  0.004  0.036  0.075  0.054  0.128
Q13_A -0.008 -0.009  0.028  0.027 -0.070 -0.020  0.030  0.029  0.012  0.144
Q14_A  0.219 -0.017 -0.062 -0.019 -0.062 -0.053  0.022  0.038  0.063  0.121
Q23_A  0.014 -0.015 -0.058 -0.019 -0.033 -0.025 -0.042 -0.037 -0.005  0.069
Q24_A  0.001 -0.049 -0.120 -0.076 -0.107 -0.102  0.001 -0.055 -0.087  0.023
Q25_A -0.016 -0.059 -0.037 -0.027 -0.061 -0.081  0.038 -0.011 -0.067  0.120
       Q16_F  Q17_F  Q18_F  Q20_S  Q21_S  Q22_S   Q8_A   Q9_A  Q12_A  Q13_A
Q5_P
Q6_P
Q7_P
Q11_P
Q19_P
Q26_P
Q3_F
Q4_F
Q10_F
Q15_F
Q16_F  0.000
Q17_F  0.006  0.000
Q18_F -0.043  0.040  0.000
Q20_S  0.031  0.012  0.011  0.000
Q21_S  0.111  0.041  0.056 -0.055  0.000
Q22_S  0.014 -0.046 -0.014  0.031 -0.041  0.000
Q8_A  -0.015 -0.029 -0.019  0.004 -0.045 -0.041  0.000
Q9_A   0.021 -0.044 -0.022 -0.025 -0.089  0.010  0.093  0.000
Q12_A  0.010 -0.059 -0.060 -0.066  0.004  0.026 -0.077 -0.074  0.000
Q13_A  0.000 -0.030 -0.056  0.019 -0.071  0.030 -0.072 -0.052  0.070  0.000
Q14_A  0.048 -0.050 -0.094 -0.025 -0.024  0.026 -0.078 -0.083  0.021  0.018
Q23_A -0.012 -0.088 -0.071  0.053 -0.006  0.123 -0.040  0.072 -0.058 -0.054
Q24_A -0.040 -0.082 -0.087 -0.012 -0.001  0.040  0.013  0.008  0.030  0.044
Q25_A  0.036 -0.054 -0.036 -0.004  0.011 -0.003  0.020 -0.021 -0.044 -0.032
       Q14_A  Q23_A  Q24_A  Q25_A
Q5_P
Q6_P
Q7_P
Q11_P
Q19_P
Q26_P
Q3_F
Q4_F
Q10_F
Q15_F
Q16_F
Q17_F
Q18_F
```

```
Q20_S
Q21_S
Q22_S
Q8_A
Q9_A
Q12_A
Q13_A
Q14_A  0.000
Q23_A -0.074  0.000
Q24_A -0.047  0.114  0.000
Q25_A -0.119  0.132  0.244  0.000

$mean
 Q5_P  Q6_P  Q7_P Q11_P Q19_P Q26_P  Q3_F  Q4_F Q10_F Q15_F Q16_F Q17_F Q18_F
    0     0     0     0     0     0     0     0     0     0     0     0     0
Q20_S Q21_S Q22_S  Q8_A  Q9_A Q12_A Q13_A Q14_A Q23_A Q24_A Q25_A
    0     0     0     0     0     0     0     0     0     0     0

$th
 Q5_P|t1  Q5_P|t2  Q5_P|t3  Q5_P|t4  Q6_P|t1  Q6_P|t2  Q6_P|t3  Q6_P|t4
       0        0        0        0        0        0        0        0
 Q7_P|t1  Q7_P|t2  Q7_P|t3  Q7_P|t4 Q11_P|t1 Q11_P|t2 Q11_P|t3 Q11_P|t4
       0        0        0        0        0        0        0        0
Q19_P|t1 Q19_P|t2 Q19_P|t3 Q26_P|t1 Q26_P|t2 Q26_P|t3 Q26_P|t4  Q3_F|t1
       0        0        0        0        0        0        0        0
 Q3_F|t2  Q3_F|t3  Q3_F|t4  Q4_F|t1  Q4_F|t2  Q4_F|t3  Q4_F|t4 Q10_F|t1
       0        0        0        0        0        0        0        0
Q10_F|t2 Q10_F|t3 Q10_F|t4 Q15_F|t1 Q15_F|t2 Q15_F|t3 Q15_F|t4 Q16_F|t1
       0        0        0        0        0        0        0        0
Q16_F|t2 Q16_F|t3 Q17_F|t1 Q17_F|t2 Q17_F|t3 Q18_F|t1 Q18_F|t2 Q18_F|t3
       0        0        0        0        0        0        0        0
Q20_S|t1 Q20_S|t2 Q20_S|t3 Q21_S|t1 Q21_S|t2 Q21_S|t3 Q22_S|t1 Q22_S|t2
       0        0        0        0        0        0        0        0
Q22_S|t3  Q8_A|t1  Q8_A|t2  Q8_A|t3  Q8_A|t4  Q9_A|t1  Q9_A|t2  Q9_A|t3
       0        0        0        0        0        0        0        0
 Q9_A|t4 Q12_A|t1 Q12_A|t2 Q12_A|t3 Q12_A|t4 Q13_A|t1 Q13_A|t2 Q13_A|t3
       0        0        0        0        0        0        0        0
Q13_A|t4 Q14_A|t1 Q14_A|t2 Q14_A|t3 Q14_A|t4 Q23_A|t1 Q23_A|t2 Q23_A|t3
       0        0        0        0        0        0        0        0
Q24_A|t1 Q24_A|t2 Q24_A|t3 Q25_A|t1 Q25_A|t2 Q25_A|t3
       0        0        0        0        0        0
```

Source: Article Notebook

```r
# Standardized residuals (values > |2| indicate poor local fit)
residuals(est_4fa, type = "standardized")$cov
```

```
         Q5_P    Q6_P    Q7_P   Q11_P   Q19_P   Q26_P    Q3_F    Q4_F   Q10_F
Q5_P    0.000
Q6_P    6.597   0.000
Q7_P   -4.309   1.271   0.000
```

```
Q11_P  -3.863  -0.144   1.532   0.000
Q19_P  -5.300   1.759  -1.248  -1.019   0.000
Q26_P  -0.978   3.114   0.484  -1.232   0.773   0.000
Q3_F   -2.893  -4.717  -0.935   2.686  -4.686   1.570   0.000
Q4_F   -1.167  -2.828  -0.340   3.329  -4.290   1.539   0.000   0.000
Q10_F  -0.101   1.103   5.068   6.631   0.456   5.594   2.529   0.854   0.000
Q15_F  -4.413  -2.874  -1.530   2.243  -5.180  -2.219   6.594   5.317  -1.901
Q16_F  -0.465  -1.397  -0.580  -2.255   2.611   3.136  -1.301   0.070  -1.180
Q17_F -11.034  -4.439   1.471  -2.537   4.719   0.134  -0.885  -1.037  -5.201
Q18_F -10.305  -3.972   4.432  -3.144   6.315  -2.328  -1.356  -1.795  -8.974
Q20_S   1.379   3.096  -1.370  -1.777   1.691  -1.450  -3.156  -3.733  -1.088
Q21_S  -0.430   0.660  -2.284   0.456   1.451   0.129   1.543  -0.543   3.530
Q22_S   2.737   1.745  -4.617  -0.536  -0.613  -3.319  -1.161  -1.999  -1.521
Q8_A    4.183   0.076   3.580   0.710  -1.174   0.195   1.097   0.464   2.572
Q9_A    2.887  -0.497   1.536   1.085  -0.423  -0.283   1.117   1.574   4.121
Q12_A   5.435  -4.153  -1.612   1.692  -3.074   0.192   1.425   3.259   3.256
Q13_A  -0.374  -0.408   1.384   1.355  -4.301  -0.859   1.123   1.088   0.688
Q14_A  15.152  -1.005  -3.407  -0.931  -4.229  -2.518   0.891   1.571   3.799
Q23_A   0.676  -0.630  -2.535  -0.795  -1.687  -0.961  -1.388  -1.210  -0.232
Q24_A   0.065  -1.970  -4.799  -3.161  -5.285  -3.903   0.041  -1.841  -3.972
Q25_A  -0.715  -2.443  -1.625  -1.171  -3.039  -3.327   1.348  -0.354  -3.130
        Q15_F   Q16_F   Q17_F   Q18_F   Q20_S   Q21_S   Q22_S    Q8_A    Q9_A
Q5_P
Q6_P
Q7_P
Q11_P
Q19_P
Q26_P
Q3_F
Q4_F
Q10_F
Q15_F   0.000
Q16_F  -3.569   0.000
Q17_F  -5.262   0.487   0.000
Q18_F  -2.879  -3.011   8.291   0.000
Q20_S  -2.269   1.418   0.961   0.806   0.000
Q21_S   0.153   4.523   2.762   3.393  -2.796   0.000
Q22_S  -1.598   0.583  -3.272  -0.904   3.510  -2.180   0.000
Q8_A    2.544  -0.580  -1.537  -0.967   0.171  -1.677  -1.846   0.000
Q9_A    3.191   0.840  -2.414  -1.114  -1.360  -3.705   0.502   4.159   0.000
Q12_A   5.281   0.445  -3.781  -3.464  -3.379   0.153   1.345  -3.691  -3.250
Q13_A   6.198  -0.004  -1.941  -3.304   1.023  -2.836   1.424  -3.146  -2.552
Q14_A   5.355   2.129  -3.373  -5.830  -1.432  -1.198   1.512  -3.926  -4.844
Q23_A   2.409  -0.460  -4.222  -3.336   2.493  -0.209   5.393  -1.542   3.270
Q24_A   0.826  -1.401  -3.897  -4.047  -0.518  -0.026   1.677   0.498   0.344
Q25_A   5.167   1.343  -2.670  -1.690  -0.185   0.377  -0.145   0.763  -1.002
        Q12_A   Q13_A   Q14_A   Q23_A   Q24_A   Q25_A
Q5_P
Q6_P
```

```
Q7_P
Q11_P
Q19_P
Q26_P
Q3_F
Q4_F
Q10_F
Q15_F
Q16_F
Q17_F
Q18_F
Q20_S
Q21_S
Q22_S
Q8_A
Q9_A
Q12_A   0.000
Q13_A   3.826   0.000
Q14_A   1.274   1.012   0.000
Q23_A  -2.624  -2.378  -3.317   0.000
Q24_A   1.472   2.042  -2.382   4.798   0.000
Q25_A  -2.178  -1.474  -5.714   5.725  10.514   0.000
```

Source: Article Notebook

0.5.3.4 Reliability Coefficients

```
# alpha = Cronbach's alpha
# omega = Composite reliability (preferred for CFA)
# omega2 = Bentler's omega
# omega3 = McDonald's omega
# avevar = Average Variance Extracted (should be > .5)
knitr::kable(semTools::reliability(est_4fa), digits = 3)
```

```
Warning in semTools::reliability(est_4fa):
The reliability() function was deprecated in 2022 and will cease to be included in future versio

It is replaced by the compRelSEM() function, which can estimate alpha and model-based reliability

The average variance extracted should never have been included because it is not a reliability co
For constructs with categorical indicators, Zumbo et al.`s (2007) "ordinal alpha" is calculated i
```

Table 4: Reliability coefficients for the 4-factor model

	psycho	physical	social	environment
alpha	0.820	0.823	0.687	0.787
alpha.ord	0.854	0.863	0.741	0.820
omega	0.821	0.801	0.707	0.794
omega2	0.821	0.801	0.707	0.794
omega3	0.823	0.811	0.723	0.797

Table 4: Reliability coefficients for the 4-factor model

	psycho	physical	social	environment
avevar	0.504	0.500	0.524	0.375

Source: Article Notebook

Note

All reliability coefficients exceed .70, indicating adequate internal consistency for all four factors. The AVE (average variance extracted) values are also acceptable, with most factors explaining substantial variance in their indicators.

0.5.3.5 Preliminary Evaluation

Based on the overall and local fit indices:

- **Overall fit is mediocre**, compatible with a Level 2 misspecification according to DFI standards
- **Local indices indicate**:
  - High correlation between physical and psychological domains (which may suggest a hierarchical structure)
  - Low factor loadings for Q4 (Physical) and Q24 (Environment)
  - **Warning**: Item Q5 (Psychological) appears problematic—it shows up in many high MIs and may not fit well in this factor structure

0.5.4 Path Diagrams

Path diagrams provide a visual representation of the CFA model, showing factor loadings and correlations.

0.5.4.1 Tree Layout

```
# Convert to semPlotModel first to avoid parsing errors with large ordinal models
sem_model_4fa <- semPlot::semPlotModel(est_4fa)

# Create path diagram with standardized estimates
semPlot::semPaths(
  sem_model_4fa,
  what = "std",              # Show standardized estimates
  style = "lisrel",          # LISREL-style layout
  layout = "tree",           # Tree layout
  intercepts = FALSE,        # Don't show intercepts
  thresholds = FALSE,        # Don't show thresholds (too cluttered)
  residuals = FALSE,         # Don't show residual variances
  edge.label.cex = 0.6,      # Edge label size
  label.cex = 1.3,           # Node label size
  weighted = FALSE,          # Don't weight edges by magnitude
  edge.color = "black",      # Black edges
```

```r
  node.width = 0.6,            # Node width
  mar = c(3, 1, 3, 1)          # Margins
)
```

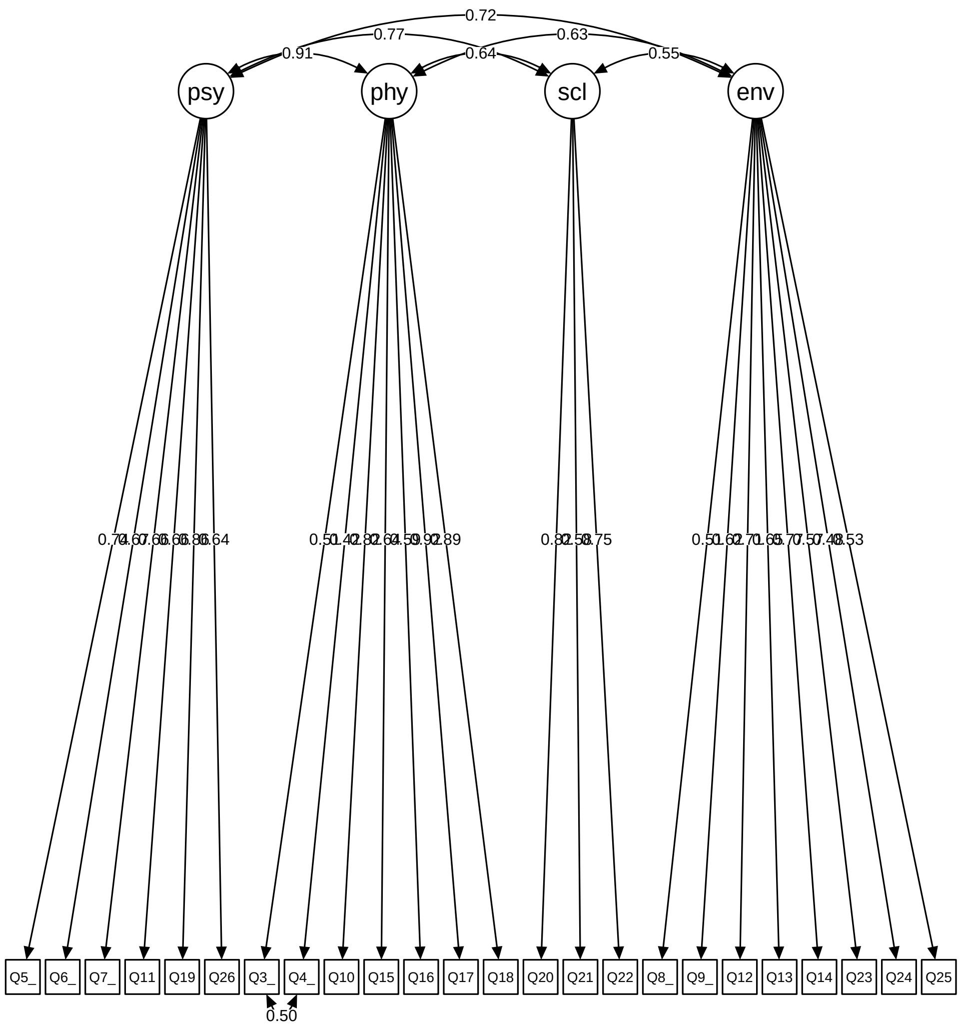


Figure 1: Path diagram of the 4-factor CFA model (tree layout)

Source: Article Notebook

0.5.4.2 Circle Layout

```r
semPlot::semPaths(
  sem_model_4fa,
```

```
  what = "std",
  style = "lisrel",
  layout = "circle",          # Circle layout shows correlations well
  intercepts = FALSE,
  thresholds = FALSE,
  residuals = FALSE,
  edge.label.cex = 1,
  label.cex = 1,
  weighted = FALSE,
  edge.color = "black",
  mar = c(1.5, 0.8, 3, 0.3)
)
```

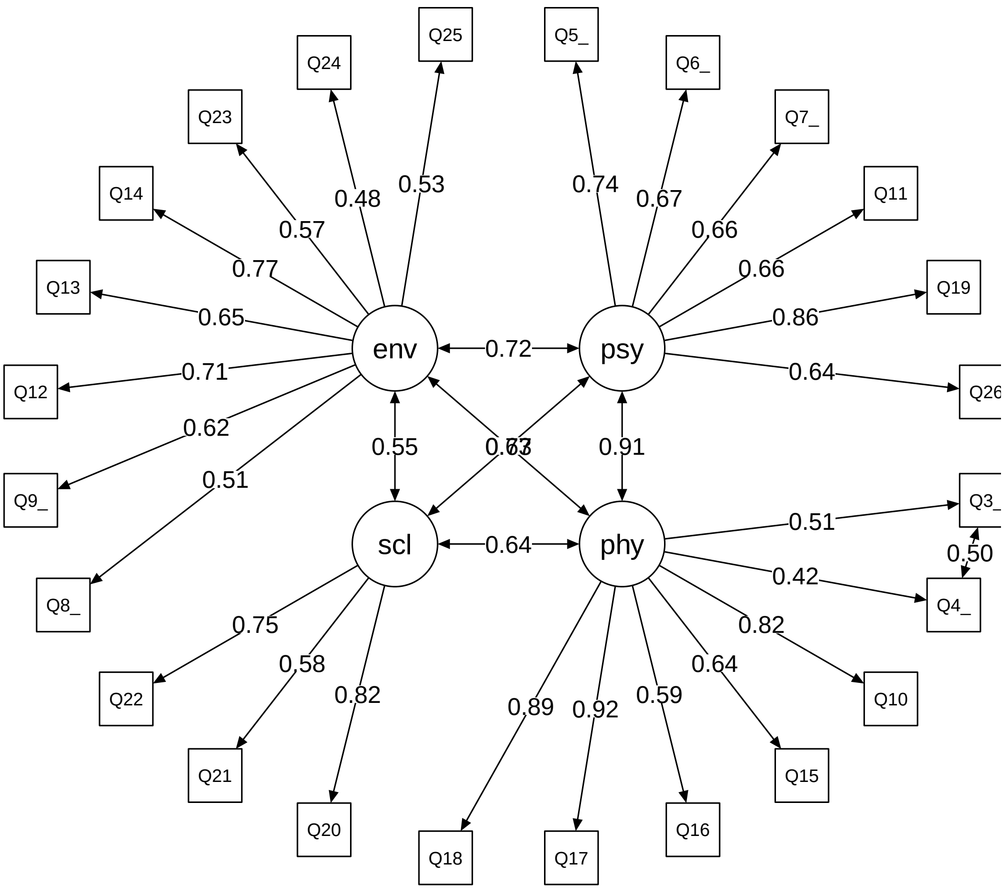


Figure 2: Path diagram of the 4-factor CFA model (circle layout)

Source: Article Notebook

0.6 4-Factor Bifactor Model

A bifactor model adds a general Quality of Life (QOL) factor loading on all items, alongside the four specific group factors (Psychological, Physical, Social, Environment). In bifactor models, all factors are orthogonal (uncorrelated).

> **Why Bifactor Models?**
>
> Bifactor models are useful when you hypothesize both a general construct (overall QOL) and specific domain constructs. However, they come with important assumptions and limitations (discussed later).

Item Q5 is excluded from this model due to convergence problems—another indication that this item is problematic.

0.6.1 Model Specification

```
# In bifactor models:
# - All factors are orthogonal (uncorrelated)
# - The general factor (QOL) must be specified FIRST
# - Q4 and Q3 correlation is removed (both load on the general factor)
spe_4f_bi <- "
QOL =~ Q6_P + Q7_P + Q11_P + Q19_P + Q26_P + Q3_F + Q4_F + Q10_F + Q15_F + Q16_F + Q17_F + Q18_F
psycho =~ Q6_P + Q7_P + Q11_P + Q19_P + Q26_P
physical =~ Q3_F + Q4_F + Q10_F + Q15_F + Q16_F + Q17_F + Q18_F
social =~ Q20_S + Q21_S + Q22_S
environment =~ Q8_A + Q9_A + Q12_A + Q13_A + Q14_A + Q23_A + Q24_A + Q25_A
"
```

Source: Article Notebook

0.6.2 Model Estimation

```
est_4f_bi <- cfa(spe_4f_bi,
  data = df0,
  orthogonal = TRUE,   # All factors are uncorrelated (required for bifactor)
  ordered = TRUE,      # All items are ordinal
  std.lv = TRUE,       # Standardize latent variables
  estimator = "WLSMV"  # DWLS estimator for ordinal data
)
```

Source: Article Notebook

0.6.3 Model Evaluation

0.6.3.1 Overall Fit and Parameter Estimates

```
summary(est_4f_bi, fit.measures = TRUE, standardized = TRUE, rsq = TRUE)
```

```
lavaan 0.6-21 ended normally after 63 iterations
```

```
  Estimator                                       DWLS
  Optimization method                           NLMINB
  Number of model parameters                       128

  Number of observations                          1047

Model Test User Model:
                                              Standard      Scaled
  Test Statistic                               862.260    1212.683
  Degrees of freedom                               207         207
  P-value (Unknown)                                 NA       0.000
  Scaling correction factor                                  0.747
  Shift parameter                                           58.000
    simple second-order correction

Model Test Baseline Model:

  Test statistic                             64072.987   24218.548
  Degrees of freedom                               253         253
  P-value                                           NA       0.000
  Scaling correction factor                                  2.663

User Model versus Baseline Model:

  Comparative Fit Index (CFI)                    0.990       0.958
  Tucker-Lewis Index (TLI)                       0.987       0.949

  Robust Comparative Fit Index (CFI)                         0.888
  Robust Tucker-Lewis Index (TLI)                            0.863

Root Mean Square Error of Approximation:

  RMSEA                                          0.055       0.068
  90 Percent confidence interval - lower         0.051       0.064
  90 Percent confidence interval - upper         0.059       0.072
  P-value H_0: RMSEA <= 0.050                    0.015       0.000
  P-value H_0: RMSEA >= 0.080                    0.000       0.000

  Robust RMSEA                                               0.078
  90 Percent confidence interval - lower                     0.073
  90 Percent confidence interval - upper                     0.082
  P-value H_0: Robust RMSEA <= 0.050                         0.000
  P-value H_0: Robust RMSEA >= 0.080                         0.225

Standardized Root Mean Square Residual:

  SRMR                                           0.049       0.049

Parameter Estimates:
```

```
  Parameterization                               Delta
  Standard errors                           Robust.sem
  Information                                 Expected
  Information saturated (h1) model        Unstructured

Latent Variables:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
  QOL =~
    Q6_P              0.632    0.022   29.265    0.000    0.632    0.632
    Q7_P              0.668    0.019   35.513    0.000    0.668    0.668
    Q11_P             0.666    0.020   33.248    0.000    0.666    0.666
    Q19_P             0.851    0.012   71.086    0.000    0.851    0.851
    Q26_P             0.634    0.022   29.242    0.000    0.634    0.634
    Q3_F              0.422    0.030   13.966    0.000    0.422    0.422
    Q4_F              0.349    0.031   11.186    0.000    0.349    0.349
    Q10_F             0.785    0.014   57.090    0.000    0.785    0.785
    Q15_F             0.584    0.027   21.563    0.000    0.584    0.584
    Q16_F             0.573    0.023   24.897    0.000    0.573    0.573
    Q17_F             0.875    0.012   74.814    0.000    0.875    0.875
    Q18_F             0.842    0.013   66.801    0.000    0.842    0.842
    Q20_S             0.598    0.021   28.122    0.000    0.598    0.598
    Q21_S             0.452    0.026   17.521    0.000    0.452    0.452
    Q22_S             0.541    0.024   22.910    0.000    0.541    0.541
    Q8_A              0.372    0.028   13.122    0.000    0.372    0.372
    Q9_A              0.454    0.026   17.366    0.000    0.454    0.454
    Q12_A             0.499    0.024   20.475    0.000    0.499    0.499
    Q13_A             0.461    0.025   18.203    0.000    0.461    0.461
    Q14_A             0.527    0.024   21.667    0.000    0.527    0.527
    Q23_A             0.380    0.029   13.230    0.000    0.380    0.380
    Q24_A             0.246    0.031    7.807    0.000    0.246    0.246
    Q25_A             0.328    0.029   11.254    0.000    0.328    0.328
  psycho =~
    Q6_P              0.696    0.479    1.454    0.146    0.696    0.696
    Q7_P              0.059    0.051    1.158    0.247    0.059    0.059
    Q11_P             0.021    0.032    0.672    0.502    0.021    0.021
    Q19_P             0.085    0.066    1.282    0.200    0.085    0.085
    Q26_P             0.135    0.102    1.330    0.184    0.135    0.135
  physical =~
    Q3_F              0.690    0.042   16.346    0.000    0.690    0.690
    Q4_F              0.606    0.039   15.419    0.000    0.606    0.606
    Q10_F             0.116    0.030    3.844    0.000    0.116    0.116
    Q15_F             0.290    0.037    7.942    0.000    0.290    0.290
    Q16_F             0.059    0.037    1.611    0.107    0.059    0.059
    Q17_F             0.268    0.031    8.771    0.000    0.268    0.268
    Q18_F             0.257    0.031    8.338    0.000    0.257    0.257
  social =~
    Q20_S             0.566    0.052   10.952    0.000    0.566    0.566
    Q21_S             0.269    0.035    7.611    0.000    0.269    0.269
```

```
    Q22_S             0.567    0.052   10.845    0.000    0.567    0.567
  environment =~
    Q8_A              0.300    0.032    9.446    0.000    0.300    0.300
    Q9_A              0.386    0.029   13.488    0.000    0.386    0.386
    Q12_A             0.485    0.026   18.608    0.000    0.485    0.485
    Q13_A             0.453    0.028   16.478    0.000    0.453    0.453
    Q14_A             0.441    0.028   15.757    0.000    0.441    0.441
    Q23_A             0.468    0.029   16.353    0.000    0.468    0.468
    Q24_A             0.603    0.028   21.426    0.000    0.603    0.603
    Q25_A             0.518    0.027   18.868    0.000    0.518    0.518

Covariances:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
  QOL ~~
    psycho            0.000                               0.000    0.000
    physical          0.000                               0.000    0.000
    social            0.000                               0.000    0.000
    environment       0.000                               0.000    0.000
  psycho ~~
    physical          0.000                               0.000    0.000
    social            0.000                               0.000    0.000
    environment       0.000                               0.000    0.000
  physical ~~
    social            0.000                               0.000    0.000
    environment       0.000                               0.000    0.000
  social ~~
    environment       0.000                               0.000    0.000

Thresholds:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
    Q6_P|t1          -1.931    0.081  -23.937    0.000   -1.931   -1.931
    Q6_P|t2          -1.319    0.054  -24.483    0.000   -1.319   -1.319
    Q6_P|t3          -0.598    0.041  -14.442    0.000   -0.598   -0.598
    Q6_P|t4           0.541    0.041   13.237    0.000    0.541    0.541
    Q7_P|t1          -2.244    0.106  -21.085    0.000   -2.244   -2.244
    Q7_P|t2          -1.238    0.052  -23.927    0.000   -1.238   -1.238
    Q7_P|t3          -0.290    0.039   -7.373    0.000   -0.290   -0.290
    Q7_P|t4           1.183    0.050   23.465    0.000    1.183    1.183
    Q11_P|t1         -2.033    0.088  -23.148    0.000   -2.033   -2.033
    Q11_P|t2         -1.296    0.053  -24.344    0.000   -1.296   -1.296
    Q11_P|t3         -0.415    0.040  -10.378    0.000   -0.415   -0.415
    Q11_P|t4          0.589    0.041   14.262    0.000    0.589    0.589
    Q19_P|t1         -1.062    0.048  -22.193    0.000   -1.062   -1.062
    Q19_P|t2         -0.405    0.040  -10.134    0.000   -0.405   -0.405
    Q19_P|t3          0.829    0.044   18.847    0.000    0.829    0.829
    Q26_P|t1         -1.846    0.075  -24.465    0.000   -1.846   -1.846
    Q26_P|t2         -1.378    0.056  -24.791    0.000   -1.378   -1.378
    Q26_P|t3         -0.796    0.044  -18.275    0.000   -0.796   -0.796
    Q26_P|t4          0.723    0.043   16.938    0.000    0.723    0.723
```

Q3_F|t1	-2.528	0.143	-17.684	0.000	-2.528	-2.528
Q3_F|t2	-1.471	0.059	-25.106	0.000	-1.471	-1.471
Q3_F|t3	-0.819	0.044	-18.676	0.000	-0.819	-0.819
Q3_F|t4	0.116	0.039	2.996	0.003	0.116	0.116
Q4_F|t1	-2.188	0.101	-21.680	0.000	-2.188	-2.188
Q4_F|t2	-1.302	0.053	-24.380	0.000	-1.302	-1.302
Q4_F|t3	-0.754	0.043	-17.522	0.000	-0.754	-0.754
Q4_F|t4	0.167	0.039	4.292	0.000	0.167	0.167
Q10_F|t1	-2.344	0.117	-19.953	0.000	-2.344	-2.344
Q10_F|t2	-1.285	0.053	-24.272	0.000	-1.285	-1.285
Q10_F|t3	-0.112	0.039	-2.872	0.004	-0.112	-0.112
Q10_F|t4	0.924	0.045	20.355	0.000	0.924	0.924
Q15_F|t1	-2.528	0.143	-17.684	0.000	-2.528	-2.528
Q15_F|t2	-1.820	0.074	-24.599	0.000	-1.820	-1.820
Q15_F|t3	-1.248	0.052	-24.007	0.000	-1.248	-1.248
Q15_F|t4	-0.167	0.039	-4.292	0.000	-0.167	-0.167
Q16_F|t1	-0.738	0.043	-17.231	0.000	-0.738	-0.738
Q16_F|t2	-0.116	0.039	-2.996	0.003	-0.116	-0.116
Q16_F|t3	0.917	0.045	20.245	0.000	0.917	0.917
Q17_F|t1	-1.070	0.048	-22.293	0.000	-1.070	-1.070
Q17_F|t2	-0.476	0.040	-11.781	0.000	-0.476	-0.476
Q17_F|t3	0.906	0.045	20.080	0.000	0.906	0.906
Q18_F|t1	-1.041	0.047	-21.941	0.000	-1.041	-1.041
Q18_F|t2	-0.479	0.040	-11.842	0.000	-0.479	-0.479
Q18_F|t3	0.793	0.044	18.217	0.000	0.793	0.793
Q20_S|t1	-1.178	0.050	-23.421	0.000	-1.178	-1.178
Q20_S|t2	-0.248	0.039	-6.326	0.000	-0.248	-0.248
Q20_S|t3	0.993	0.047	21.321	0.000	0.993	0.993
Q21_S|t1	-0.783	0.043	-18.044	0.000	-0.783	-0.783
Q21_S|t2	-0.035	0.039	-0.896	0.370	-0.035	-0.035
Q21_S|t3	0.985	0.046	21.215	0.000	0.985	0.985
Q22_S|t1	-1.150	0.050	-23.152	0.000	-1.150	-1.150
Q22_S|t2	-0.153	0.039	-3.922	0.000	-0.153	-0.153
Q22_S|t3	1.070	0.048	22.293	0.000	1.070	1.070
Q8_A|t1	-1.416	0.057	-24.945	0.000	-1.416	-1.416
Q8_A|t2	-0.492	0.041	-12.146	0.000	-0.492	-0.492
Q8_A|t3	0.558	0.041	13.600	0.000	0.558	0.558
Q8_A|t4	1.728	0.069	24.978	0.000	1.728	1.728
Q9_A|t1	-1.859	0.076	-24.391	0.000	-1.859	-1.859
Q9_A|t2	-1.096	0.049	-22.587	0.000	-1.096	-1.096
Q9_A|t3	-0.141	0.039	-3.613	0.000	-0.141	-0.141
Q9_A|t4	1.372	0.055	24.763	0.000	1.372	1.372
Q12_A|t1	-1.648	0.065	-25.169	0.000	-1.648	-1.648
Q12_A|t2	-0.947	0.046	-20.681	0.000	-0.947	-0.947
Q12_A|t3	0.080	0.039	2.069	0.039	0.080	0.080
Q12_A|t4	0.826	0.044	18.790	0.000	0.826	0.826
Q13_A|t1	-2.668	0.168	-15.900	0.000	-2.668	-2.668
Q13_A|t2	-1.603	0.064	-25.218	0.000	-1.603	-1.603
Q13_A|t3	-0.653	0.042	-15.578	0.000	-0.653	-0.653

```
    Q13_A|t4          0.692    0.042   16.349    0.000    0.692    0.692
    Q14_A|t1         -1.820    0.074  -24.599    0.000   -1.820   -1.820
    Q14_A|t2         -0.913    0.045  -20.190    0.000   -0.913   -0.913
    Q14_A|t3          0.109    0.039    2.810    0.005    0.109    0.109
    Q14_A|t4          1.291    0.053   24.309    0.000    1.291    1.291
    Q23_A|t1         -1.212    0.051  -23.722    0.000   -1.212   -1.212
    Q23_A|t2         -0.665    0.042  -15.816    0.000   -0.665   -0.665
    Q23_A|t3          0.410    0.040   10.256    0.000    0.410    0.410
    Q24_A|t1         -1.062    0.048  -22.193    0.000   -1.062   -1.062
    Q24_A|t2         -0.386    0.040   -9.705    0.000   -0.386   -0.386
    Q24_A|t3          0.748    0.043   17.406    0.000    0.748    0.748
    Q25_A|t1         -1.118    0.049  -22.826    0.000   -1.118   -1.118
    Q25_A|t2         -0.487    0.040  -12.025    0.000   -0.487   -0.487
    Q25_A|t3          0.606    0.041   14.622    0.000    0.606    0.606

Variances:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
   .Q6_P              0.116                               0.116    0.116
   .Q7_P              0.551                               0.551    0.551
   .Q11_P             0.557                               0.557    0.557
   .Q19_P             0.268                               0.268    0.268
   .Q26_P             0.580                               0.580    0.580
   .Q3_F              0.345                               0.345    0.345
   .Q4_F              0.511                               0.511    0.511
   .Q10_F             0.370                               0.370    0.370
   .Q15_F             0.575                               0.575    0.575
   .Q16_F             0.668                               0.668    0.668
   .Q17_F             0.162                               0.162    0.162
   .Q18_F             0.224                               0.224    0.224
   .Q20_S             0.322                               0.322    0.322
   .Q21_S             0.724                               0.724    0.724
   .Q22_S             0.386                               0.386    0.386
   .Q8_A              0.772                               0.772    0.772
   .Q9_A              0.645                               0.645    0.645
   .Q12_A             0.515                               0.515    0.515
   .Q13_A             0.582                               0.582    0.582
   .Q14_A             0.528                               0.528    0.528
   .Q23_A             0.636                               0.636    0.636
   .Q24_A             0.576                               0.576    0.576
   .Q25_A             0.625                               0.625    0.625
    QOL               1.000                               1.000    1.000
    psycho            1.000                               1.000    1.000
    physical          1.000                               1.000    1.000
    social            1.000                               1.000    1.000
    environment       1.000                               1.000    1.000

R-Square:
                   Estimate
    Q6_P              0.884
```

```
Q7_P                 0.449
Q11_P                0.443
Q19_P                0.732
Q26_P                0.420
Q3_F                 0.655
Q4_F                 0.489
Q10_F                0.630
Q15_F                0.425
Q16_F                0.332
Q17_F                0.838
Q18_F                0.776
Q20_S                0.678
Q21_S                0.276
Q22_S                0.614
Q8_A                 0.228
Q9_A                 0.355
Q12_A                0.485
Q13_A                0.418
Q14_A                0.472
Q23_A                0.364
Q24_A                0.424
Q25_A                0.375
```

Source: Article Notebook

0.6.3.2 Modification Indices

```
knitr::kable(modificationindices(est_4f_bi,
  sort. = TRUE, minimum.value = 3.84,
  maximum.number = 20
), digits = 3)
```

Table 5: Top modification indices for the bifactor model

	lhs	op	rhs	mi	epc	sepc.lv	sepc.all	sepc.nox
462	Q17_F	~~	Q18_F	155.946	0.215	0.215	1.128	1.128
280	environment	=~	Q15_F	114.813	0.300	0.300	0.300	0.300
387	Q3_F	~~	Q4_F	58.434	0.819	0.819	1.951	1.951
283	environment	=~	Q18_F	46.170	-0.177	-0.177	-0.177	-0.177
286	environment	=~	Q22_S	45.651	0.170	0.170	0.170	0.170
539	Q24_A	~~	Q25_A	44.606	0.231	0.231	0.385	0.385
282	environment	=~	Q17_F	42.692	-0.170	-0.170	-0.170	-0.170
166	social	~~	environment	38.239	0.196	0.196	0.196	0.196
269	social	=~	Q23_A	36.821	0.232	0.232	0.232	0.232
164	physical	~~	social	34.825	-0.274	-0.274	-0.274	-0.274
162	psycho	~~	social	33.459	0.280	0.280	0.280	0.280
252	social	=~	Q6_P	31.214	0.184	0.184	0.184	0.184
526	Q12_A	~~	Q14_A	31.078	0.152	0.152	0.292	0.292
509	Q22_S	~~	Q23_A	30.368	0.165	0.165	0.333	0.333
512	Q8_A	~~	Q9_A	29.681	0.156	0.156	0.221	0.221

Table 5: Top modification indices for the bifactor model

	lhs	op	rhs	mi	epc	sepc.lv	sepc.all	sepc.nox
391	Q3_F	~~	Q17_F	29.502	-0.244	-0.244	-1.032	-1.032
392	Q3_F	~~	Q18_F	28.622	-0.239	-0.239	-0.860	-0.860
448	Q15_F	~~	Q25_A	26.451	0.154	0.154	0.258	0.258
241	physical	=~	Q20_S	23.444	-0.174	-0.174	-0.174	-0.174
408	Q4_F	~~	Q18_F	23.223	-0.198	-0.198	-0.585	-0.585

Source: Article Notebook

0.6.3.3 Residual Correlations

```
residuals(est_4f_bi, type = "cor")
```

```
$type
[1] "cor.bollen"

$cov
        Q6_P   Q7_P  Q11_P  Q19_P  Q26_P   Q3_F   Q4_F  Q10_F  Q15_F  Q16_F
Q6_P   0.000
Q7_P   0.001  0.000
Q11_P  0.004  0.020  0.000
Q19_P  0.000 -0.018 -0.014  0.000
Q26_P -0.001  0.004 -0.026  0.010  0.000
Q3_F  -0.060  0.001  0.088 -0.053  0.067  0.000
Q4_F  -0.031  0.015  0.100 -0.038  0.062  0.037  0.000
Q10_F  0.018  0.034  0.059 -0.025  0.073  0.048  0.016  0.000
Q15_F -0.043 -0.037  0.042 -0.091 -0.053  0.041  0.027 -0.001  0.000
Q16_F -0.031 -0.036 -0.072  0.014  0.052 -0.013  0.017  0.010 -0.061  0.000
Q17_F -0.052 -0.007 -0.067  0.010 -0.016 -0.103 -0.096 -0.010 -0.085  0.035
Q18_F -0.042  0.037 -0.072  0.024 -0.054 -0.111 -0.107 -0.053 -0.055 -0.016
Q20_S  0.086 -0.008 -0.015  0.059 -0.005 -0.059 -0.078 -0.061 -0.071 -0.003
Q21_S  0.028 -0.050  0.005  0.024  0.004  0.040 -0.015  0.011 -0.022  0.072
Q22_S  0.072 -0.060  0.010  0.028 -0.034 -0.015 -0.036 -0.060 -0.051 -0.012
Q8_A   0.015  0.072  0.013 -0.021  0.006  0.038  0.019  0.021  0.064 -0.038
Q9_A   0.006  0.034  0.021 -0.005 -0.005  0.039  0.052  0.045  0.075 -0.006
Q12_A -0.050 -0.021  0.041 -0.028  0.018  0.053  0.091  0.029  0.124 -0.010
Q13_A  0.016  0.033  0.031 -0.057 -0.011  0.043  0.041 -0.016  0.137 -0.022
Q14_A  0.022 -0.045 -0.003 -0.032 -0.031  0.044  0.058  0.043  0.122  0.033
Q23_A  0.023 -0.037  0.001  0.000 -0.001 -0.020 -0.017 -0.010  0.077 -0.017
Q24_A  0.030 -0.052 -0.010 -0.016 -0.034  0.052 -0.012 -0.032  0.074 -0.001
Q25_A -0.010 -0.002  0.008 -0.010 -0.044  0.069  0.016 -0.052  0.142  0.046
       Q17_F  Q18_F  Q20_S  Q21_S  Q22_S   Q8_A   Q9_A  Q12_A  Q13_A  Q14_A
Q6_P
Q7_P
Q11_P
Q19_P
Q26_P
```

```
Q3_F
Q4_F
Q10_F
Q15_F
Q16_F
Q17_F  0.000
Q18_F  0.051  0.000
Q20_S -0.031 -0.031  0.000
Q21_S -0.011  0.005  0.000  0.000
Q22_S -0.078 -0.046  0.000  0.000  0.000
Q8_A  -0.058 -0.047  0.011 -0.049 -0.032  0.000
Q9_A  -0.079 -0.056 -0.016 -0.093  0.022  0.127  0.000
Q12_A -0.083 -0.082 -0.044  0.007  0.050 -0.045 -0.043  0.000
Q13_A -0.056 -0.082  0.035 -0.071  0.048 -0.048 -0.031  0.083  0.000
Q14_A -0.066 -0.110  0.004 -0.016  0.058 -0.015 -0.015  0.089  0.072  0.000
Q23_A -0.090 -0.073  0.082  0.005  0.153 -0.030  0.075 -0.068 -0.071 -0.044
Q24_A -0.018 -0.025  0.058  0.043  0.106 -0.014 -0.035 -0.041 -0.030 -0.074
Q25_A -0.034 -0.017  0.038  0.032  0.038  0.013 -0.040 -0.082 -0.074 -0.115
       Q23_A  Q24_A  Q25_A
Q6_P
Q7_P
Q11_P
Q19_P
Q26_P
Q3_F
Q4_F
Q10_F
Q15_F
Q16_F
Q17_F
Q18_F
Q20_S
Q21_S
Q22_S
Q8_A
Q9_A
Q12_A
Q13_A
Q14_A
Q23_A  0.000
Q24_A  0.013  0.000
Q25_A  0.067  0.107  0.000

$mean
 Q6_P  Q7_P Q11_P Q19_P Q26_P  Q3_F  Q4_F Q10_F Q15_F Q16_F Q17_F Q18_F Q20_S
    0     0     0     0     0     0     0     0     0     0     0     0     0
Q21_S Q22_S  Q8_A  Q9_A Q12_A Q13_A Q14_A Q23_A Q24_A Q25_A
    0     0     0     0     0     0     0     0     0     0
```

```
$th
 Q6_P|t1  Q6_P|t2  Q6_P|t3  Q6_P|t4  Q7_P|t1  Q7_P|t2  Q7_P|t3  Q7_P|t4 
       0        0        0        0        0        0        0        0 
Q11_P|t1 Q11_P|t2 Q11_P|t3 Q11_P|t4 Q19_P|t1 Q19_P|t2 Q19_P|t3 Q26_P|t1 
       0        0        0        0        0        0        0        0 
Q26_P|t2 Q26_P|t3 Q26_P|t4  Q3_F|t1  Q3_F|t2  Q3_F|t3  Q3_F|t4  Q4_F|t1 
       0        0        0        0        0        0        0        0 
 Q4_F|t2  Q4_F|t3  Q4_F|t4 Q10_F|t1 Q10_F|t2 Q10_F|t3 Q10_F|t4 Q15_F|t1 
       0        0        0        0        0        0        0        0 
Q15_F|t2 Q15_F|t3 Q15_F|t4 Q16_F|t1 Q16_F|t2 Q16_F|t3 Q17_F|t1 Q17_F|t2 
       0        0        0        0        0        0        0        0 
Q17_F|t3 Q18_F|t1 Q18_F|t2 Q18_F|t3 Q20_S|t1 Q20_S|t2 Q20_S|t3 Q21_S|t1 
       0        0        0        0        0        0        0        0 
Q21_S|t2 Q21_S|t3 Q22_S|t1 Q22_S|t2 Q22_S|t3  Q8_A|t1  Q8_A|t2  Q8_A|t3 
       0        0        0        0        0        0        0        0 
 Q8_A|t4  Q9_A|t1  Q9_A|t2  Q9_A|t3  Q9_A|t4 Q12_A|t1 Q12_A|t2 Q12_A|t3 
       0        0        0        0        0        0        0        0 
Q12_A|t4 Q13_A|t1 Q13_A|t2 Q13_A|t3 Q13_A|t4 Q14_A|t1 Q14_A|t2 Q14_A|t3 
       0        0        0        0        0        0        0        0 
Q14_A|t4 Q23_A|t1 Q23_A|t2 Q23_A|t3 Q24_A|t1 Q24_A|t2 Q24_A|t3 Q25_A|t1 
       0        0        0        0        0        0        0        0 
Q25_A|t2 Q25_A|t3 
       0        0 
```

Source: Article Notebook

```
residuals(est_4f_bi, type = "standardized")$cov
```

```
        Q6_P   Q7_P  Q11_P  Q19_P  Q26_P   Q3_F   Q4_F  Q10_F  Q15_F  Q16_F
Q6_P   0.000                                                               
Q7_P   0.465  0.000                                                        
Q11_P  1.218  1.088  0.000                                                 
Q19_P -0.070 -1.439 -1.092  0.000                                          
Q26_P -1.352  0.221 -1.417  1.139  0.000                                   
Q3_F  -2.793  0.036  3.730 -2.654  2.745  0.000                            
Q4_F  -1.286  0.602  4.300 -2.216  2.616  8.207  0.000                     
Q10_F  1.259  2.746  4.290 -2.879  4.405  4.470  1.261  0.000              
Q15_F -1.988 -1.774  2.035 -5.232 -2.179  3.461  1.969 -0.033  0.000       
Q16_F -1.495 -1.891 -3.502  1.026  2.339 -0.886  1.006  0.648 -2.613  0.000
Q17_F -3.871 -0.534 -4.729  1.495 -1.101 -9.217 -7.516 -1.401 -5.647  2.961
Q18_F -3.444  2.708 -5.526  3.548 -3.481 -8.571 -7.937 -6.265 -3.027 -1.159
Q20_S  5.526 -0.389 -0.819  3.706 -0.249 -2.639 -3.298 -4.095 -2.979 -0.141
Q21_S  1.443 -2.434  0.231  1.530  0.192  1.471 -0.546  0.677 -0.891  3.003
Q22_S  4.306 -3.296  0.472  1.819 -1.816 -0.616 -1.488 -3.734 -2.003 -0.538
Q8_A   0.747  3.640  0.580 -1.194  0.258  1.356  0.708  1.205  2.186 -1.475
Q9_A   0.279  1.481  0.911 -0.300 -0.213  1.463  1.916  2.599  2.719 -0.253
Q12_A -2.635 -1.100  2.139 -1.952  0.871  2.189  4.008  1.940  5.145 -0.424
Q13_A  0.755  1.650  1.566 -3.750 -0.487  1.641  1.592 -1.009  6.101 -0.966
Q14_A  1.248 -2.503 -0.168 -2.479 -1.588  1.854  2.435  2.923  5.406  1.436
Q23_A  1.003 -1.657  0.038 -0.022 -0.035 -0.682 -0.579 -0.481  2.732 -0.654
Q24_A  1.187 -2.138 -0.404 -0.869 -1.396  1.709 -0.411 -1.602  2.618 -0.019
```

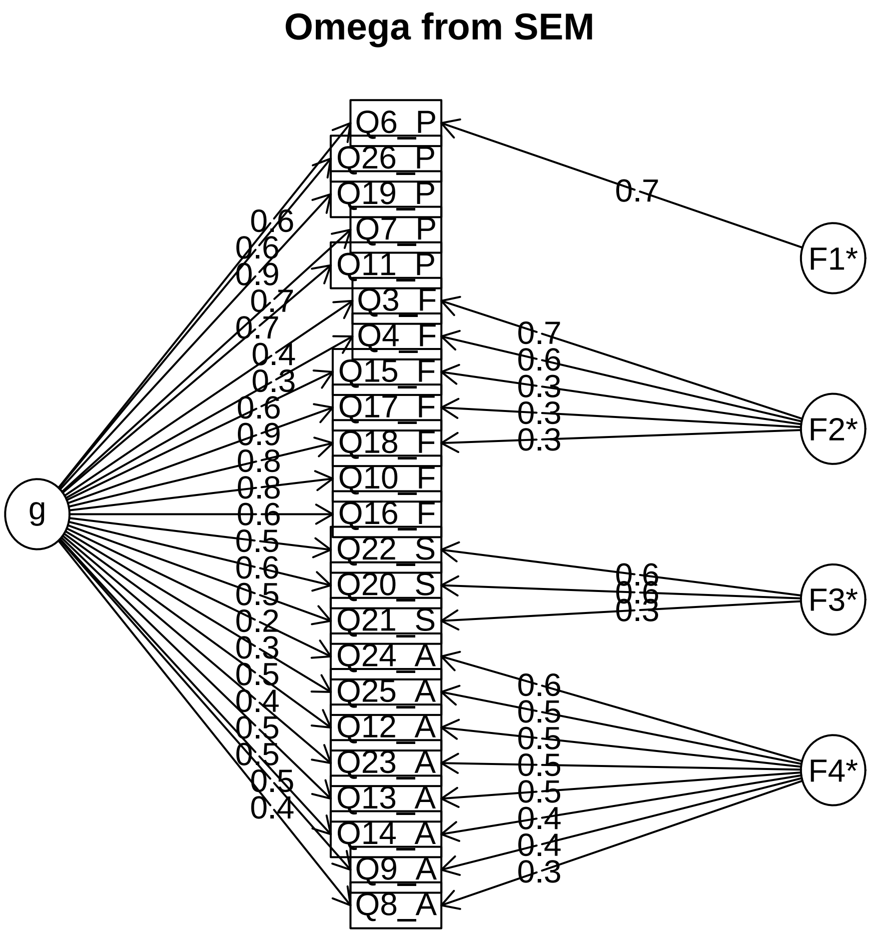
```
Q25_A -0.400 -0.070  0.350 -0.560 -1.851  2.510  0.542 -2.784  6.072  1.720
       Q17_F  Q18_F  Q20_S  Q21_S  Q22_S   Q8_A   Q9_A  Q12_A  Q13_A  Q14_A
Q6_P
Q7_P
Q11_P
Q19_P
Q26_P
Q3_F
Q4_F
Q10_F
Q15_F
Q16_F
Q17_F  0.000
Q18_F 11.030  0.000
Q20_S -2.390 -2.183  0.000
Q21_S -0.751  0.283  0.000  0.000
Q22_S -5.314 -2.783  0.000  0.000  0.000
Q8_A  -3.698 -2.818  0.510 -1.866 -1.383  0.000
Q9_A  -4.975 -3.288 -0.803 -3.876  1.037  6.354  0.000
Q12_A -5.788 -5.266 -2.155  0.282  2.465 -2.411 -2.352  0.000
Q13_A -4.028 -5.157  1.780 -2.800  2.159 -2.394 -1.789  5.859  0.000
Q14_A -5.230 -7.764  0.232 -0.759  3.017 -0.819 -0.972  7.061  4.755  0.000
Q23_A -4.872 -3.939  3.651  0.190  6.286 -1.309  4.200 -3.736 -3.758 -2.321
Q24_A -0.969 -1.359  2.377  1.477  4.233 -0.725 -2.084 -2.898 -1.977 -4.878
Q25_A -1.978 -0.932  1.583  1.142  1.512  0.611 -2.327 -4.968 -4.168 -6.460
       Q23_A  Q24_A  Q25_A
Q6_P
Q7_P
Q11_P
Q19_P
Q26_P
Q3_F
Q4_F
Q10_F
Q15_F
Q16_F
Q17_F
Q18_F
Q20_S
Q21_S
Q22_S
Q8_A
Q9_A
Q12_A
Q13_A
Q14_A
Q23_A  0.000
Q24_A  0.783  0.000
Q25_A  3.889  7.009  0.000
```

Source: Article Notebook

0.6.3.4 Reliability Coefficients

For bifactor models, omega coefficients from `psych::omegaFromSem()` provide the most appropriate reliability estimates, particularly omega hierarchical (H), which estimates the reliability attributable to the general factor alone.

```r
# Omega coefficients for bifactor model
psych::omegaFromSem(fit = est_4f_bi)
```




```
 The following analyses were done using the  lavaan  package 

 Omega Hierarchical from a confirmatory model using sem =  0.84
 Omega Total  from a confirmatory model using sem =  0.94 
With loadings of 
         g  F1*  F2*  F3*  F4*   h2   u2   p2
Q6_P  0.63 0.70                0.88 0.12 0.45
Q7_P  0.67                     0.45 0.55 1.00
```

```
Q11_P 0.67                     0.44 0.56 1.02
Q19_P 0.85                     0.73 0.27 0.99
Q26_P 0.63                     0.42 0.58 0.95
Q3_F  0.42      0.69           0.65 0.35 0.27
Q4_F  0.35      0.61           0.49 0.51 0.25
Q10_F 0.79                     0.63 0.37 0.99
Q15_F 0.58      0.29           0.42 0.58 0.80
Q16_F 0.57                     0.33 0.67 0.98
Q17_F 0.88      0.27           0.84 0.16 0.92
Q18_F 0.84      0.26           0.78 0.22 0.90
Q20_S 0.60           0.57      0.68 0.32 0.53
Q21_S 0.45           0.27      0.28 0.72 0.72
Q22_S 0.54           0.57      0.61 0.39 0.48
Q8_A  0.37                0.30 0.23 0.77 0.60
Q9_A  0.45                0.39 0.36 0.64 0.56
Q12_A 0.50                0.49 0.48 0.52 0.52
Q13_A 0.46                0.45 0.42 0.58 0.50
Q14_A 0.53                0.44 0.47 0.53 0.60
Q23_A 0.38                0.47 0.36 0.64 0.40
Q24_A 0.25                0.60 0.42 0.58 0.15
Q25_A 0.33                0.52 0.38 0.62 0.29

With sum of squared loadings of:
   g  F1*  F2*  F3*  F4*
7.73 0.51 1.08 0.71 1.72

The degrees of freedom of the confirmatory model are  207  and the fit is  862.2597  with p =  NA
general/max  4.48   max/min =   3.36
mean percent general =  0.65    with sd =  0.28 and cv of  0.43
Explained Common Variance of the general factor =  0.66

Measures of factor score adequacy
                                                 g F1*  F2*  F3*  F4*
Correlation of scores with factors            0.97 0.9 0.87 0.83 0.91
Multiple R square of scores with factors      0.94 0.8 0.76 0.69 0.83
Minimum correlation of factor score estimates 0.89 0.6 0.52 0.37 0.67

 Total, General and Subset omega for each subset
                                                 g  F1*  F2*  F3*  F4*
Omega total for total scores and subscales    0.94 0.86 0.93 0.76 0.85
Omega general for total scores and subscales  0.84 0.80 0.73 0.43 0.38
Omega group for total scores and subscales    0.11 0.07 0.19 0.33 0.47

To get the standard sem fit statistics, ask for summary on the fitted object
```

Source: Article Notebook

0.6.3.5 Preliminary Evaluation

- **Overall fit has improved** compared to the original 4-factor model, but remains mediocre (Level 2 of DFI)

- The **general (G) factor is statistically relevant** and explains substantial variance in the observed variables
- The **Psychological specific factor loses importance** when the general factor is included—most of its variance is captured by the general QOL factor
- This pattern suggests that quality of life may be better conceptualized as a hierarchical construct

0.6.4 Path Diagram

```
# Convert to semPlotModel first
sem_model_bi <- semPlot::semPlotModel(est_4f_bi)

# Bifactor diagrams use tree3 layout for best visualization
semPlot::semPaths(
  sem_model_bi,
  what = "std",
  style = "lisrel",
  layout = "tree3",          # tree3 layout is best for bifactor models
  intercepts = FALSE,
  thresholds = FALSE,
  residuals = FALSE,
  edge.label.cex = 1.1,
  label.cex = 1.2,
  weighted = FALSE,
  edge.color = "black",
  bifactor = "QOL",          # Specify the general factor
  edge.label.position = 0.7,
  mar = c(3, 0.8, 3, 1)
)
```

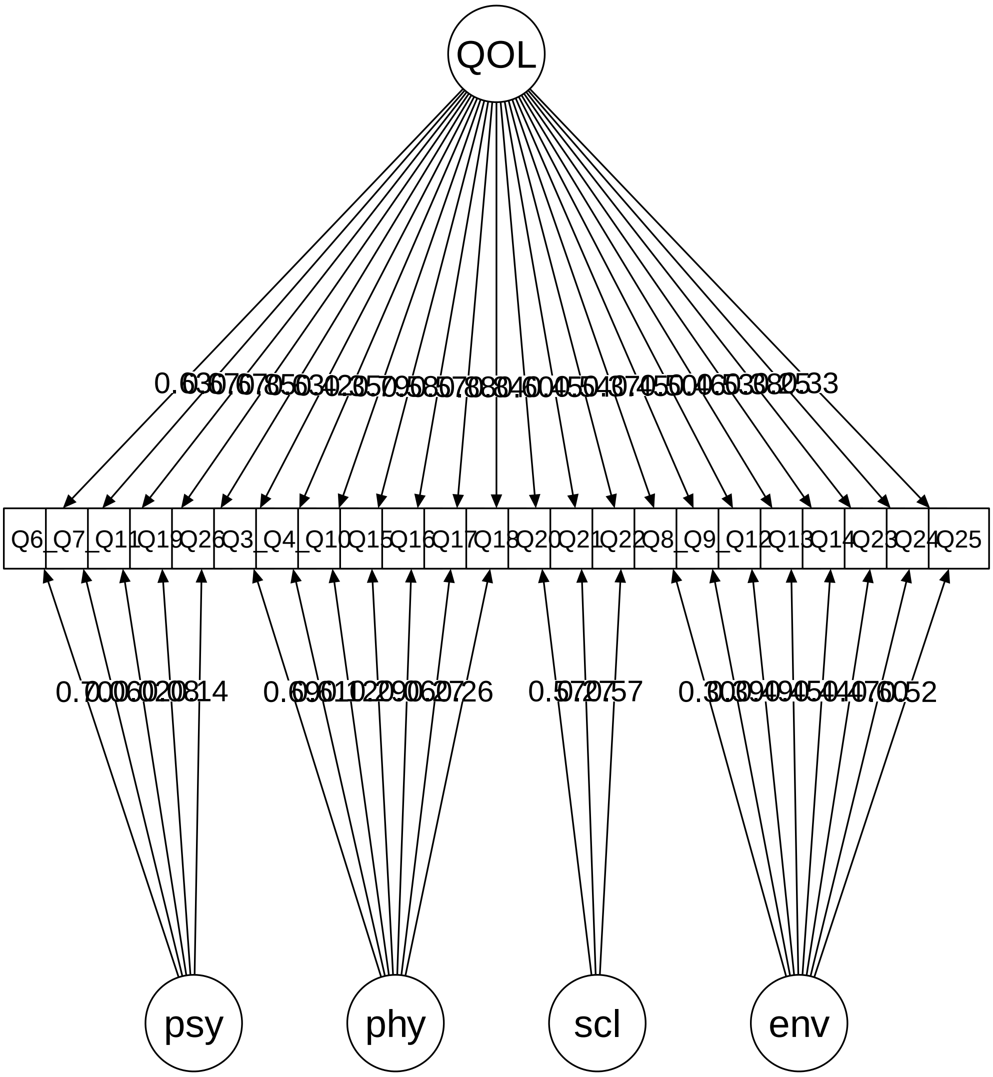


Figure 3: Path diagram of the bifactor CFA model

Source: Article Notebook

0.7 Second-Order CFA Model

A second-order (hierarchical) model specifies that a general QOL factor loads on the four first-order factors (Psychological, Physical, Social, Environment), which in turn load on the observed items. This differs from the bifactor model where the general factor directly loads on the items.

Item Q5 is excluded for comparability with the bifactor model. The psychological factor variance must be fixed to zero due to convergence problems (empirical underidentification).

0.7.1 Model Specification

```
# Second-order model:
# - First-order factors load on items
# - General QOL factor loads on first-order factors
# - First-order factors can be correlated (but this is absorbed in the second-order structure)
spe_4f_2or <- "
psycho =~ Q6_P + Q7_P + Q11_P + Q19_P + Q26_P
physical =~ Q3_F + Q4_F + Q10_F + Q15_F + Q16_F + Q17_F + Q18_F
social =~ Q20_S + Q21_S + Q22_S
environment =~ Q8_A + Q9_A + Q12_A + Q13_A + Q14_A + Q23_A + Q24_A + Q25_A
QOL =~ psycho + physical + social + environment
Q3_F ~~ Q4_F
psycho ~~ 0*psycho  # Fix psycho variance to 0 due to convergence issues
"
```

Source: Article Notebook

> ⚠ Convergence Issues
>
> Fixing the psychological factor variance to zero is necessary for model convergence. This suggests that nearly all variance in the psychological domain is explained by the general QOL factor, consistent with our bifactor results where the psychological specific factor had minimal importance.

0.7.2 Model Estimation

```
est_4f_2or <- cfa(
  spe_4f_2or,
  data = df0,
  ordered = TRUE,       # All items are ordinal
  estimator = "WLSMV"   # DWLS estimator for ordinal data
  # Note: std.lv not used here due to the second-order structure
)
```

Source: Article Notebook

0.7.3 Model Evaluation

0.7.3.1 Overall Fit and Parameter Estimates

```
summary(est_4f_2or, fit.measures = TRUE, standardized = TRUE, rsq = TRUE)
```

```
lavaan 0.6-21 ended normally after 44 iterations
```

```
  Estimator                                         DWLS
  Optimization method                             NLMINB
  Number of model parameters                         109

  Number of observations                            1047

Model Test User Model:
                                              Standard      Scaled
  Test Statistic                              1007.178    1285.170
  Degrees of freedom                               226         226
  P-value (Unknown)                                 NA       0.000
  Scaling correction factor                                  0.827
  Shift parameter                                           66.876
    simple second-order correction

Model Test Baseline Model:

  Test statistic                             64072.987   24218.548
  Degrees of freedom                               253         253
  P-value                                           NA       0.000
  Scaling correction factor                                  2.663

User Model versus Baseline Model:

  Comparative Fit Index (CFI)                    0.988       0.956
  Tucker-Lewis Index (TLI)                       0.986       0.951

  Robust Comparative Fit Index (CFI)                         0.888
  Robust Tucker-Lewis Index (TLI)                            0.875

Root Mean Square Error of Approximation:

  RMSEA                                          0.057       0.067
  90 Percent confidence interval - lower         0.054       0.063
  90 Percent confidence interval - upper         0.061       0.071
  P-value H_0: RMSEA <= 0.050                    0.000       0.000
  P-value H_0: RMSEA >= 0.080                    0.000       0.000

  Robust RMSEA                                               0.075
  90 Percent confidence interval - lower                     0.070
  90 Percent confidence interval - upper                     0.079
  P-value H_0: Robust RMSEA <= 0.050                         0.000
  P-value H_0: Robust RMSEA >= 0.080                         0.018

Standardized Root Mean Square Residual:

  SRMR                                           0.054       0.054

Parameter Estimates:
```

```
  Parameterization                               Delta
  Standard errors                           Robust.sem
  Information                                 Expected
  Information saturated (h1) model        Unstructured

Latent Variables:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
  psycho =~
    Q6_P              1.000                               0.660    0.660
    Q7_P              1.027    0.039   26.007    0.000    0.678    0.678
    Q11_P             1.017    0.039   26.076    0.000    0.671    0.671
    Q19_P             1.322    0.041   32.224    0.000    0.872    0.872
    Q26_P             0.982    0.038   25.960    0.000    0.648    0.648
  physical =~
    Q3_F              1.000                               0.509    0.509
    Q4_F              0.828    0.057   14.643    0.000    0.422    0.422
    Q10_F             1.591    0.080   19.823    0.000    0.811    0.811
    Q15_F             1.261    0.071   17.707    0.000    0.642    0.642
    Q16_F             1.157    0.070   16.418    0.000    0.590    0.590
    Q17_F             1.811    0.092   19.626    0.000    0.923    0.923
    Q18_F             1.743    0.089   19.580    0.000    0.888    0.888
  social =~
    Q20_S             1.000                               0.818    0.818
    Q21_S             0.718    0.038   19.118    0.000    0.588    0.588
    Q22_S             0.913    0.036   25.194    0.000    0.747    0.747
  environment =~
    Q8_A              1.000                               0.510    0.510
    Q9_A              1.234    0.078   15.732    0.000    0.630    0.630
    Q12_A             1.401    0.088   15.882    0.000    0.715    0.715
    Q13_A             1.298    0.086   15.037    0.000    0.663    0.663
    Q14_A             1.433    0.090   15.929    0.000    0.731    0.731
    Q23_A             1.135    0.081   13.958    0.000    0.579    0.579
    Q24_A             0.963    0.075   12.769    0.000    0.492    0.492
    Q25_A             1.060    0.077   13.750    0.000    0.541    0.541
  QOL =~
    psycho            1.000                               1.000    1.000
    physical          0.704    0.043   16.526    0.000    0.912    0.912
    social            0.918    0.038   23.864    0.000    0.740    0.740
    environment       0.529    0.036   14.566    0.000    0.685    0.685

Covariances:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
 .Q3_F ~~
   .Q4_F              0.387    0.025   15.655    0.000    0.387    0.497

Thresholds:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
    Q6_P|t1          -1.931    0.081  -23.937    0.000   -1.931   -1.931
```

Q6_P|t2	-1.319	0.054	-24.483	0.000	-1.319	-1.319
Q6_P|t3	-0.598	0.041	-14.442	0.000	-0.598	-0.598
Q6_P|t4	0.541	0.041	13.237	0.000	0.541	0.541
Q7_P|t1	-2.244	0.106	-21.085	0.000	-2.244	-2.244
Q7_P|t2	-1.238	0.052	-23.927	0.000	-1.238	-1.238
Q7_P|t3	-0.290	0.039	-7.373	0.000	-0.290	-0.290
Q7_P|t4	1.183	0.050	23.465	0.000	1.183	1.183
Q11_P|t1	-2.033	0.088	-23.148	0.000	-2.033	-2.033
Q11_P|t2	-1.296	0.053	-24.344	0.000	-1.296	-1.296
Q11_P|t3	-0.415	0.040	-10.378	0.000	-0.415	-0.415
Q11_P|t4	0.589	0.041	14.262	0.000	0.589	0.589
Q19_P|t1	-1.062	0.048	-22.193	0.000	-1.062	-1.062
Q19_P|t2	-0.405	0.040	-10.134	0.000	-0.405	-0.405
Q19_P|t3	0.829	0.044	18.847	0.000	0.829	0.829
Q26_P|t1	-1.846	0.075	-24.465	0.000	-1.846	-1.846
Q26_P|t2	-1.378	0.056	-24.791	0.000	-1.378	-1.378
Q26_P|t3	-0.796	0.044	-18.275	0.000	-0.796	-0.796
Q26_P|t4	0.723	0.043	16.938	0.000	0.723	0.723
Q3_F|t1	-2.528	0.143	-17.684	0.000	-2.528	-2.528
Q3_F|t2	-1.471	0.059	-25.106	0.000	-1.471	-1.471
Q3_F|t3	-0.819	0.044	-18.676	0.000	-0.819	-0.819
Q3_F|t4	0.116	0.039	2.996	0.003	0.116	0.116
Q4_F|t1	-2.188	0.101	-21.680	0.000	-2.188	-2.188
Q4_F|t2	-1.302	0.053	-24.380	0.000	-1.302	-1.302
Q4_F|t3	-0.754	0.043	-17.522	0.000	-0.754	-0.754
Q4_F|t4	0.167	0.039	4.292	0.000	0.167	0.167
Q10_F|t1	-2.344	0.117	-19.953	0.000	-2.344	-2.344
Q10_F|t2	-1.285	0.053	-24.272	0.000	-1.285	-1.285
Q10_F|t3	-0.112	0.039	-2.872	0.004	-0.112	-0.112
Q10_F|t4	0.924	0.045	20.355	0.000	0.924	0.924
Q15_F|t1	-2.528	0.143	-17.684	0.000	-2.528	-2.528
Q15_F|t2	-1.820	0.074	-24.599	0.000	-1.820	-1.820
Q15_F|t3	-1.248	0.052	-24.007	0.000	-1.248	-1.248
Q15_F|t4	-0.167	0.039	-4.292	0.000	-0.167	-0.167
Q16_F|t1	-0.738	0.043	-17.231	0.000	-0.738	-0.738
Q16_F|t2	-0.116	0.039	-2.996	0.003	-0.116	-0.116
Q16_F|t3	0.917	0.045	20.245	0.000	0.917	0.917
Q17_F|t1	-1.070	0.048	-22.293	0.000	-1.070	-1.070
Q17_F|t2	-0.476	0.040	-11.781	0.000	-0.476	-0.476
Q17_F|t3	0.906	0.045	20.080	0.000	0.906	0.906
Q18_F|t1	-1.041	0.047	-21.941	0.000	-1.041	-1.041
Q18_F|t2	-0.479	0.040	-11.842	0.000	-0.479	-0.479
Q18_F|t3	0.793	0.044	18.217	0.000	0.793	0.793
Q20_S|t1	-1.178	0.050	-23.421	0.000	-1.178	-1.178
Q20_S|t2	-0.248	0.039	-6.326	0.000	-0.248	-0.248
Q20_S|t3	0.993	0.047	21.321	0.000	0.993	0.993
Q21_S|t1	-0.783	0.043	-18.044	0.000	-0.783	-0.783
Q21_S|t2	-0.035	0.039	-0.896	0.370	-0.035	-0.035
Q21_S|t3	0.985	0.046	21.215	0.000	0.985	0.985

```
    Q22_S|t1          -1.150    0.050  -23.152    0.000   -1.150   -1.150
    Q22_S|t2          -0.153    0.039   -3.922    0.000   -0.153   -0.153
    Q22_S|t3           1.070    0.048   22.293    0.000    1.070    1.070
    Q8_A|t1           -1.416    0.057  -24.945    0.000   -1.416   -1.416
    Q8_A|t2           -0.492    0.041  -12.146    0.000   -0.492   -0.492
    Q8_A|t3            0.558    0.041   13.600    0.000    0.558    0.558
    Q8_A|t4            1.728    0.069   24.978    0.000    1.728    1.728
    Q9_A|t1           -1.859    0.076  -24.391    0.000   -1.859   -1.859
    Q9_A|t2           -1.096    0.049  -22.587    0.000   -1.096   -1.096
    Q9_A|t3           -0.141    0.039   -3.613    0.000   -0.141   -0.141
    Q9_A|t4            1.372    0.055   24.763    0.000    1.372    1.372
    Q12_A|t1          -1.648    0.065  -25.169    0.000   -1.648   -1.648
    Q12_A|t2          -0.947    0.046  -20.681    0.000   -0.947   -0.947
    Q12_A|t3           0.080    0.039    2.069    0.039    0.080    0.080
    Q12_A|t4           0.826    0.044   18.790    0.000    0.826    0.826
    Q13_A|t1          -2.668    0.168  -15.900    0.000   -2.668   -2.668
    Q13_A|t2          -1.603    0.064  -25.218    0.000   -1.603   -1.603
    Q13_A|t3          -0.653    0.042  -15.578    0.000   -0.653   -0.653
    Q13_A|t4           0.692    0.042   16.349    0.000    0.692    0.692
    Q14_A|t1          -1.820    0.074  -24.599    0.000   -1.820   -1.820
    Q14_A|t2          -0.913    0.045  -20.190    0.000   -0.913   -0.913
    Q14_A|t3           0.109    0.039    2.810    0.005    0.109    0.109
    Q14_A|t4           1.291    0.053   24.309    0.000    1.291    1.291
    Q23_A|t1          -1.212    0.051  -23.722    0.000   -1.212   -1.212
    Q23_A|t2          -0.665    0.042  -15.816    0.000   -0.665   -0.665
    Q23_A|t3           0.410    0.040   10.256    0.000    0.410    0.410
    Q24_A|t1          -1.062    0.048  -22.193    0.000   -1.062   -1.062
    Q24_A|t2          -0.386    0.040   -9.705    0.000   -0.386   -0.386
    Q24_A|t3           0.748    0.043   17.406    0.000    0.748    0.748
    Q25_A|t1          -1.118    0.049  -22.826    0.000   -1.118   -1.118
    Q25_A|t2          -0.487    0.040  -12.025    0.000   -0.487   -0.487
    Q25_A|t3           0.606    0.041   14.622    0.000    0.606    0.606

Variances:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
   .psycho            0.000                               0.000    0.000
   .Q6_P              0.564                               0.564    0.564
   .Q7_P              0.540                               0.540    0.540
   .Q11_P             0.550                               0.550    0.550
   .Q19_P             0.239                               0.239    0.239
   .Q26_P             0.580                               0.580    0.580
   .Q3_F              0.740                               0.740    0.740
   .Q4_F              0.822                               0.822    0.822
   .Q10_F             0.343                               0.343    0.343
   .Q15_F             0.587                               0.587    0.587
   .Q16_F             0.652                               0.652    0.652
   .Q17_F             0.148                               0.148    0.148
   .Q18_F             0.211                               0.211    0.211
   .Q20_S             0.330                               0.330    0.330
```

```
   .Q21_S             0.654                               0.654    0.654
   .Q22_S             0.442                               0.442    0.442
   .Q8_A              0.739                               0.739    0.739
   .Q9_A              0.603                               0.603    0.603
   .Q12_A             0.489                               0.489    0.489
   .Q13_A             0.561                               0.561    0.561
   .Q14_A             0.465                               0.465    0.465
   .Q23_A             0.664                               0.664    0.664
   .Q24_A             0.758                               0.758    0.758
   .Q25_A             0.707                               0.707    0.707
   .physical          0.044    0.006    7.418    0.000    0.168    0.168
   .social            0.303    0.025   12.191    0.000    0.452    0.452
   .environment       0.138    0.017    8.255    0.000    0.531    0.531
    QOL               0.436    0.025   17.121    0.000    1.000    1.000

R-Square:
                   Estimate
    psycho            1.000
    Q6_P              0.436
    Q7_P              0.460
    Q11_P             0.450
    Q19_P             0.761
    Q26_P             0.420
    Q3_F              0.260
    Q4_F              0.178
    Q10_F             0.657
    Q15_F             0.413
    Q16_F             0.348
    Q17_F             0.852
    Q18_F             0.789
    Q20_S             0.670
    Q21_S             0.346
    Q22_S             0.558
    Q8_A              0.261
    Q9_A              0.397
    Q12_A             0.511
    Q13_A             0.439
    Q14_A             0.535
    Q23_A             0.336
    Q24_A             0.242
    Q25_A             0.293
    physical          0.832
    social            0.548
    environment       0.469
```

Source: Article Notebook

0.7.3.2 Modification Indices

```r
knitr::kable(modificationindices(est_4f_2or,
  sort. = TRUE, minimum.value = 3.84,
  maximum.number = 20
), digits = 3)
```

Table 6: Top modification indices for the second-order model

	lhs	op	rhs	mi	epc	sepc.lv	sepc.all	sepc.nox
533	Q24_A	~~	Q25_A	115.469	0.274	0.274	0.374	0.374
252	environment	=~	Q15_F	94.778	0.673	0.343	0.343	0.343
456	Q17_F	~~	Q18_F	82.991	0.156	0.156	0.880	0.880
206	psycho	=~	Q24_A	74.190	-0.449	-0.296	-0.296	-0.296
280	QOL	=~	Q24_A	74.190	-0.449	-0.296	-0.296	-0.296
222	physical	=~	Q24_A	72.314	-0.523	-0.267	-0.267	-0.267
192	psycho	=~	Q10_F	54.814	1.152	0.760	0.760	0.760
266	QOL	=~	Q10_F	54.814	1.152	0.760	0.760	0.760
255	environment	=~	Q18_F	36.927	-0.394	-0.201	-0.201	-0.201
254	environment	=~	Q17_F	36.682	-0.395	-0.202	-0.202	-0.202
417	Q10_F	~~	Q18_F	32.049	-0.119	-0.119	-0.441	-0.441
383	Q3_F	~~	Q15_F	31.584	0.179	0.179	0.271	0.271
195	psycho	=~	Q17_F	28.227	-0.929	-0.613	-0.613	-0.613
269	QOL	=~	Q17_F	28.227	-0.929	-0.613	-0.613	-0.613
242	social	=~	Q24_A	27.741	-0.206	-0.169	-0.169	-0.169
503	Q22_S	~~	Q23_A	24.588	0.147	0.147	0.270	0.270
532	Q23_A	~~	Q25_A	24.194	0.141	0.141	0.205	0.205
208	physical	=~	Q6_P	22.864	-1.054	-0.537	-0.537	-0.537
215	physical	=~	Q22_S	22.662	-0.599	-0.305	-0.305	-0.305
438	Q15_F	~~	Q13_A	22.272	0.148	0.148	0.258	0.258

Source: Article Notebook

0.7.3.3 Residual Correlations

```r
residuals(est_4f_2or, type = "cor")
```

```
$type
[1] "cor.bollen"

$cov
        Q6_P   Q7_P  Q11_P  Q19_P  Q26_P   Q3_F   Q4_F  Q10_F  Q15_F  Q16_F
Q6_P   0.000
Q7_P   0.017  0.000
Q11_P -0.003  0.011  0.000
Q19_P  0.021 -0.036 -0.031  0.000
Q26_P  0.066 -0.004 -0.036 -0.004  0.000
Q3_F  -0.100 -0.032  0.057 -0.099  0.033  0.000
Q4_F  -0.065 -0.014  0.073 -0.077  0.034  0.000  0.000
Q10_F  0.026  0.056  0.085 -0.002  0.092  0.047  0.018  0.000
```

```
Q15_F -0.061 -0.044  0.037 -0.105 -0.063  0.161  0.136 -0.029  0.000
Q16_F -0.024 -0.018 -0.052  0.033  0.066 -0.030  0.004 -0.011 -0.088  0.000
Q17_F -0.054  0.006 -0.050  0.020 -0.007 -0.019 -0.017 -0.040 -0.089  0.008
Q18_F -0.045  0.050 -0.055  0.034 -0.045 -0.030 -0.033 -0.082 -0.059 -0.041
Q20_S  0.064 -0.019 -0.024  0.039 -0.019 -0.088 -0.103 -0.040 -0.077  0.014
Q21_S  0.026 -0.043  0.013  0.029  0.008  0.029 -0.025  0.044 -0.013  0.097
Q22_S  0.049 -0.074 -0.001  0.006 -0.049 -0.044 -0.061 -0.044 -0.059  0.000
Q8_A   0.019  0.083  0.026 -0.009  0.015  0.032  0.014  0.055  0.076 -0.013
Q9_A   0.008  0.045  0.033  0.005  0.003  0.031  0.044  0.082  0.087  0.022
Q12_A -0.058 -0.019  0.045 -0.030  0.018  0.036  0.077  0.059  0.128  0.013
Q13_A  0.008  0.033  0.033 -0.060 -0.012  0.027  0.027  0.011  0.140 -0.002
Q14_A  0.024 -0.033  0.011 -0.021 -0.022  0.034  0.049  0.087  0.136  0.065
Q23_A  0.001 -0.052 -0.012 -0.023 -0.017 -0.044 -0.037 -0.005  0.066 -0.013
Q24_A -0.037 -0.116 -0.072 -0.100 -0.097 -0.001 -0.056 -0.088  0.020 -0.041
Q25_A -0.047 -0.034 -0.022 -0.054 -0.076  0.035 -0.012 -0.069  0.116  0.035
       Q17_F  Q18_F  Q20_S  Q21_S  Q22_S   Q8_A   Q9_A  Q12_A  Q13_A  Q14_A
Q6_P
Q7_P
Q11_P
Q19_P
Q26_P
Q3_F
Q4_F
Q10_F
Q15_F
Q16_F
Q17_F  0.000
Q18_F  0.037  0.000
Q20_S -0.018 -0.018  0.000
Q21_S  0.018  0.033 -0.059  0.000
Q22_S -0.070 -0.038  0.033 -0.043  0.000
Q8_A  -0.027 -0.017  0.021 -0.033 -0.024  0.000
Q9_A  -0.044 -0.023 -0.006 -0.076  0.029  0.090  0.000
Q12_A -0.058 -0.058 -0.042  0.019  0.050 -0.079 -0.079  0.000
Q13_A -0.035 -0.061  0.036 -0.060  0.047 -0.079 -0.064  0.059  0.000
Q14_A -0.027 -0.072  0.016  0.004  0.066 -0.061 -0.066  0.044  0.031  0.000
Q23_A -0.091 -0.074  0.069  0.004  0.139 -0.045  0.063 -0.066 -0.068 -0.061
Q24_A -0.086 -0.091  0.001  0.007  0.053  0.007 -0.001  0.022  0.030 -0.038
Q25_A -0.058 -0.041  0.009  0.019  0.010  0.014 -0.032 -0.054 -0.047 -0.109
       Q23_A  Q24_A  Q25_A
Q6_P
Q7_P
Q11_P
Q19_P
Q26_P
Q3_F
Q4_F
Q10_F
Q15_F
```

```
Q16_F
Q17_F
Q18_F
Q20_S
Q21_S
Q22_S
Q8_A
Q9_A
Q12_A
Q13_A
Q14_A
Q23_A  0.000
Q24_A  0.104  0.000
Q25_A  0.120  0.233  0.000

$mean
 Q6_P  Q7_P Q11_P Q19_P Q26_P  Q3_F  Q4_F Q10_F Q15_F Q16_F Q17_F Q18_F Q20_S
    0     0     0     0     0     0     0     0     0     0     0     0     0
Q21_S Q22_S  Q8_A  Q9_A Q12_A Q13_A Q14_A Q23_A Q24_A Q25_A
    0     0     0     0     0     0     0     0     0     0

$th
 Q6_P|t1  Q6_P|t2  Q6_P|t3  Q6_P|t4  Q7_P|t1  Q7_P|t2  Q7_P|t3  Q7_P|t4
       0        0        0        0        0        0        0        0
Q11_P|t1 Q11_P|t2 Q11_P|t3 Q11_P|t4 Q19_P|t1 Q19_P|t2 Q19_P|t3 Q26_P|t1
       0        0        0        0        0        0        0        0
Q26_P|t2 Q26_P|t3 Q26_P|t4  Q3_F|t1  Q3_F|t2  Q3_F|t3  Q3_F|t4  Q4_F|t1
       0        0        0        0        0        0        0        0
 Q4_F|t2  Q4_F|t3  Q4_F|t4 Q10_F|t1 Q10_F|t2 Q10_F|t3 Q10_F|t4 Q15_F|t1
       0        0        0        0        0        0        0        0
Q15_F|t2 Q15_F|t3 Q15_F|t4 Q16_F|t1 Q16_F|t2 Q16_F|t3 Q17_F|t1 Q17_F|t2
       0        0        0        0        0        0        0        0
Q17_F|t3 Q18_F|t1 Q18_F|t2 Q18_F|t3 Q20_S|t1 Q20_S|t2 Q20_S|t3 Q21_S|t1
       0        0        0        0        0        0        0        0
Q21_S|t2 Q21_S|t3 Q22_S|t1 Q22_S|t2 Q22_S|t3  Q8_A|t1  Q8_A|t2  Q8_A|t3
       0        0        0        0        0        0        0        0
 Q8_A|t4  Q9_A|t1  Q9_A|t2  Q9_A|t3  Q9_A|t4 Q12_A|t1 Q12_A|t2 Q12_A|t3
       0        0        0        0        0        0        0        0
Q12_A|t4 Q13_A|t1 Q13_A|t2 Q13_A|t3 Q13_A|t4 Q14_A|t1 Q14_A|t2 Q14_A|t3
       0        0        0        0        0        0        0        0
Q14_A|t4 Q23_A|t1 Q23_A|t2 Q23_A|t3 Q24_A|t1 Q24_A|t2 Q24_A|t3 Q25_A|t1
       0        0        0        0        0        0        0        0
Q25_A|t2 Q25_A|t3
       0        0
```

Source: Article Notebook

```
residuals(est_4f_2or, type = "standardized")$cov
```

```
        Q6_P   Q7_P  Q11_P  Q19_P  Q26_P   Q3_F   Q4_F  Q10_F  Q15_F  Q16_F
Q6_P   0.000
```

```
Q7_P   1.087  0.000
Q11_P -0.210  0.581  0.000
Q19_P  1.723 -2.708 -2.458  0.000
Q26_P  3.178 -0.184 -1.958 -0.325  0.000
Q3_F  -4.511 -1.274  2.374 -5.163  1.340  0.000
Q4_F  -2.624 -0.556  3.139 -4.568  1.412  0.000  0.000
Q10_F  1.732  4.501  6.150 -0.181  5.332  2.632  1.063  0.000
Q15_F -2.706 -2.050  1.773 -5.824 -2.512  6.506  5.308 -1.819  0.000
Q16_F -1.089 -0.911 -2.523  2.225  2.967 -1.280  0.156 -0.764 -3.590  0.000
Q17_F -4.101  0.471 -3.491  2.923 -0.481 -1.047 -1.025 -4.929 -5.606  0.701
Q18_F -3.681  3.539 -4.183  4.727 -2.995 -1.519 -1.815 -8.973 -3.127 -2.969
Q20_S  4.143 -0.972 -1.282  2.410 -0.893 -3.796 -4.216 -2.469 -3.123  0.643
Q21_S  1.362 -2.101  0.652  1.835  0.387  1.042 -0.918  2.491 -0.526  3.966
Q22_S  2.914 -4.053 -0.044  0.397 -2.593 -1.671 -2.386 -2.617 -2.276  0.014
Q8_A   0.904  4.001  1.093 -0.472  0.619  1.119  0.520  2.754  2.567 -0.475
Q9_A   0.360  1.873  1.427  0.282  0.129  1.072  1.575  4.188  3.126  0.875
Q12_A -2.822 -0.978  2.236 -1.968  0.797  1.410  3.272  3.517  5.247  0.554
Q13_A  0.350  1.628  1.617 -3.645 -0.529  0.990  1.024  0.604  5.998 -0.068
Q14_A  1.296 -1.758  0.559 -1.435 -1.017  1.358  2.004  5.139  5.860  2.797
Q23_A  0.037 -2.248 -0.514 -1.163 -0.654 -1.444 -1.226 -0.221  2.306 -0.463
Q24_A -1.440 -4.573 -2.932 -4.830 -3.664 -0.037 -1.868 -3.943  0.709 -1.424
Q25_A -1.861 -1.452 -0.978 -2.647 -3.072  1.237 -0.403 -3.189  4.974  1.274
       Q17_F  Q18_F  Q20_S  Q21_S  Q22_S   Q8_A   Q9_A  Q12_A  Q13_A  Q14_A
Q6_P
Q7_P
Q11_P
Q19_P
Q26_P
Q3_F
Q4_F
Q10_F
Q15_F
Q16_F
Q17_F  0.000
Q18_F  8.346  0.000
Q20_S -1.348 -1.226  0.000
Q21_S  1.165  1.943 -3.022  0.000
Q22_S -4.819 -2.294  3.746 -2.250  0.000
Q8_A  -1.418 -0.883  0.975 -1.208 -0.999  0.000
Q9_A  -2.411 -1.151 -0.281 -3.035  1.382  4.044  0.000
Q12_A -3.593 -3.322 -1.998  0.794  2.396 -3.764 -3.552  0.000
Q13_A -2.207 -3.541  1.796 -2.298  2.078 -3.486 -3.192  3.236  0.000
Q14_A -1.782 -4.408  0.829  0.163  3.430 -3.028 -3.822  2.615  1.715  0.000
Q23_A -4.264 -3.473  3.020  0.152  5.745 -1.735  2.898 -2.996 -3.017 -2.710
Q24_A -4.016 -4.169  0.040  0.245  2.083  0.294 -0.041  1.073  1.445 -1.912
Q25_A -2.890 -1.915  0.392  0.662  0.413  0.526 -1.518 -2.679 -2.197 -5.237
       Q23_A  Q24_A  Q25_A
Q6_P
Q7_P
```

```
Q11_P
Q19_P
Q26_P
Q3_F
Q4_F
Q10_F
Q15_F
Q16_F
Q17_F
Q18_F
Q20_S
Q21_S
Q22_S
Q8_A
Q9_A
Q12_A
Q13_A
Q14_A
Q23_A  0.000
Q24_A  4.414  0.000
Q25_A  5.267 10.056  0.000
```

Source: Article Notebook

0.7.3.4 Reliability Coefficients

```r
knitr::kable(semTools::reliability(est_4f_2or), digits = 3)
```

```
Warning in semTools::reliability(est_4f_2or):
The reliability() function was deprecated in 2022 and will cease to be included in future versio

It is replaced by the compRelSEM() function, which can estimate alpha and model-based reliability

The average variance extracted should never have been included because it is not a reliability co

For constructs with categorical indicators, Zumbo et al.`s (2007) "ordinal alpha" is calculated i
```

Table 7: Reliability coefficients for first-order factors

	psycho	physical	social	environment
alpha	0.793	0.823	0.687	0.787
alpha.ord	0.831	0.863	0.741	0.820
omega	0.794	0.801	0.708	0.795
omega2	0.794	0.801	0.708	0.795
omega3	0.794	0.810	0.725	0.802
avevar	0.505	0.499	0.525	0.377

Source: Article Notebook

For second-order models, we also need to examine the reliability of the second-order factor:

```
knitr::kable(as.data.frame(t(semTools::reliabilityL2(est_4f_2or, "QOL"))), digits = 3)
```

```
Warning in semTools::reliabilityL2(est_4f_2or, "QOL"):
The reliabilityL2() function was deprecated in 2022 and will cease to be included in future versi

It is replaced by the compRelSEM() function, which can estimate alpha and model-based reliability
```

Table 8: Second-order factor reliability

omegaL1	omegaL2	partialOmegaL1
0.838	0.899	0.926

Source: Article Notebook

0.7.4 Path Diagram

```
# Convert to semPlotModel first
sem_model_2or <- semPlot::semPlotModel(est_4f_2or)

semPlot::semPaths(
  sem_model_2or,
  what = "std",
  style = "lisrel",
  layout = "tree",
  intercepts = FALSE,
  thresholds = FALSE,
  residuals = FALSE,
  edge.label.cex = 0.8,
  label.cex = 1.2,
  weighted = FALSE,
  edge.color = "black",
  node.width = 0.8,
  mar = c(4, 1, 3, 1)
)
```

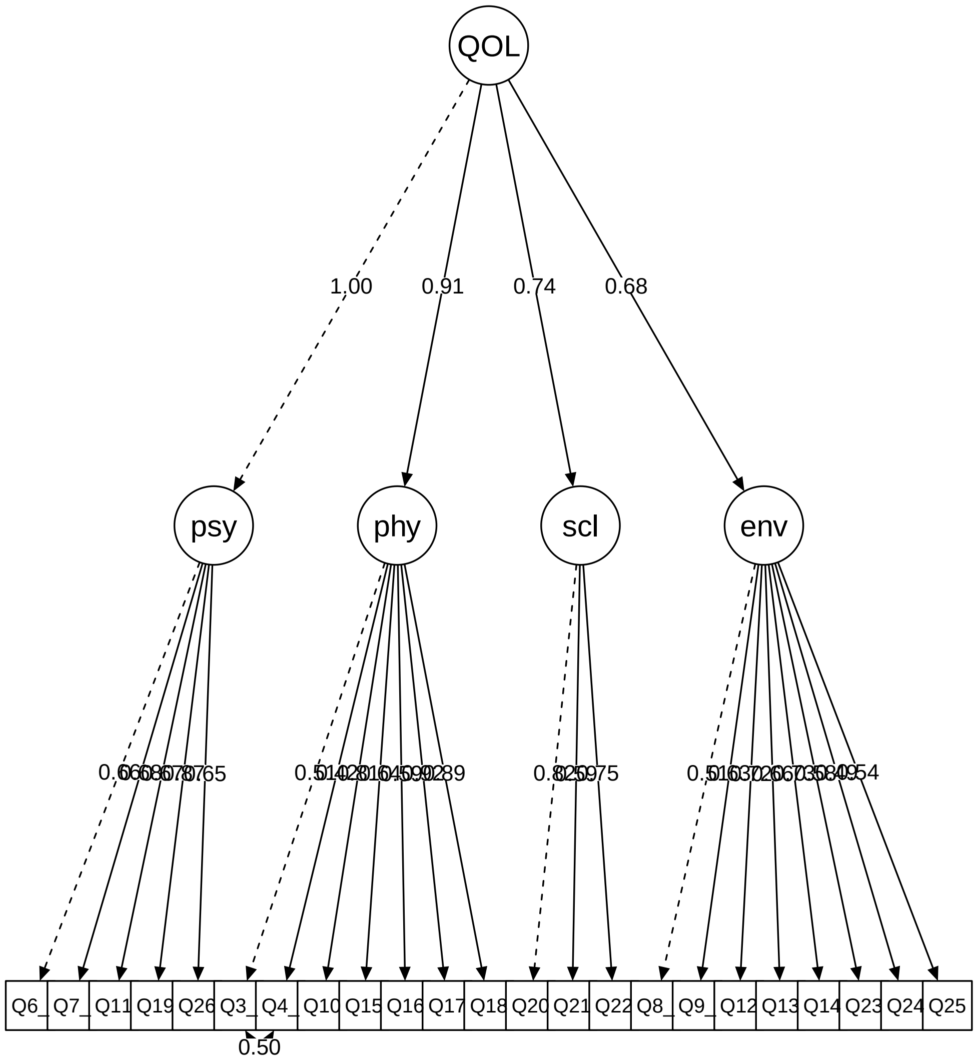


Figure 4: Path diagram of the second-order CFA model

Source: Article Notebook

0.8 4-Factor Model Without Item Q5

Based on the preliminary evaluation of the original 4-factor model, item Q5 was identified as problematic. It appeared in many high modification indices, caused convergence problems in the bifactor model, and showed inconsistent behavior across models.

This section presents the revised 4-factor model excluding Q5, which will serve as our recommended model for comparison purposes.

0.8.1 Model Specification

```
# Same as original 4-factor model, but without Q5
spe_4fb <- "
psycho =~ Q6_P + Q7_P + Q11_P + Q19_P + Q26_P
physical =~ Q3_F + Q4_F + Q10_F + Q15_F + Q16_F + Q17_F + Q18_F
social =~ Q20_S + Q21_S + Q22_S
environment =~ Q8_A + Q9_A + Q12_A + Q13_A + Q14_A + Q23_A + Q24_A + Q25_A
Q4_F ~~ Q3_F
"
```

Source: Article Notebook

0.8.2 Model Estimation

```
est_4fb <- cfa(
  spe_4fb,
  data = df0,
  ordered = TRUE,       # All items are ordinal
  std.lv = TRUE,        # Standardize latent variables
  estimator = "WLSMV"  # DWLS estimator for ordinal data
)
```

Source: Article Notebook

0.8.3 Model Evaluation

What to Look For

When evaluating this revised model, compare:

1. Overall fit to the original model (should improve)
2. Whether modification indices still suggest problems
3. Whether factor loadings are all acceptable (> .40)
4. Whether reliability remains adequate

0.8.3.1 Overall Fit and Parameter Estimates

```
summary(est_4fb, fit.measures = TRUE, standardized = TRUE, rsq = TRUE)
```

```
lavaan 0.6-21 ended normally after 33 iterations

  Estimator                                         DWLS
  Optimization method                             NLMINB
  Number of model parameters                         112
```

```
  Number of observations                          1047

Model Test User Model:
                                              Standard      Scaled
  Test Statistic                               977.820    1285.457
  Degrees of freedom                               223         223
  P-value (Unknown)                                 NA       0.000
  Scaling correction factor                                  0.801
  Shift parameter                                           64.145
    simple second-order correction

Model Test Baseline Model:

  Test statistic                             64072.987   24218.548
  Degrees of freedom                               253         253
  P-value                                           NA       0.000
  Scaling correction factor                                  2.663

User Model versus Baseline Model:

  Comparative Fit Index (CFI)                    0.988       0.956
  Tucker-Lewis Index (TLI)                       0.987       0.950

  Robust Comparative Fit Index (CFI)                         0.889
  Robust Tucker-Lewis Index (TLI)                            0.874

Root Mean Square Error of Approximation:

  RMSEA                                          0.057       0.067
  90 Percent confidence interval - lower         0.053       0.064
  90 Percent confidence interval - upper         0.061       0.071
  P-value H_0: RMSEA <= 0.050                    0.001       0.000
  P-value H_0: RMSEA >= 0.080                    0.000       0.000

  Robust RMSEA                                               0.075
  90 Percent confidence interval - lower                     0.070
  90 Percent confidence interval - upper                     0.079
  P-value H_0: Robust RMSEA <= 0.050                         0.000
  P-value H_0: Robust RMSEA >= 0.080                         0.021

Standardized Root Mean Square Residual:

  SRMR                                           0.053       0.053

Parameter Estimates:

  Parameterization                               Delta
  Standard errors                           Robust.sem
  Information                                 Expected
```

```
  Information saturated (h1) model        Unstructured

Latent Variables:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
  psycho =~
    Q6_P              0.659    0.019   34.689    0.000    0.659    0.659
    Q7_P              0.677    0.018   37.254    0.000    0.677    0.677
    Q11_P             0.671    0.019   34.426    0.000    0.671    0.671
    Q19_P             0.869    0.012   71.132    0.000    0.869    0.869
    Q26_P             0.648    0.020   32.730    0.000    0.648    0.648
  physical =~
    Q3_F              0.510    0.026   19.526    0.000    0.510    0.510
    Q4_F              0.423    0.029   14.737    0.000    0.423    0.423
    Q10_F             0.811    0.012   66.021    0.000    0.811    0.811
    Q15_F             0.643    0.025   25.889    0.000    0.643    0.643
    Q16_F             0.590    0.023   25.647    0.000    0.590    0.590
    Q17_F             0.923    0.007  125.428    0.000    0.923    0.923
    Q18_F             0.888    0.009  103.944    0.000    0.888    0.888
  social =~
    Q20_S             0.819    0.019   43.875    0.000    0.819    0.819
    Q21_S             0.586    0.027   21.558    0.000    0.586    0.586
    Q22_S             0.748    0.020   37.362    0.000    0.748    0.748
  environment =~
    Q8_A              0.510    0.028   17.913    0.000    0.510    0.510
    Q9_A              0.629    0.025   25.608    0.000    0.629    0.629
    Q12_A             0.715    0.021   34.196    0.000    0.715    0.715
    Q13_A             0.663    0.023   28.432    0.000    0.663    0.663
    Q14_A             0.732    0.021   35.663    0.000    0.732    0.732
    Q23_A             0.580    0.027   21.770    0.000    0.580    0.580
    Q24_A             0.492    0.028   17.504    0.000    0.492    0.492
    Q25_A             0.542    0.027   19.777    0.000    0.542    0.542

Covariances:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
 .Q3_F ~~
   .Q4_F              0.387    0.025   15.605    0.000    0.387    0.496
  psycho ~~
    physical          0.922    0.010   96.703    0.000    0.922    0.922
    social            0.755    0.019   39.028    0.000    0.755    0.755
    environment       0.659    0.024   27.049    0.000    0.659    0.659
  physical ~~
    social            0.638    0.024   26.528    0.000    0.638    0.638
    environment       0.630    0.022   28.640    0.000    0.630    0.630
  social ~~
    environment       0.550    0.028   19.517    0.000    0.550    0.550

Thresholds:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
    Q6_P|t1          -1.931    0.081  -23.937    0.000   -1.931   -1.931
```

Q6_P|t2	-1.319	0.054	-24.483	0.000	-1.319	-1.319
Q6_P|t3	-0.598	0.041	-14.442	0.000	-0.598	-0.598
Q6_P|t4	0.541	0.041	13.237	0.000	0.541	0.541
Q7_P|t1	-2.244	0.106	-21.085	0.000	-2.244	-2.244
Q7_P|t2	-1.238	0.052	-23.927	0.000	-1.238	-1.238
Q7_P|t3	-0.290	0.039	-7.373	0.000	-0.290	-0.290
Q7_P|t4	1.183	0.050	23.465	0.000	1.183	1.183
Q11_P|t1	-2.033	0.088	-23.148	0.000	-2.033	-2.033
Q11_P|t2	-1.296	0.053	-24.344	0.000	-1.296	-1.296
Q11_P|t3	-0.415	0.040	-10.378	0.000	-0.415	-0.415
Q11_P|t4	0.589	0.041	14.262	0.000	0.589	0.589
Q19_P|t1	-1.062	0.048	-22.193	0.000	-1.062	-1.062
Q19_P|t2	-0.405	0.040	-10.134	0.000	-0.405	-0.405
Q19_P|t3	0.829	0.044	18.847	0.000	0.829	0.829
Q26_P|t1	-1.846	0.075	-24.465	0.000	-1.846	-1.846
Q26_P|t2	-1.378	0.056	-24.791	0.000	-1.378	-1.378
Q26_P|t3	-0.796	0.044	-18.275	0.000	-0.796	-0.796
Q26_P|t4	0.723	0.043	16.938	0.000	0.723	0.723
Q3_F|t1	-2.528	0.143	-17.684	0.000	-2.528	-2.528
Q3_F|t2	-1.471	0.059	-25.106	0.000	-1.471	-1.471
Q3_F|t3	-0.819	0.044	-18.676	0.000	-0.819	-0.819
Q3_F|t4	0.116	0.039	2.996	0.003	0.116	0.116
Q4_F|t1	-2.188	0.101	-21.680	0.000	-2.188	-2.188
Q4_F|t2	-1.302	0.053	-24.380	0.000	-1.302	-1.302
Q4_F|t3	-0.754	0.043	-17.522	0.000	-0.754	-0.754
Q4_F|t4	0.167	0.039	4.292	0.000	0.167	0.167
Q10_F|t1	-2.344	0.117	-19.953	0.000	-2.344	-2.344
Q10_F|t2	-1.285	0.053	-24.272	0.000	-1.285	-1.285
Q10_F|t3	-0.112	0.039	-2.872	0.004	-0.112	-0.112
Q10_F|t4	0.924	0.045	20.355	0.000	0.924	0.924
Q15_F|t1	-2.528	0.143	-17.684	0.000	-2.528	-2.528
Q15_F|t2	-1.820	0.074	-24.599	0.000	-1.820	-1.820
Q15_F|t3	-1.248	0.052	-24.007	0.000	-1.248	-1.248
Q15_F|t4	-0.167	0.039	-4.292	0.000	-0.167	-0.167
Q16_F|t1	-0.738	0.043	-17.231	0.000	-0.738	-0.738
Q16_F|t2	-0.116	0.039	-2.996	0.003	-0.116	-0.116
Q16_F|t3	0.917	0.045	20.245	0.000	0.917	0.917
Q17_F|t1	-1.070	0.048	-22.293	0.000	-1.070	-1.070
Q17_F|t2	-0.476	0.040	-11.781	0.000	-0.476	-0.476
Q17_F|t3	0.906	0.045	20.080	0.000	0.906	0.906
Q18_F|t1	-1.041	0.047	-21.941	0.000	-1.041	-1.041
Q18_F|t2	-0.479	0.040	-11.842	0.000	-0.479	-0.479
Q18_F|t3	0.793	0.044	18.217	0.000	0.793	0.793
Q20_S|t1	-1.178	0.050	-23.421	0.000	-1.178	-1.178
Q20_S|t2	-0.248	0.039	-6.326	0.000	-0.248	-0.248
Q20_S|t3	0.993	0.047	21.321	0.000	0.993	0.993
Q21_S|t1	-0.783	0.043	-18.044	0.000	-0.783	-0.783
Q21_S|t2	-0.035	0.039	-0.896	0.370	-0.035	-0.035
Q21_S|t3	0.985	0.046	21.215	0.000	0.985	0.985

```
    Q22_S|t1          -1.150    0.050  -23.152    0.000   -1.150   -1.150
    Q22_S|t2          -0.153    0.039   -3.922    0.000   -0.153   -0.153
    Q22_S|t3           1.070    0.048   22.293    0.000    1.070    1.070
    Q8_A|t1           -1.416    0.057  -24.945    0.000   -1.416   -1.416
    Q8_A|t2           -0.492    0.041  -12.146    0.000   -0.492   -0.492
    Q8_A|t3            0.558    0.041   13.600    0.000    0.558    0.558
    Q8_A|t4            1.728    0.069   24.978    0.000    1.728    1.728
    Q9_A|t1           -1.859    0.076  -24.391    0.000   -1.859   -1.859
    Q9_A|t2           -1.096    0.049  -22.587    0.000   -1.096   -1.096
    Q9_A|t3           -0.141    0.039   -3.613    0.000   -0.141   -0.141
    Q9_A|t4            1.372    0.055   24.763    0.000    1.372    1.372
    Q12_A|t1          -1.648    0.065  -25.169    0.000   -1.648   -1.648
    Q12_A|t2          -0.947    0.046  -20.681    0.000   -0.947   -0.947
    Q12_A|t3           0.080    0.039    2.069    0.039    0.080    0.080
    Q12_A|t4           0.826    0.044   18.790    0.000    0.826    0.826
    Q13_A|t1          -2.668    0.168  -15.900    0.000   -2.668   -2.668
    Q13_A|t2          -1.603    0.064  -25.218    0.000   -1.603   -1.603
    Q13_A|t3          -0.653    0.042  -15.578    0.000   -0.653   -0.653
    Q13_A|t4           0.692    0.042   16.349    0.000    0.692    0.692
    Q14_A|t1          -1.820    0.074  -24.599    0.000   -1.820   -1.820
    Q14_A|t2          -0.913    0.045  -20.190    0.000   -0.913   -0.913
    Q14_A|t3           0.109    0.039    2.810    0.005    0.109    0.109
    Q14_A|t4           1.291    0.053   24.309    0.000    1.291    1.291
    Q23_A|t1          -1.212    0.051  -23.722    0.000   -1.212   -1.212
    Q23_A|t2          -0.665    0.042  -15.816    0.000   -0.665   -0.665
    Q23_A|t3           0.410    0.040   10.256    0.000    0.410    0.410
    Q24_A|t1          -1.062    0.048  -22.193    0.000   -1.062   -1.062
    Q24_A|t2          -0.386    0.040   -9.705    0.000   -0.386   -0.386
    Q24_A|t3           0.748    0.043   17.406    0.000    0.748    0.748
    Q25_A|t1          -1.118    0.049  -22.826    0.000   -1.118   -1.118
    Q25_A|t2          -0.487    0.040  -12.025    0.000   -0.487   -0.487
    Q25_A|t3           0.606    0.041   14.622    0.000    0.606    0.606

Variances:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
   .Q6_P              0.565                               0.565    0.565
   .Q7_P              0.541                               0.541    0.541
   .Q11_P             0.550                               0.550    0.550
   .Q19_P             0.245                               0.245    0.245
   .Q26_P             0.580                               0.580    0.580
   .Q3_F              0.740                               0.740    0.740
   .Q4_F              0.821                               0.821    0.821
   .Q10_F             0.343                               0.343    0.343
   .Q15_F             0.586                               0.586    0.586
   .Q16_F             0.652                               0.652    0.652
   .Q17_F             0.149                               0.149    0.149
   .Q18_F             0.212                               0.212    0.212
   .Q20_S             0.330                               0.330    0.330
   .Q21_S             0.656                               0.656    0.656
```

```
    .Q22_S          0.441                              0.441    0.441
    .Q8_A           0.740                              0.740    0.740
    .Q9_A           0.605                              0.605    0.605
    .Q12_A          0.489                              0.489    0.489
    .Q13_A          0.561                              0.561    0.561
    .Q14_A          0.465                              0.465    0.465
    .Q23_A          0.663                              0.663    0.663
    .Q24_A          0.758                              0.758    0.758
    .Q25_A          0.707                              0.707    0.707
     psycho         1.000                              1.000    1.000
     physical       1.000                              1.000    1.000
     social         1.000                              1.000    1.000
     environment    1.000                              1.000    1.000

R-Square:
                   Estimate
    Q6_P            0.435
    Q7_P            0.459
    Q11_P           0.450
    Q19_P           0.755
    Q26_P           0.420
    Q3_F            0.260
    Q4_F            0.179
    Q10_F           0.657
    Q15_F           0.414
    Q16_F           0.348
    Q17_F           0.851
    Q18_F           0.788
    Q20_S           0.670
    Q21_S           0.344
    Q22_S           0.559
    Q8_A            0.260
    Q9_A            0.395
    Q12_A           0.511
    Q13_A           0.439
    Q14_A           0.535
    Q23_A           0.337
    Q24_A           0.242
    Q25_A           0.293
```

Source: Article Notebook

0.8.3.2 Modification Indices

```
knitr::kable(modificationindices(est_4fb,
  sort. = TRUE, minimum.value = 3.84,
  maximum.number = 20
), digits = 3)
```

Table 9: Top modification indices for the revised 4-factor model

	lhs	op	rhs	mi	epc	sepc.lv	sepc.all	sepc.nox
510	Q24_A	~~	Q25_A	114.934	0.273	0.273	0.373	0.373
252	environment	=~	Q15_F	94.407	0.351	0.351	0.351	0.351
433	Q17_F	~~	Q18_F	84.659	0.157	0.157	0.887	0.887
222	physical	=~	Q24_A	78.505	-0.281	-0.281	-0.281	-0.281
206	psycho	=~	Q24_A	77.213	-0.289	-0.289	-0.289	-0.289
254	environment	=~	Q17_F	43.449	-0.232	-0.232	-0.232	-0.232
255	environment	=~	Q18_F	42.877	-0.228	-0.228	-0.228	-0.228
192	psycho	=~	Q10_F	42.388	0.704	0.704	0.704	0.704
208	physical	=~	Q6_P	36.087	-0.700	-0.700	-0.700	-0.700
242	social	=~	Q24_A	33.243	-0.203	-0.203	-0.203	-0.203
394	Q10_F	~~	Q18_F	31.926	-0.118	-0.118	-0.440	-0.440
360	Q3_F	~~	Q15_F	31.211	0.178	0.178	0.270	0.270
214	physical	=~	Q21_S	31.008	0.235	0.235	0.235	0.235
198	psycho	=~	Q21_S	26.547	0.277	0.277	0.277	0.277
509	Q23_A	~~	Q25_A	23.889	0.140	0.140	0.204	0.204
415	Q15_F	~~	Q13_A	21.582	0.146	0.146	0.255	0.255
376	Q4_F	~~	Q15_F	21.431	0.147	0.147	0.212	0.212
416	Q15_F	~~	Q14_A	20.568	0.143	0.143	0.273	0.273
480	Q22_S	~~	Q23_A	18.885	0.131	0.131	0.242	0.242
224	social	=~	Q6_P	18.702	0.220	0.220	0.220	0.220

Source: Article Notebook

0.8.3.3 Residual Correlations

```
residuals(est_4fb, type = "cor")
```

```
$type
[1] "cor.bollen"

$cov
        Q6_P   Q7_P  Q11_P  Q19_P  Q26_P   Q3_F   Q4_F  Q10_F  Q15_F  Q16_F
Q6_P   0.000
Q7_P   0.017  0.000
Q11_P -0.003  0.011  0.000
Q19_P  0.024 -0.033 -0.028  0.000
Q26_P  0.066 -0.003 -0.035 -0.002  0.000
Q3_F  -0.104 -0.036  0.053 -0.102  0.030  0.000
Q4_F  -0.068 -0.017  0.070 -0.080  0.031  0.000  0.000
Q10_F  0.021  0.051  0.080 -0.006  0.087  0.046  0.017  0.000
Q15_F -0.065 -0.049  0.032 -0.109 -0.067  0.160  0.135 -0.030  0.000
Q16_F -0.028 -0.022 -0.055  0.029  0.063 -0.031  0.003 -0.011 -0.089  0.000
Q17_F -0.060  0.001 -0.055  0.015 -0.013 -0.020 -0.018 -0.040 -0.090  0.008
Q18_F -0.050  0.044 -0.060  0.029 -0.050 -0.031 -0.033 -0.081 -0.059 -0.041
Q20_S  0.057 -0.027 -0.032  0.031 -0.026 -0.073 -0.090 -0.015 -0.058  0.032
Q21_S  0.022 -0.048  0.009  0.024  0.004  0.040 -0.015  0.063  0.001  0.111
```

```
Q22_S  0.042 -0.081 -0.008 -0.002 -0.056 -0.030 -0.049 -0.021 -0.041  0.017
Q8_A   0.028  0.093  0.035  0.004  0.024  0.031  0.013  0.053  0.074 -0.014
Q9_A   0.020  0.057  0.045  0.022  0.014  0.029  0.043  0.080  0.085  0.021
Q12_A -0.046 -0.006  0.058 -0.012  0.030  0.034  0.075  0.056  0.125  0.011
Q13_A  0.019  0.045  0.045 -0.044 -0.001  0.025  0.025  0.008  0.137 -0.004
Q14_A  0.037 -0.020  0.024 -0.003 -0.010  0.031  0.047  0.083  0.133  0.063
Q23_A  0.011 -0.042 -0.003 -0.009 -0.008 -0.046 -0.039 -0.008  0.063 -0.015
Q24_A -0.029 -0.108 -0.064 -0.088 -0.089 -0.003 -0.058 -0.091  0.018 -0.043
Q25_A -0.038 -0.024 -0.013 -0.041 -0.067  0.033 -0.014 -0.071  0.114  0.033
       Q17_F  Q18_F  Q20_S  Q21_S  Q22_S   Q8_A   Q9_A  Q12_A  Q13_A  Q14_A
Q6_P
Q7_P
Q11_P
Q19_P
Q26_P
Q3_F
Q4_F
Q10_F
Q15_F
Q16_F
Q17_F  0.000
Q18_F  0.038  0.000
Q20_S  0.010  0.010  0.000
Q21_S  0.039  0.053 -0.058  0.000
Q22_S -0.044 -0.013  0.032 -0.042  0.000
Q8_A  -0.029 -0.019  0.004 -0.045 -0.040  0.000
Q9_A  -0.047 -0.025 -0.028 -0.091  0.009  0.091  0.000
Q12_A -0.061 -0.062 -0.068  0.001  0.026 -0.078 -0.079  0.000
Q13_A -0.038 -0.064  0.013 -0.077  0.026 -0.079 -0.063  0.059  0.000
Q14_A -0.030 -0.076 -0.010 -0.014  0.042 -0.060 -0.065  0.043  0.031  0.000
Q23_A -0.094 -0.077  0.048 -0.010  0.120 -0.045  0.063 -0.066 -0.068 -0.062
Q24_A -0.089 -0.093 -0.017 -0.005  0.037  0.007 -0.001  0.022  0.030 -0.039
Q25_A -0.061 -0.043 -0.010  0.006 -0.008  0.014 -0.032 -0.055 -0.047 -0.110
       Q23_A  Q24_A  Q25_A
Q6_P
Q7_P
Q11_P
Q19_P
Q26_P
Q3_F
Q4_F
Q10_F
Q15_F
Q16_F
Q17_F
Q18_F
Q20_S
Q21_S
Q22_S
```

```
Q8_A
Q9_A
Q12_A
Q13_A
Q14_A
Q23_A  0.000
Q24_A  0.103  0.000
Q25_A  0.119  0.233  0.000

$mean
 Q6_P   Q7_P  Q11_P  Q19_P  Q26_P   Q3_F   Q4_F  Q10_F  Q15_F  Q16_F  Q17_F  Q18_F  Q20_S
    0      0      0      0      0      0      0      0      0      0      0      0      0
Q21_S  Q22_S   Q8_A   Q9_A  Q12_A  Q13_A  Q14_A  Q23_A  Q24_A  Q25_A
    0      0      0      0      0      0      0      0      0      0

$th
 Q6_P|t1  Q6_P|t2  Q6_P|t3  Q6_P|t4  Q7_P|t1  Q7_P|t2  Q7_P|t3  Q7_P|t4
       0        0        0        0        0        0        0        0
Q11_P|t1 Q11_P|t2 Q11_P|t3 Q11_P|t4 Q19_P|t1 Q19_P|t2 Q19_P|t3 Q26_P|t1
       0        0        0        0        0        0        0        0
Q26_P|t2 Q26_P|t3 Q26_P|t4  Q3_F|t1  Q3_F|t2  Q3_F|t3  Q3_F|t4  Q4_F|t1
       0        0        0        0        0        0        0        0
 Q4_F|t2  Q4_F|t3  Q4_F|t4 Q10_F|t1 Q10_F|t2 Q10_F|t3 Q10_F|t4 Q15_F|t1
       0        0        0        0        0        0        0        0
Q15_F|t2 Q15_F|t3 Q15_F|t4 Q16_F|t1 Q16_F|t2 Q16_F|t3 Q17_F|t1 Q17_F|t2
       0        0        0        0        0        0        0        0
Q17_F|t3 Q18_F|t1 Q18_F|t2 Q18_F|t3 Q20_S|t1 Q20_S|t2 Q20_S|t3 Q21_S|t1
       0        0        0        0        0        0        0        0
Q21_S|t2 Q21_S|t3 Q22_S|t1 Q22_S|t2 Q22_S|t3  Q8_A|t1  Q8_A|t2  Q8_A|t3
       0        0        0        0        0        0        0        0
 Q8_A|t4  Q9_A|t1  Q9_A|t2  Q9_A|t3  Q9_A|t4 Q12_A|t1 Q12_A|t2 Q12_A|t3
       0        0        0        0        0        0        0        0
Q12_A|t4 Q13_A|t1 Q13_A|t2 Q13_A|t3 Q13_A|t4 Q14_A|t1 Q14_A|t2 Q14_A|t3
       0        0        0        0        0        0        0        0
Q14_A|t4 Q23_A|t1 Q23_A|t2 Q23_A|t3 Q24_A|t1 Q24_A|t2 Q24_A|t3 Q25_A|t1
       0        0        0        0        0        0        0        0
Q25_A|t2 Q25_A|t3
       0        0
```

Source: Article Notebook

```
residuals(est_4fb, type = "standardized")$cov
```

```
        Q6_P   Q7_P  Q11_P  Q19_P  Q26_P   Q3_F   Q4_F  Q10_F  Q15_F  Q16_F
Q6_P   0.000
Q7_P   1.197  0.000
Q11_P -0.174  0.640  0.000
Q19_P  2.191 -2.801 -2.412  0.000
Q26_P  3.202 -0.165 -1.976 -0.171  0.000
Q3_F  -4.731 -1.431  2.235 -5.388  1.200  0.000
Q4_F  -2.773 -0.693  3.016 -4.794  1.284  0.000  0.000
```

```
Q10_F  1.425  4.164  5.817 -0.671  5.107  2.599  1.015  0.000
Q15_F -2.924 -2.297  1.574 -6.106 -2.723  6.469  5.268 -1.870  0.000
Q16_F -1.260 -1.111 -2.734  2.025  2.814 -1.302  0.131 -0.770 -3.614  0.000
Q17_F -4.458  0.062 -3.952  2.339 -0.874 -1.083 -1.075 -4.923 -5.651  0.702
Q18_F -3.999  3.276 -4.668  4.175 -3.376 -1.550 -1.860 -8.966 -3.165 -2.965
Q20_S  4.118 -1.385 -1.735  2.094 -1.307 -3.232 -3.742 -0.925 -2.366  1.478
Q21_S  1.168 -2.363  0.430  1.631  0.178  1.472 -0.572  3.555  0.045  4.521
Q22_S  2.703 -4.566 -0.400 -0.122 -3.085 -1.171 -1.956 -1.255 -1.619  0.692
Q8_A   1.356  4.480  1.489  0.192  0.979  1.078  0.475  2.674  2.526 -0.531
Q9_A   0.882  2.390  1.947  1.140  0.571  1.028  1.536  4.115  3.093  0.820
Q12_A -2.284 -0.330  2.910 -0.785  1.348  1.340  3.221  3.305  5.192  0.463
Q13_A  0.907  2.213  2.202 -2.693 -0.043  0.928  0.962  0.442  5.949 -0.160
Q14_A  2.045 -1.081  1.197 -0.190 -0.453  1.275  1.932  4.969  5.857  2.721
Q23_A  0.446 -1.832 -0.107 -0.465 -0.293 -1.529 -1.295 -0.352  2.233 -0.553
Q24_A -1.140 -4.286 -2.621 -4.344 -3.379 -0.096 -1.935 -4.110  0.638 -1.503
Q25_A -1.506 -1.046 -0.573 -2.020 -2.707  1.179 -0.461 -3.320  4.937  1.211
       Q17_F  Q18_F  Q20_S  Q21_S  Q22_S   Q8_A   Q9_A  Q12_A  Q13_A  Q14_A
Q6_P
Q7_P
Q11_P
Q19_P
Q26_P
Q3_F
Q4_F
Q10_F
Q15_F
Q16_F
Q17_F  0.000
Q18_F  8.366  0.000
Q20_S  0.866  0.708  0.000
Q21_S  2.648  3.270 -2.965  0.000
Q22_S -3.239 -0.848  3.674 -2.210  0.000
Q8_A  -1.550 -0.981  0.172 -1.699 -1.789  0.000
Q9_A  -2.608 -1.276 -1.478 -3.813  0.465  4.088  0.000
Q12_A -3.931 -3.574 -3.445  0.065  1.347 -3.736 -3.511  0.000
Q13_A -2.517 -3.816  0.688 -3.063  1.223 -3.459 -3.149  3.243  0.000
Q14_A -2.054 -4.655 -0.563 -0.703  2.360 -3.007 -3.779  2.605  1.709  0.000
Q23_A -4.569 -3.656  2.277 -0.355  5.264 -1.738  2.902 -3.027 -3.042 -2.747
Q24_A -4.246 -4.370 -0.737 -0.170  1.534  0.295 -0.033  1.050  1.428 -1.941
Q25_A -3.071 -2.055 -0.447  0.203 -0.323  0.532 -1.500 -2.698 -2.206 -5.260
       Q23_A  Q24_A  Q25_A
Q6_P
Q7_P
Q11_P
Q19_P
Q26_P
Q3_F
Q4_F
Q10_F
```

```
Q15_F
Q16_F
Q17_F
Q18_F
Q20_S
Q21_S
Q22_S
Q8_A
Q9_A
Q12_A
Q13_A
Q14_A
Q23_A  0.000
Q24_A  4.385  0.000
Q25_A  5.231 10.047  0.000
```

Source: Article Notebook

0.8.3.4 Reliability Coefficients

```
knitr::kable(semTools::reliability(est_4fb), digits = 3)
```

```
Warning in semTools::reliability(est_4fb):
The reliability() function was deprecated in 2022 and will cease to be included in future versio

It is replaced by the compRelSEM() function, which can estimate alpha and model-based reliabilit

The average variance extracted should never have been included because it is not a reliability c

For constructs with categorical indicators, Zumbo et al.`s (2007) "ordinal alpha" is calculated i
```

Table 10: Reliability coefficients for the revised 4-factor model

	psycho	physical	social	environment
alpha	0.793	0.823	0.687	0.787
alpha.ord	0.831	0.863	0.741	0.820
omega	0.793	0.801	0.708	0.795
omega2	0.793	0.801	0.708	0.795
omega3	0.792	0.811	0.725	0.802
avevar	0.504	0.500	0.524	0.377

Source: Article Notebook

> **Note**
>
> Notice that reliability for the Psychological domain remains adequate even after removing Q5, confirming that Q5 was not essential for measuring this construct.

0.8.4 Path Diagram

```
# Convert to semPlotModel first
sem_model_4fb <- semPlot::semPlotModel(est_4fb)

semPlot::semPaths(
  sem_model_4fb,
  what = "std",
  style = "lisrel",
  layout = "circle",
  intercepts = FALSE,
  thresholds = FALSE,
  residuals = FALSE,
  edge.label.cex = 1,
  label.cex = 1,
  weighted = FALSE,
  edge.color = "black",
  mar = c(1.5, 0.8, 3, 0.3)
)
```

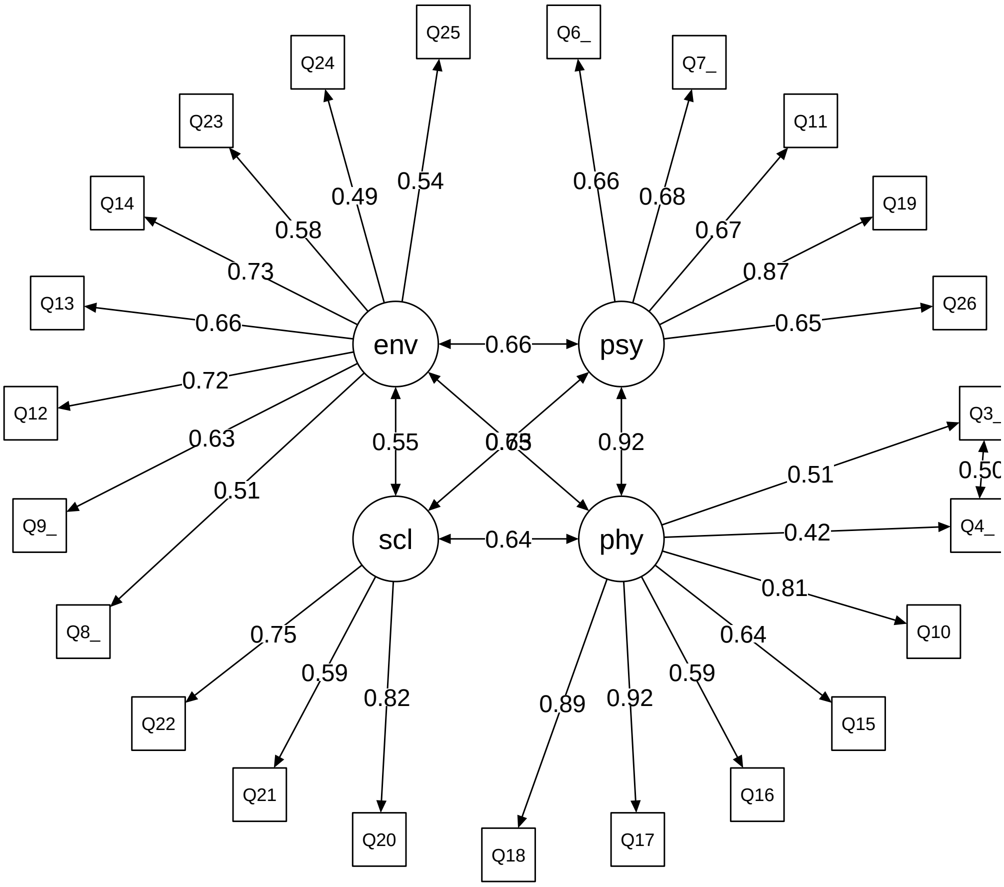


Figure 5: Path diagram of the revised 4-factor CFA model (without Q5)

Source: [Article Notebook]

0.9 Model Comparison

Now we compare fit indices across all four CFA models to determine which best represents the WHOQOL-BREF factor structure.

0.9.1 Extract Fit Indices

```
# Extract key fit indices for each model
fit_4fa    <- fitmeasures(est_4fa,    c("srmr", "rmsea.scaled", "cfi.scaled"))
fit_4f_bi  <- fitmeasures(est_4f_bi,  c("srmr", "rmsea.scaled", "cfi.scaled"))
fit_4f_2or <- fitmeasures(est_4f_2or, c("srmr", "rmsea.scaled", "cfi.scaled"))
fit_4fb    <- fitmeasures(est_4fb,    c("srmr", "rmsea.scaled", "cfi.scaled"))
```

```r
# Display each model's fit
cat("4-Factor (all items):\n")
```

```
4-Factor (all items):
```

```r
print(fit_4fa)
```

```
        srmr rmsea.scaled   cfi.scaled
       0.057        0.076        0.941
```

```r
cat("\nBifactor:\n")
```

```
Bifactor:
```

```r
print(fit_4f_bi)
```

```
        srmr rmsea.scaled   cfi.scaled
       0.049        0.068        0.958
```

```r
cat("\nSecond-Order:\n")
```

```
Second-Order:
```

```r
print(fit_4f_2or)
```

```
        srmr rmsea.scaled   cfi.scaled
       0.054        0.067        0.956
```

```r
cat("\n4-Factor (no Q5):\n")
```

```
4-Factor (no Q5):
```

```r
print(fit_4fb)
```

```
        srmr rmsea.scaled   cfi.scaled
       0.053        0.067        0.956
```

Source: Article Notebook

0.9.2 Comparison Table

```r
# Create comparison table
comparison_df <- data.frame(
  Model = c("4-Factor (all items)", "Bifactor", "Second-Order", "4-Factor (no Q5)"),
  srmr = c(fit_4fa["srmr"], fit_4f_bi["srmr"],
           fit_4f_2or["srmr"], fit_4fb["srmr"]),
  rmsea.scaled = c(fit_4fa["rmsea.scaled"], fit_4f_bi["rmsea.scaled"],
                   fit_4f_2or["rmsea.scaled"], fit_4fb["rmsea.scaled"]),
  cfi.scaled = c(fit_4fa["cfi.scaled"], fit_4f_bi["cfi.scaled"],
                 fit_4f_2or["cfi.scaled"], fit_4fb["cfi.scaled"])
)
rownames(comparison_df) <- NULL

knitr::kable(comparison_df, digits = 3)
```

Table 11: Fit indices comparison across all CFA models

Model	srmr	rmsea.scaled	cfi.scaled
4-Factor (all items)	0.057	0.076	0.941
Bifactor	0.049	0.068	0.958
Second-Order	0.054	0.067	0.956
4-Factor (no Q5)	0.053	0.067	0.956

Source: Article Notebook

0.9.3 General Considerations

The models exhibit a similar level of misspecification, although the original 4-factor model (with all items) performs slightly worse. The bifactor model shows a marginally better fit, which aligns with findings in the literature. This is likely due to the greater number of parameters in bifactor structures, which generally lead to improved model fit.

Some authors argue that bifactor models have been used to "salvage" traditional scales that are no longer sustainable under modern estimation and factor selection techniques. The improved fit comes at the cost of strong assumptions and complex interpretation.

! Model Selection Recommendation

A known limitation of bifactor models is the assumption that specific factors are uncorrelated—an assumption that often does not hold in practice. Additionally, bifactor models can suffer from empirical underidentification and may capitalize on chance variations in the data.
The second-order model also showed convergence problems (requiring the psychological factor variance to be fixed to zero).
Based on empirical evidence and practical considerations, the four correlated factor model excluding item Q5 is recommended for measuring WHOQOL-BREF in this sample. This model:

- Has acceptable fit
- Avoids strong orthogonality assumptions
- Is easier to interpret
- Has no convergence issues
- Maintains adequate reliability across all domains

0.10 Post Hoc Power Analysis

Power analysis assesses whether our sample size (N = 1,047) provides adequate statistical power to detect misfit and estimate parameters precisely. We use Monte Carlo simulation via the `simsem` package, simulating data from the fitted model and re-estimating it many times.

ℹ Why Post Hoc Power Analysis?

While *a priori* power analysis is ideal for study planning, *post hoc* power analysis helps us understand:

1. Whether our estimates are stable and replicable
2. Whether we have adequate power to detect misfit
3. Which parameters may need larger samples for precise estimation

We use the revised four-factor model (without Q5) as our generating model. To preserve the ordinal nature of items, we generate data from the fitted lavaan object rather than a simplified population model.

0.10.1 Define Cutoff Values

```
# Rule-of-thumb cutoffs for fit indices
rule_of_thumb <- c(srmr = 0.06, rmsea.scaled = 0.06, cfi.scaled = 0.95)
```

Source: Article Notebook

0.10.2 Run Simulation

⚠ Warning

To reduce computational time, we run only 100 replications. For final results, increase to 1,000 or even 5,000 replications. More replications provide more stable power estimates but take much longer to run.

```
my_modelci <- simsem::sim(
  100,                          # Number of replications (increase for final analysis)
  n = nrow(df0),               # Sample size (1,047)
  model = spe_4fb,             # Model specification
  generate = est_4fb,          # Generate data from fitted model (preserves ordinal structure)
  lavaanfun = "cfa",           # Use cfa() function for estimation
  ordered = TRUE,              # Items are ordinal
  std.lv = TRUE,               # Standardize latent variables
  estimator = "WLSMV",         # DWLS estimator
  seed = 123456                # For reproducibility
)
```

```
The RNGkind() was changed from "Mersenne-Twister" to RNGkind("L'Ecuyer-CMRG") in order to enable
To set a RNG seed using the previous RNGkind(), you can use
    RNGkind("Mersenne-Twister","Inversion","Rejection")
 or
    set.seed(seed, kind = "Mersenne-Twister")
```

```
Progress: 1 / 100
Progress: 2 / 100
Progress: 3 / 100
Progress: 4 / 100
Progress: 5 / 100
```

```
Progress: 6 / 100
Progress: 7 / 100
Progress: 8 / 100
Progress: 9 / 100
Progress: 10 / 100
Progress: 11 / 100
Progress: 12 / 100
Progress: 13 / 100
Progress: 14 / 100
Progress: 15 / 100
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Progress: 54 / 100
```

```
Progress: 55 / 100
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Progress: 92 / 100
Progress: 93 / 100
Progress: 94 / 100
Progress: 95 / 100
Progress: 96 / 100
Progress: 97 / 100
Progress: 98 / 100
Progress: 99 / 100
Progress: 100 / 100
```

```
Warning in (function (..., deparse.level = 1) : number of columns of result is
not a multiple of vector length (arg 13)
```

```
Warning in (function (..., deparse.level = 1) : number of columns of result is
not a multiple of vector length (arg 13)

Warning in (function (..., deparse.level = 1) : number of columns of result is
not a multiple of vector length (arg 13)

Warning in (function (..., deparse.level = 1) : number of columns of result is
not a multiple of vector length (arg 13)

Warning in (function (..., deparse.level = 1) : number of columns of result is
not a multiple of vector length (arg 13)

Warning in (function (..., deparse.level = 1) : number of columns of result is
not a multiple of vector length (arg 13)
```

Source: Article Notebook

0.10.3 Simulation Results

0.10.3.1 Population Model Summary

```
# Shows the "true" population values used for data generation
summaryPopulation(my_modelci)
```

```
                 psycho=~Q6_P psycho=~Q7_P psycho=~Q11_P psycho=~Q19_P
Population Value 0.6594432    0.6773329    0.6707227     0.8690565
                 psycho=~Q26_P physical=~Q3_F physical=~Q4_F physical=~Q10_F
Population Value 0.6478382     0.5102898      0.423065       0.8108556
                 physical=~Q15_F physical=~Q16_F physical=~Q17_F
Population Value 0.6432376       0.5896579       0.9226506
                 physical=~Q18_F social=~Q20_S social=~Q21_S social=~Q22_S
Population Value 0.8877922       0.8187287     0.5864209     0.7477769
                 environment=~Q8_A environment=~Q9_A environment=~Q12_A
Population Value 0.5097653         0.62881           0.7150961
                 environment=~Q13_A environment=~Q14_A environment=~Q23_A
Population Value 0.662578           0.7315944          0.5804092
                 environment=~Q24_A environment=~Q25_A Q3_F~~Q4_F Q6_P|t1
Population Value 0.4923491          0.5415261          0.3866074  -1.930975
                 Q6_P|t2   Q6_P|t3    Q6_P|t4  Q7_P|t1   Q7_P|t2   Q7_P|t3
Population Value -1.318903 -0.5975455 0.541232 -2.243993 -1.237626 -0.2901144
                 Q7_P|t4  Q11_P|t1  Q11_P|t2  Q11_P|t3  Q11_P|t4  Q19_P|t1
Population Value 1.182853 -2.033274 -1.296386 -0.415019 0.5889811 -1.061543
                 Q19_P|t2   Q19_P|t3  Q26_P|t1 Q26_P|t2  Q26_P|t3   Q26_P|t4
Population Value -0.4046037 0.8294023 -1.84562 -1.378337 -0.7960913 0.7226189
                 Q3_F|t1  Q3_F|t2   Q3_F|t3    Q3_F|t4   Q4_F|t1   Q4_F|t2
Population Value -2.52831 -1.470759 -0.8193137 0.1163763 -2.188224 -1.301954
                 Q4_F|t3    Q4_F|t4   Q10_F|t1  Q10_F|t2 Q10_F|t3   Q10_F|t4
Population Value -0.7540623 0.1671656 -2.343531 -1.28537 -0.1115569 0.9243563
                 Q15_F|t1 Q15_F|t2  Q15_F|t3  Q15_F|t4   Q16_F|t1   Q16_F|t2
Population Value -2.52831 -1.819942 -1.247991 -0.1671656 -0.7382493 -0.1163763
                 Q16_F|t3  Q17_F|t1  Q17_F|t2   Q17_F|t3  Q18_F|t1  Q18_F|t2
Population Value 0.9170407 -1.069992 -0.4758336 0.9061585 -1.040744 -0.4785164
                 Q18_F|t3  Q20_S|t1  Q20_S|t2  Q20_S|t3  Q21_S|t1   Q21_S|t2
```

```
Population Value 0.7928088 -1.178048 -0.247912 0.9925754 -0.7830123 -0.0347215
                 Q21_S|t3  Q22_S|t1 Q22_S|t2   Q22_S|t3 Q8_A|t1  Q8_A|t2
Population Value 0.9847694 -1.14977 -0.1526161 1.069992 -1.41647 -0.4919826
                 Q8_A|t3   Q8_A|t4  Q9_A|t1   Q9_A|t2   Q9_A|t3    Q9_A|t4
Population Value 0.5579386 1.727657 -1.858929 -1.095811 -0.1405162 1.372173
                 Q12_A|t1  Q12_A|t2   Q12_A|t3   Q12_A|t4  Q13_A|t1 Q13_A|t2
Population Value -1.648104 -0.9466061 0.08028871 0.8260301 -2.66754 -1.603236
                 Q13_A|t3  Q13_A|t4  Q14_A|t1  Q14_A|t2   Q14_A|t3  Q14_A|t4
Population Value -0.652856 0.6918744 -1.819942 -0.9134013 0.1091481 1.290859
                 Q23_A|t1  Q23_A|t2   Q23_A|t3  Q24_A|t1  Q24_A|t2   Q24_A|t3
Population Value -1.212278 -0.6647534 0.4098058 -1.061543 -0.3864811 0.7477147
                 Q25_A|t1  Q25_A|t2   Q25_A|t3  Q6_P~~Q6_P Q7_P~~Q7_P
Population Value -1.117899 -0.4865856 0.6061538 0.5651347  0.5412202
                 Q11_P~~Q11_P Q19_P~~Q19_P Q26_P~~Q26_P Q3_F~~Q3_F Q4_F~~Q4_F
Population Value 0.5501311    0.2447408    0.5803056    0.7396043  0.821016
                 Q10_F~~Q10_F Q15_F~~Q15_F Q16_F~~Q16_F Q17_F~~Q17_F
Population Value 0.3425131    0.5862454    0.6523036    0.1487158
                 Q18_F~~Q18_F Q20_S~~Q20_S Q21_S~~Q21_S Q22_S~~Q22_S Q8_A~~Q8_A
Population Value 0.211825     0.3296833    0.6561106    0.4408296    0.7401393
                 Q9_A~~Q9_A Q12_A~~Q12_A Q13_A~~Q13_A Q14_A~~Q14_A Q23_A~~Q23_A
Population Value 0.6045979  0.4886376    0.5609904    0.4647697    0.6631252
                 Q24_A~~Q24_A Q25_A~~Q25_A psycho~~psycho physical~~physical
Population Value 0.7575924    0.7067495    1              1
                 social~~social environment~~environment psycho~~physical
Population Value 1              1                        0.922137
                 psycho~~social psycho~~environment physical~~social
Population Value 0.7546453      0.6585704           0.6378955
                 physical~~environment social~~environment Q6_P~*~Q6_P
Population Value 0.629862              0.5500233           1
                 Q7_P~*~Q7_P Q11_P~*~Q11_P Q19_P~*~Q19_P Q26_P~*~Q26_P
Population Value 1           1             1             1
                 Q3_F~*~Q3_F Q4_F~*~Q4_F Q10_F~*~Q10_F Q15_F~*~Q15_F
Population Value 1           1           1             1
                 Q16_F~*~Q16_F Q17_F~*~Q17_F Q18_F~*~Q18_F Q20_S~*~Q20_S
Population Value 1             1             1             1
                 Q21_S~*~Q21_S Q22_S~*~Q22_S Q8_A~*~Q8_A Q9_A~*~Q9_A
Population Value 1             1             1           1
                 Q12_A~*~Q12_A Q13_A~*~Q13_A Q14_A~*~Q14_A Q23_A~*~Q23_A
Population Value 1             1             1             1
                 Q24_A~*~Q24_A Q25_A~*~Q25_A Q6_P~1 Q7_P~1 Q11_P~1 Q19_P~1
Population Value 1             1             0      0      0       0
                 Q26_P~1 Q3_F~1 Q4_F~1 Q10_F~1 Q15_F~1 Q16_F~1 Q17_F~1 Q18_F~1
Population Value 0       0      0      0       0       0       0       0
                 Q20_S~1 Q21_S~1 Q22_S~1 Q8_A~1 Q9_A~1 Q12_A~1 Q13_A~1 Q14_A~1
Population Value 0       0       0       0      0      0       0       0
                 Q23_A~1 Q24_A~1 Q25_A~1 psycho~1 physical~1 social~1
Population Value 0       0       0       0        0          0
                 environment~1
Population Value 0
```

Source: Article Notebook

0.10.3.2 Overall Summary

```r
# Summary of simulation results
summary(my_modelci)
```

```
RESULT OBJECT
Model Type
[1] "lavaan"
========= Fit Indices Cutoffs ============
              Alpha
Fit Indices         0.1    0.05    0.01   0.001    Mean     SD
  chisq.scaled 251.568 255.332 267.794 271.527 225.610 18.460
  rmsea.scaled   0.011   0.012   0.014   0.014   0.004  0.005
  cfi.scaled     1.000   1.000   1.000   1.000   1.000  0.001
  tli.scaled     1.001   1.001   1.003   1.003   1.000  0.001
  srmr           0.024   0.024   0.025   0.025   0.022  0.001
========= Parameter Estimates and Standard Errors ============
                      Estimate Average Estimate SD Average SE
psycho=~Q6_P                     0.660       0.023      0.022
psycho=~Q7_P                     0.676       0.020      0.021
psycho=~Q11_P                    0.676       0.023      0.021
psycho=~Q19_P                    0.871       0.013      0.013
psycho=~Q26_P                    0.648       0.024      0.023
physical=~Q3_F                   0.512       0.033      0.029
physical=~Q4_F                   0.420       0.033      0.031
physical=~Q10_F                  0.813       0.014      0.015
physical=~Q15_F                  0.648       0.026      0.025
physical=~Q16_F                  0.592       0.026      0.025
physical=~Q17_F                  0.922       0.010      0.010
physical=~Q18_F                  0.888       0.013      0.012
social=~Q20_S                    0.821       0.023      0.022
social=~Q21_S                    0.586       0.027      0.030
social=~Q22_S                    0.749       0.023      0.024
environment=~Q8_A                0.504       0.034      0.030
environment=~Q9_A                0.631       0.028      0.027
environment=~Q12_A               0.719       0.023      0.023
environment=~Q13_A               0.662       0.025      0.026
environment=~Q14_A               0.729       0.023      0.023
environment=~Q23_A               0.579       0.028      0.029
environment=~Q24_A               0.489       0.032      0.032
environment=~Q25_A               0.541       0.031      0.030
Q3_F~~Q4_F                       0.390       0.026      0.029
Q6_P|t1                         -1.933       0.083      0.081
Q6_P|t2                         -1.315       0.057      0.054
Q6_P|t3                         -0.588       0.038      0.041
Q6_P|t4                          0.546       0.043      0.041
Q7_P|t1                         -2.267       0.118      0.110
Q7_P|t2                         -1.229       0.047      0.052
```

Q7_P|t3	-0.287	0.033	0.039
Q7_P|t4	1.182	0.048	0.050
Q11_P|t1	-2.040	0.095	0.089
Q11_P|t2	-1.297	0.054	0.053
Q11_P|t3	-0.414	0.044	0.040
Q11_P|t4	0.591	0.041	0.041
Q19_P|t1	-1.055	0.052	0.048
Q19_P|t2	-0.404	0.039	0.040
Q19_P|t3	0.831	0.047	0.044
Q26_P|t1	-1.849	0.083	0.076
Q26_P|t2	-1.377	0.057	0.056
Q26_P|t3	-0.790	0.046	0.044
Q26_P|t4	0.722	0.043	0.043
Q3_F|t1	-2.552	0.159	0.151
Q3_F|t2	-1.470	0.063	0.059
Q3_F|t3	-0.818	0.049	0.044
Q3_F|t4	0.108	0.039	0.039
Q4_F|t1	-2.200	0.101	0.103
Q4_F|t2	-1.303	0.053	0.053
Q4_F|t3	-0.754	0.050	0.043
Q4_F|t4	0.167	0.039	0.039
Q10_F|t1	-2.359	0.129	0.121
Q10_F|t2	-1.283	0.058	0.053
Q10_F|t3	-0.108	0.036	0.039
Q10_F|t4	0.924	0.052	0.045
Q15_F|t1	-2.542	0.192	0.150
Q15_F|t2	-1.801	0.102	0.073
Q15_F|t3	-1.221	0.153	0.052
Q15_F|t4	-0.182	0.092	0.039
Q16_F|t1	-0.724	0.095	0.043
Q16_F|t2	-0.086	0.147	0.039
Q16_F|t3	0.886	0.288	0.046
Q17_F|t1	-1.053	0.093	0.048
Q17_F|t2	-0.442	0.191	0.040
Q17_F|t3	0.872	0.286	0.045
Q18_F|t1	-1.029	0.094	0.047
Q18_F|t2	-0.453	0.171	0.040
Q18_F|t3	0.756	0.280	0.044
Q20_S|t1	-1.167	0.138	0.050
Q20_S|t2	-0.225	0.173	0.039
Q20_S|t3	0.954	0.254	0.046
Q21_S|t1	-0.772	0.107	0.043
Q21_S|t2	-0.020	0.143	0.039
Q21_S|t3	0.938	0.305	0.046
Q22_S|t1	-1.135	0.145	0.050
Q22_S|t2	-0.125	0.171	0.039
Q22_S|t3	1.012	0.365	0.048
Q8_A|t1	-1.404	0.136	0.057
Q8_A|t2	-0.473	0.141	0.041

```
Q8_A|t3                          0.587        0.170      0.042
Q8_A|t4                          1.653        0.513      0.069
Q9_A|t1                         -1.849        0.137      0.076
Q9_A|t2                         -1.076        0.142      0.048
Q9_A|t3                         -0.108        0.215      0.039
Q9_A|t4                          1.322        0.441      0.056
Q12_A|t1                        -1.634        0.117      0.065
Q12_A|t2                        -0.929        0.154      0.046
Q12_A|t3                         0.098        0.109      0.039
Q12_A|t4                         0.758        0.496      0.047
Q13_A|t1                        -2.623        0.302      0.170
Q13_A|t2                        -1.555        0.215      0.063
Q13_A|t3                        -0.584        0.298      0.042
Q13_A|t4                         0.570        0.555      0.044
Q14_A|t1                        -1.783        0.201      0.073
Q14_A|t2                        -0.865        0.229      0.045
Q14_A|t3                         0.165        0.255      0.040
Q14_A|t4                         1.166        0.548      0.053
Q23_A|t1                        -1.188        0.128      0.051
Q23_A|t2                        -0.610        0.231      0.042
Q23_A|t3                         0.342        0.329      0.040
Q24_A|t1                        -1.033        0.156      0.048
Q24_A|t2                        -0.334        0.245      0.040
Q24_A|t3                         0.654        0.410      0.043
Q25_A|t1                        -1.087        0.144      0.049
Q25_A|t2                        -0.431        0.244      0.041
Q25_A|t3                         0.625        0.084      0.040
psycho~~physical                 0.914        0.035      0.013
psycho~~social                   0.750        0.031      0.024
psycho~~environment              0.663        0.025      0.025
physical~~social                 0.636        0.025      0.027
physical~~environment            0.631        0.028      0.025
social~~environment              0.557        0.040      0.031
                      Power (Not equal 0) Std Est Std Est SD Std Ave SE
psycho=~Q6_P                         1.00   0.660        0.023        0.022
psycho=~Q7_P                         1.00   0.676        0.020        0.021
psycho=~Q11_P                        1.00   0.676        0.023        0.021
psycho=~Q19_P                        1.00   0.871        0.013        0.013
psycho=~Q26_P                        1.00   0.648        0.024        0.023
physical=~Q3_F                       1.00   0.512        0.033        0.029
physical=~Q4_F                       1.00   0.420        0.033        0.031
physical=~Q10_F                      1.00   0.813        0.014        0.015
physical=~Q15_F                      1.00   0.648        0.026        0.025
physical=~Q16_F                      1.00   0.592        0.026        0.025
physical=~Q17_F                      1.00   0.922        0.010        0.010
physical=~Q18_F                      1.00   0.888        0.013        0.012
social=~Q20_S                        1.00   0.821        0.023        0.022
social=~Q21_S                        1.00   0.586        0.027        0.030
social=~Q22_S                        1.00   0.749        0.023        0.024
```

environment=~Q8_A	1.00	0.504	0.034	0.030
environment=~Q9_A	1.00	0.631	0.028	0.027
environment=~Q12_A	1.00	0.719	0.023	0.023
environment=~Q13_A	1.00	0.662	0.025	0.026
environment=~Q14_A	1.00	0.729	0.023	0.023
environment=~Q23_A	1.00	0.579	0.028	0.029
environment=~Q24_A	1.00	0.489	0.032	0.032
environment=~Q25_A	1.00	0.541	0.031	0.030
Q3_F~~Q4_F	1.00	0.502	0.030	0.031
Q6_P|t1	1.00	-1.933	0.083	0.081
Q6_P|t2	1.00	-1.315	0.057	0.054
Q6_P|t3	1.00	-0.588	0.038	0.041
Q6_P|t4	1.00	0.546	0.043	0.041
Q7_P|t1	1.00	-2.267	0.118	0.110
Q7_P|t2	1.00	-1.229	0.047	0.052
Q7_P|t3	1.00	-0.287	0.033	0.039
Q7_P|t4	1.00	1.182	0.048	0.050
Q11_P|t1	1.00	-2.040	0.095	0.089
Q11_P|t2	1.00	-1.297	0.054	0.053
Q11_P|t3	1.00	-0.414	0.044	0.040
Q11_P|t4	1.00	0.591	0.041	0.041
Q19_P|t1	1.00	-1.055	0.052	0.048
Q19_P|t2	1.00	-0.404	0.039	0.040
Q19_P|t3	1.00	0.831	0.047	0.044
Q26_P|t1	1.00	-1.849	0.083	0.076
Q26_P|t2	1.00	-1.377	0.057	0.056
Q26_P|t3	1.00	-0.790	0.046	0.044
Q26_P|t4	1.00	0.722	0.043	0.043
Q3_F|t1	1.00	-2.552	0.159	0.151
Q3_F|t2	1.00	-1.470	0.063	0.059
Q3_F|t3	1.00	-0.818	0.049	0.044
Q3_F|t4	0.79	0.108	0.039	0.039
Q4_F|t1	1.00	-2.200	0.101	0.103
Q4_F|t2	1.00	-1.303	0.053	0.053
Q4_F|t3	1.00	-0.754	0.050	0.043
Q4_F|t4	1.00	0.167	0.039	0.039
Q10_F|t1	1.00	-2.359	0.129	0.121
Q10_F|t2	1.00	-1.283	0.058	0.053
Q10_F|t3	0.76	-0.108	0.036	0.039
Q10_F|t4	1.00	0.924	0.052	0.045
Q15_F|t1	1.00	-2.542	0.192	0.150
Q15_F|t2	1.00	-1.801	0.102	0.073
Q15_F|t3	1.00	-1.221	0.153	0.052
Q15_F|t4	0.98	-0.182	0.092	0.039
Q16_F|t1	1.00	-0.724	0.095	0.043
Q16_F|t2	0.78	-0.086	0.147	0.039
Q16_F|t3	1.00	0.886	0.288	0.046
Q17_F|t1	1.00	-1.053	0.093	0.048
Q17_F|t2	1.00	-0.442	0.191	0.040

Q17_F|t3	1.00	0.872	0.286	0.045
Q18_F|t1	1.00	-1.029	0.094	0.047
Q18_F|t2	1.00	-0.453	0.171	0.040
Q18_F|t3	1.00	0.756	0.280	0.044
Q20_S|t1	1.00	-1.167	0.138	0.050
Q20_S|t2	1.00	-0.225	0.173	0.039
Q20_S|t3	1.00	0.954	0.254	0.046
Q21_S|t1	1.00	-0.772	0.107	0.043
Q21_S|t2	0.15	-0.020	0.143	0.039
Q21_S|t3	1.00	0.938	0.305	0.046
Q22_S|t1	1.00	-1.135	0.145	0.050
Q22_S|t2	0.99	-0.125	0.171	0.039
Q22_S|t3	1.00	1.012	0.365	0.048
Q8_A|t1	1.00	-1.404	0.136	0.057
Q8_A|t2	1.00	-0.473	0.141	0.041
Q8_A|t3	1.00	0.587	0.170	0.042
Q8_A|t4	1.00	1.653	0.513	0.069
Q9_A|t1	1.00	-1.849	0.137	0.076
Q9_A|t2	1.00	-1.076	0.142	0.048
Q9_A|t3	0.95	-0.108	0.215	0.039
Q9_A|t4	1.00	1.322	0.441	0.056
Q12_A|t1	1.00	-1.634	0.117	0.065
Q12_A|t2	0.99	-0.929	0.154	0.046
Q12_A|t3	0.59	0.098	0.109	0.039
Q12_A|t4	1.00	0.758	0.496	0.047
Q13_A|t1	1.00	-2.623	0.302	0.170
Q13_A|t2	1.00	-1.555	0.215	0.063
Q13_A|t3	1.00	-0.584	0.298	0.042
Q13_A|t4	1.00	0.570	0.555	0.044
Q14_A|t1	1.00	-1.783	0.201	0.073
Q14_A|t2	0.99	-0.865	0.229	0.045
Q14_A|t3	0.80	0.165	0.255	0.040
Q14_A|t4	1.00	1.166	0.548	0.053
Q23_A|t1	1.00	-1.188	0.128	0.051
Q23_A|t2	1.00	-0.610	0.231	0.042
Q23_A|t3	1.00	0.342	0.329	0.040
Q24_A|t1	1.00	-1.033	0.156	0.048
Q24_A|t2	1.00	-0.334	0.245	0.040
Q24_A|t3	1.00	0.654	0.410	0.043
Q25_A|t1	1.00	-1.087	0.144	0.049
Q25_A|t2	1.00	-0.431	0.244	0.041
Q25_A|t3	1.00	0.625	0.084	0.040
psycho~~physical	1.00	0.914	0.035	0.013
psycho~~social	1.00	0.750	0.031	0.024
psycho~~environment	1.00	0.663	0.025	0.025
physical~~social	1.00	0.636	0.025	0.027
physical~~environment	1.00	0.631	0.028	0.025
social~~environment	1.00	0.557	0.040	0.031

Average Param Average Bias Coverage

psycho=~Q6_P	0.659	0.000	0.94
psycho=~Q7_P	0.677	-0.001	0.96
psycho=~Q11_P	0.671	0.005	0.92
psycho=~Q19_P	0.869	0.002	0.95
psycho=~Q26_P	0.648	0.000	0.94
physical=~Q3_F	0.510	0.001	0.89
physical=~Q4_F	0.423	-0.003	0.94
physical=~Q10_F	0.811	0.002	0.94
physical=~Q15_F	0.643	0.005	0.93
physical=~Q16_F	0.590	0.003	0.94
physical=~Q17_F	0.923	0.000	0.97
physical=~Q18_F	0.888	0.000	0.95
social=~Q20_S	0.819	0.002	0.95
social=~Q21_S	0.586	0.000	0.96
social=~Q22_S	0.748	0.002	0.95
environment=~Q8_A	0.510	-0.006	0.94
environment=~Q9_A	0.629	0.003	0.92
environment=~Q12_A	0.715	0.004	0.96
environment=~Q13_A	0.663	-0.001	0.95
environment=~Q14_A	0.732	-0.002	0.94
environment=~Q23_A	0.580	-0.001	0.96
environment=~Q24_A	0.492	-0.003	0.96
environment=~Q25_A	0.542	-0.001	0.97
Q3_F~~Q4_F	0.387	0.004	0.99
Q6_P|t1	-1.931	-0.002	0.93
Q6_P|t2	-1.319	0.004	0.95
Q6_P|t3	-0.598	0.009	0.96
Q6_P|t4	0.541	0.005	0.94
Q7_P|t1	-2.244	-0.023	0.94
Q7_P|t2	-1.238	0.008	0.96
Q7_P|t3	-0.290	0.003	0.99
Q7_P|t4	1.183	-0.001	0.98
Q11_P|t1	-2.033	-0.007	0.94
Q11_P|t2	-1.296	0.000	0.97
Q11_P|t3	-0.415	0.001	0.93
Q11_P|t4	0.589	0.002	0.97
Q19_P|t1	-1.062	0.006	0.93
Q19_P|t2	-0.405	0.000	0.98
Q19_P|t3	0.829	0.001	0.94
Q26_P|t1	-1.846	-0.003	0.93
Q26_P|t2	-1.378	0.001	0.94
Q26_P|t3	-0.796	0.006	0.91
Q26_P|t4	0.723	-0.001	0.95
Q3_F|t1	-2.528	-0.023	0.99
Q3_F|t2	-1.471	0.001	0.94
Q3_F|t3	-0.819	0.001	0.90
Q3_F|t4	0.116	-0.008	0.93
Q4_F|t1	-2.188	-0.012	0.97
Q4_F|t2	-1.302	-0.001	0.97

Q4_F|t3	-0.754	0.001	0.91
Q4_F|t4	0.167	0.000	1.00
Q10_F|t1	-2.344	-0.015	0.95
Q10_F|t2	-1.285	0.002	0.93
Q10_F|t3	-0.112	0.003	0.96
Q10_F|t4	0.924	0.000	0.94
Q15_F|t1	-2.528	-0.013	0.93
Q15_F|t2	-1.820	0.019	0.92
Q15_F|t3	-1.248	0.027	0.91
Q15_F|t4	-0.167	-0.015	0.90
Q16_F|t1	-0.738	0.014	0.93
Q16_F|t2	-0.116	0.030	0.94
Q16_F|t3	0.917	-0.031	0.93
Q17_F|t1	-1.070	0.017	0.91
Q17_F|t2	-0.476	0.034	0.91
Q17_F|t3	0.906	-0.034	0.92
Q18_F|t1	-1.041	0.012	0.92
Q18_F|t2	-0.479	0.026	0.92
Q18_F|t3	0.793	-0.037	0.91
Q20_S|t1	-1.178	0.011	0.92
Q20_S|t2	-0.248	0.022	0.94
Q20_S|t3	0.993	-0.038	0.91
Q21_S|t1	-0.783	0.011	0.96
Q21_S|t2	-0.035	0.014	0.92
Q21_S|t3	0.985	-0.047	0.94
Q22_S|t1	-1.150	0.015	0.95
Q22_S|t2	-0.153	0.027	0.96
Q22_S|t3	1.070	-0.058	0.91
Q8_A|t1	-1.416	0.012	0.95
Q8_A|t2	-0.492	0.019	0.93
Q8_A|t3	0.558	0.029	0.91
Q8_A|t4	1.728	-0.075	0.97
Q9_A|t1	-1.859	0.010	0.92
Q9_A|t2	-1.096	0.019	0.96
Q9_A|t3	-0.141	0.032	0.93
Q9_A|t4	1.372	-0.051	0.94
Q12_A|t1	-1.648	0.014	0.88
Q12_A|t2	-0.947	0.018	0.89
Q12_A|t3	0.080	0.018	0.94
Q12_A|t4	0.826	-0.068	0.93
Q13_A|t1	-2.668	0.045	0.88
Q13_A|t2	-1.603	0.048	0.90
Q13_A|t3	-0.653	0.069	0.91
Q13_A|t4	0.692	-0.122	0.90
Q14_A|t1	-1.820	0.037	0.91
Q14_A|t2	-0.913	0.048	0.89
Q14_A|t3	0.109	0.056	0.90
Q14_A|t4	1.291	-0.125	0.92
Q23_A|t1	-1.212	0.024	0.92

```
Q23_A|t2                          -0.665       0.054     0.88
Q23_A|t3                           0.410      -0.067     0.92
Q24_A|t1                          -1.062       0.029     0.88
Q24_A|t2                          -0.386       0.053     0.91
Q24_A|t3                           0.748      -0.093     0.92
Q25_A|t1                          -1.118       0.031     0.90
Q25_A|t2                          -0.487       0.056     0.90
Q25_A|t3                           0.606       0.019     0.87
psycho~~physical                   0.922      -0.008     0.93
psycho~~social                     0.755      -0.005     0.94
psycho~~environment                0.659       0.004     0.91
physical~~social                   0.638      -0.002     0.98
physical~~environment              0.630       0.001     0.95
social~~environment                0.550       0.007     0.88
========= Correlation between Fit Indices ============
             chisq.scaled rmsea.scaled cfi.scaled tli.scaled   srmr
chisq.scaled        1.000        0.939     -0.906     -0.998  0.942
rmsea.scaled        0.939        1.000     -0.950     -0.933  0.883
cfi.scaled         -0.906       -0.950      1.000      0.907 -0.853
tli.scaled         -0.998       -0.933      0.907      1.000 -0.940
srmr                0.942        0.883     -0.853     -0.940  1.000
================== Replications =====================
Number of replications = 100
Number of converged replications = 100
Number of nonconverged replications:
   1. Nonconvergent Results = 0
   2. Nonconvergent results from multiple imputation = 0
   3. At least one SE were negative or NA = 0
   4. Nonpositive-definite latent or observed (residual) covariance matrix
      (e.g., Heywood case or linear dependency) = 0
```

Source: Article Notebook

> **Note**
>
> The summary shows power for detecting parameter differences from zero. Parameters with power < .80 may need larger samples for reliable detection. Note that only two threshold parameters have power < 0.8, and thresholds are generally not parameters of interest in CFA.

0.10.3.3 Short Summary

```r
# Condensed summary
summaryShort(my_modelci)
```

```
RESULT OBJECT
[1] "lavaan"
Model Type: lavaan
Convergence 100 / 100
Sample size: 1047
```

```
Percent Completely Missing at Random: 0
Percent Missing at Random: 0
========= Fit Indices Cutoffs ============
              Alpha
Fit Indices        0.05    Mean     SD
  chisq.scaled 255.332 225.610 18.460
  rmsea.scaled   0.012   0.004  0.005
  cfi.scaled     1.000   1.000  0.001
  tli.scaled     1.001   1.000  0.001
  srmr           0.024   0.022  0.001
```

Source: Article Notebook

0.10.4 Fit Index Cutoffs

Simulation-based cutoffs at = .05 provide an alternative to rule-of-thumb values, tailored to this specific model and sample size.

```r
knitr::kable(as.data.frame(t(getCutoff(my_modelci, 0.05,
  usedFit = c("srmr", "rmsea.scaled", "cfi.scaled")))), digits = 4)
```

Table 12: Fit index cutoffs at alpha = 0.05

	95%
srmr	0.0243
rmsea.scaled	0.0118
cfi.scaled	1.0000

Source: Article Notebook

0.10.5 Power to Detect Misfit

This table shows the proportion of replications where fit indices exceeded rule-of-thumb cutoffs, indicating that the model would be rejected as having poor fit.

```r
knitr::kable(as.data.frame(t(getPowerFit(my_modelci, cutoff = rule_of_thumb,
  usedFit = c("srmr", "rmsea.scaled", "cfi.scaled")))), digits = 4)
```

Table 13: Proportion of replications indicating worse fit than rule-of-thumb cutoffs

cfi.scaled	rmsea.scaled	srmr
1	0	0

Source: Article Notebook

Interpreting Power to Detect Misfit

Low values (near 0) indicate that the model fits well in most replications—we have low power to reject it, which is actually what we want for a well-fitting model. High values would indicate we have high power to detect that the model doesn't fit well.

0.10.6 Sampling Distributions of Fit Indices

Visual representation of fit index distributions helps understand the variability and precision of fit assessment.

```
plotPowerFit(my_modelci, cutoff = rule_of_thumb, logistic = FALSE,
             usedFit = c("srmr", "rmsea.scaled", "cfi.scaled"))
```

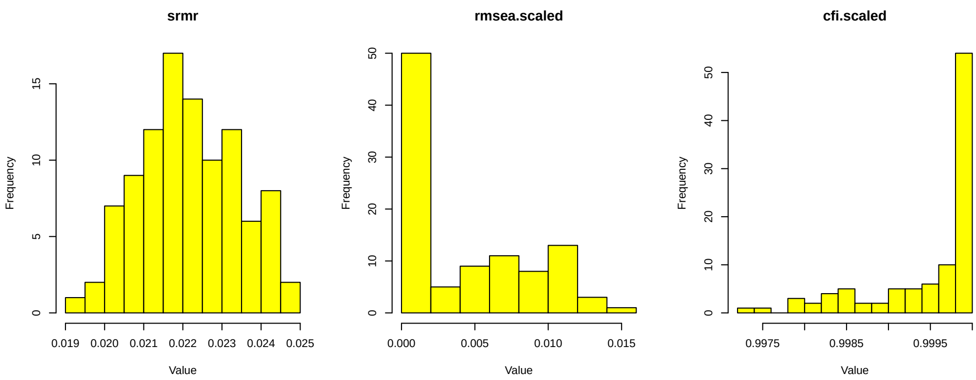

Figure 6: Sampling distributions of fit indices — Power of rejecting misspecified models

Source: Article Notebook

```
plotCutoff(my_modelci, 0.05, usedFit = c("srmr", "rmsea.scaled", "cfi.scaled"))
```

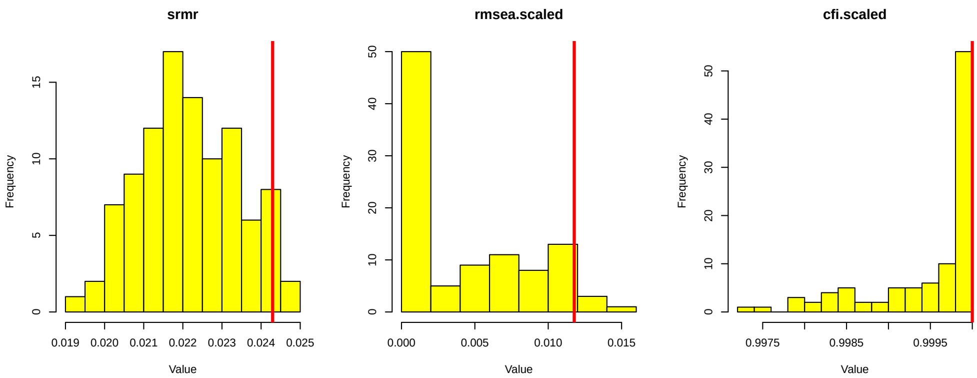

Figure 7: Sampling distributions of fit indices with cutoffs at alpha = 0.05

Source: Article Notebook

0.10.7 Detailed Parameter Summary

This table provides comprehensive information about each parameter, including:

- **Population values** (true generating values)
- **Average estimates** across replications
- **Standard deviation** of estimates
- **Average standard error** from model output
- **Coverage** (proportion of 95% CIs containing true value, should be ~.95)
- **Power** (proportion rejecting H : parameter = 0)
- **Relative bias** in estimates and standard errors

```r
summaryParam(my_modelci, detail = TRUE, digits = 3)
```

```
                      Estimate Average Estimate SD Average SE
psycho=~Q6_P                     0.660       0.023      0.022
psycho=~Q7_P                     0.676       0.020      0.021
psycho=~Q11_P                    0.676       0.023      0.021
psycho=~Q19_P                    0.871       0.013      0.013
psycho=~Q26_P                    0.648       0.024      0.023
physical=~Q3_F                   0.512       0.033      0.029
physical=~Q4_F                   0.420       0.033      0.031
physical=~Q10_F                  0.813       0.014      0.015
physical=~Q15_F                  0.648       0.026      0.025
physical=~Q16_F                  0.592       0.026      0.025
physical=~Q17_F                  0.922       0.010      0.010
physical=~Q18_F                  0.888       0.013      0.012
social=~Q20_S                    0.821       0.023      0.022
social=~Q21_S                    0.586       0.027      0.030
social=~Q22_S                    0.749       0.023      0.024
environment=~Q8_A                0.504       0.034      0.030
environment=~Q9_A                0.631       0.028      0.027
environment=~Q12_A               0.719       0.023      0.023
environment=~Q13_A               0.662       0.025      0.026
environment=~Q14_A               0.729       0.023      0.023
environment=~Q23_A               0.579       0.028      0.029
environment=~Q24_A               0.489       0.032      0.032
environment=~Q25_A               0.541       0.031      0.030
Q3_F~~Q4_F                       0.390       0.026      0.029
Q6_P|t1                         -1.933       0.083      0.081
Q6_P|t2                         -1.315       0.057      0.054
Q6_P|t3                         -0.588       0.038      0.041
Q6_P|t4                          0.546       0.043      0.041
Q7_P|t1                         -2.267       0.118      0.110
Q7_P|t2                         -1.229       0.047      0.052
Q7_P|t3                         -0.287       0.033      0.039
Q7_P|t4                          1.182       0.048      0.050
Q11_P|t1                        -2.040       0.095      0.089
Q11_P|t2                        -1.297       0.054      0.053
```

Q11_P|t3	-0.414	0.044	0.040
Q11_P|t4	0.591	0.041	0.041
Q19_P|t1	-1.055	0.052	0.048
Q19_P|t2	-0.404	0.039	0.040
Q19_P|t3	0.831	0.047	0.044
Q26_P|t1	-1.849	0.083	0.076
Q26_P|t2	-1.377	0.057	0.056
Q26_P|t3	-0.790	0.046	0.044
Q26_P|t4	0.722	0.043	0.043
Q3_F|t1	-2.552	0.159	0.151
Q3_F|t2	-1.470	0.063	0.059
Q3_F|t3	-0.818	0.049	0.044
Q3_F|t4	0.108	0.039	0.039
Q4_F|t1	-2.200	0.101	0.103
Q4_F|t2	-1.303	0.053	0.053
Q4_F|t3	-0.754	0.050	0.043
Q4_F|t4	0.167	0.039	0.039
Q10_F|t1	-2.359	0.129	0.121
Q10_F|t2	-1.283	0.058	0.053
Q10_F|t3	-0.108	0.036	0.039
Q10_F|t4	0.924	0.052	0.045
Q15_F|t1	-2.542	0.192	0.150
Q15_F|t2	-1.801	0.102	0.073
Q15_F|t3	-1.221	0.153	0.052
Q15_F|t4	-0.182	0.092	0.039
Q16_F|t1	-0.724	0.095	0.043
Q16_F|t2	-0.086	0.147	0.039
Q16_F|t3	0.886	0.288	0.046
Q17_F|t1	-1.053	0.093	0.048
Q17_F|t2	-0.442	0.191	0.040
Q17_F|t3	0.872	0.286	0.045
Q18_F|t1	-1.029	0.094	0.047
Q18_F|t2	-0.453	0.171	0.040
Q18_F|t3	0.756	0.280	0.044
Q20_S|t1	-1.167	0.138	0.050
Q20_S|t2	-0.225	0.173	0.039
Q20_S|t3	0.954	0.254	0.046
Q21_S|t1	-0.772	0.107	0.043
Q21_S|t2	-0.020	0.143	0.039
Q21_S|t3	0.938	0.305	0.046
Q22_S|t1	-1.135	0.145	0.050
Q22_S|t2	-0.125	0.171	0.039
Q22_S|t3	1.012	0.365	0.048
Q8_A|t1	-1.404	0.136	0.057
Q8_A|t2	-0.473	0.141	0.041
Q8_A|t3	0.587	0.170	0.042
Q8_A|t4	1.653	0.513	0.069
Q9_A|t1	-1.849	0.137	0.076
Q9_A|t2	-1.076	0.142	0.048

```
Q9_A|t3                        -0.108        0.215       0.039
Q9_A|t4                         1.322        0.441       0.056
Q12_A|t1                       -1.634        0.117       0.065
Q12_A|t2                       -0.929        0.154       0.046
Q12_A|t3                        0.098        0.109       0.039
Q12_A|t4                        0.758        0.496       0.047
Q13_A|t1                       -2.623        0.302       0.170
Q13_A|t2                       -1.555        0.215       0.063
Q13_A|t3                       -0.584        0.298       0.042
Q13_A|t4                        0.570        0.555       0.044
Q14_A|t1                       -1.783        0.201       0.073
Q14_A|t2                       -0.865        0.229       0.045
Q14_A|t3                        0.165        0.255       0.040
Q14_A|t4                        1.166        0.548       0.053
Q23_A|t1                       -1.188        0.128       0.051
Q23_A|t2                       -0.610        0.231       0.042
Q23_A|t3                        0.342        0.329       0.040
Q24_A|t1                       -1.033        0.156       0.048
Q24_A|t2                       -0.334        0.245       0.040
Q24_A|t3                        0.654        0.410       0.043
Q25_A|t1                       -1.087        0.144       0.049
Q25_A|t2                       -0.431        0.244       0.041
Q25_A|t3                        0.625        0.084       0.040
psycho~~physical                0.914        0.035       0.013
psycho~~social                  0.750        0.031       0.024
psycho~~environment             0.663        0.025       0.025
physical~~social                0.636        0.025       0.027
physical~~environment           0.631        0.028       0.025
social~~environment             0.557        0.040       0.031
                      Power (Not equal 0) Std Est Std Est SD Std Ave SE
psycho=~Q6_P                          1.00   0.660       0.023       0.022
psycho=~Q7_P                          1.00   0.676       0.020       0.021
psycho=~Q11_P                         1.00   0.676       0.023       0.021
psycho=~Q19_P                         1.00   0.871       0.013       0.013
psycho=~Q26_P                         1.00   0.648       0.024       0.023
physical=~Q3_F                        1.00   0.512       0.033       0.029
physical=~Q4_F                        1.00   0.420       0.033       0.031
physical=~Q10_F                       1.00   0.813       0.014       0.015
physical=~Q15_F                       1.00   0.648       0.026       0.025
physical=~Q16_F                       1.00   0.592       0.026       0.025
physical=~Q17_F                       1.00   0.922       0.010       0.010
physical=~Q18_F                       1.00   0.888       0.013       0.012
social=~Q20_S                         1.00   0.821       0.023       0.022
social=~Q21_S                         1.00   0.586       0.027       0.030
social=~Q22_S                         1.00   0.749       0.023       0.024
environment=~Q8_A                     1.00   0.504       0.034       0.030
environment=~Q9_A                     1.00   0.631       0.028       0.027
environment=~Q12_A                    1.00   0.719       0.023       0.023
environment=~Q13_A                    1.00   0.662       0.025       0.026
```

environment=~Q14_A	1.00	0.729	0.023	0.023
environment=~Q23_A	1.00	0.579	0.028	0.029
environment=~Q24_A	1.00	0.489	0.032	0.032
environment=~Q25_A	1.00	0.541	0.031	0.030
Q3_F~~Q4_F	1.00	0.502	0.030	0.031
Q6_P|t1	1.00	-1.933	0.083	0.081
Q6_P|t2	1.00	-1.315	0.057	0.054
Q6_P|t3	1.00	-0.588	0.038	0.041
Q6_P|t4	1.00	0.546	0.043	0.041
Q7_P|t1	1.00	-2.267	0.118	0.110
Q7_P|t2	1.00	-1.229	0.047	0.052
Q7_P|t3	1.00	-0.287	0.033	0.039
Q7_P|t4	1.00	1.182	0.048	0.050
Q11_P|t1	1.00	-2.040	0.095	0.089
Q11_P|t2	1.00	-1.297	0.054	0.053
Q11_P|t3	1.00	-0.414	0.044	0.040
Q11_P|t4	1.00	0.591	0.041	0.041
Q19_P|t1	1.00	-1.055	0.052	0.048
Q19_P|t2	1.00	-0.404	0.039	0.040
Q19_P|t3	1.00	0.831	0.047	0.044
Q26_P|t1	1.00	-1.849	0.083	0.076
Q26_P|t2	1.00	-1.377	0.057	0.056
Q26_P|t3	1.00	-0.790	0.046	0.044
Q26_P|t4	1.00	0.722	0.043	0.043
Q3_F|t1	1.00	-2.552	0.159	0.151
Q3_F|t2	1.00	-1.470	0.063	0.059
Q3_F|t3	1.00	-0.818	0.049	0.044
Q3_F|t4	0.79	0.108	0.039	0.039
Q4_F|t1	1.00	-2.200	0.101	0.103
Q4_F|t2	1.00	-1.303	0.053	0.053
Q4_F|t3	1.00	-0.754	0.050	0.043
Q4_F|t4	1.00	0.167	0.039	0.039
Q10_F|t1	1.00	-2.359	0.129	0.121
Q10_F|t2	1.00	-1.283	0.058	0.053
Q10_F|t3	0.76	-0.108	0.036	0.039
Q10_F|t4	1.00	0.924	0.052	0.045
Q15_F|t1	1.00	-2.542	0.192	0.150
Q15_F|t2	1.00	-1.801	0.102	0.073
Q15_F|t3	1.00	-1.221	0.153	0.052
Q15_F|t4	0.98	-0.182	0.092	0.039
Q16_F|t1	1.00	-0.724	0.095	0.043
Q16_F|t2	0.78	-0.086	0.147	0.039
Q16_F|t3	1.00	0.886	0.288	0.046
Q17_F|t1	1.00	-1.053	0.093	0.048
Q17_F|t2	1.00	-0.442	0.191	0.040
Q17_F|t3	1.00	0.872	0.286	0.045
Q18_F|t1	1.00	-1.029	0.094	0.047
Q18_F|t2	1.00	-0.453	0.171	0.040
Q18_F|t3	1.00	0.756	0.280	0.044

Q20_S|t1	1.00	-1.167	0.138	0.050
Q20_S|t2	1.00	-0.225	0.173	0.039
Q20_S|t3	1.00	0.954	0.254	0.046
Q21_S|t1	1.00	-0.772	0.107	0.043
Q21_S|t2	0.15	-0.020	0.143	0.039
Q21_S|t3	1.00	0.938	0.305	0.046
Q22_S|t1	1.00	-1.135	0.145	0.050
Q22_S|t2	0.99	-0.125	0.171	0.039
Q22_S|t3	1.00	1.012	0.365	0.048
Q8_A|t1	1.00	-1.404	0.136	0.057
Q8_A|t2	1.00	-0.473	0.141	0.041
Q8_A|t3	1.00	0.587	0.170	0.042
Q8_A|t4	1.00	1.653	0.513	0.069
Q9_A|t1	1.00	-1.849	0.137	0.076
Q9_A|t2	1.00	-1.076	0.142	0.048
Q9_A|t3	0.95	-0.108	0.215	0.039
Q9_A|t4	1.00	1.322	0.441	0.056
Q12_A|t1	1.00	-1.634	0.117	0.065
Q12_A|t2	0.99	-0.929	0.154	0.046
Q12_A|t3	0.59	0.098	0.109	0.039
Q12_A|t4	1.00	0.758	0.496	0.047
Q13_A|t1	1.00	-2.623	0.302	0.170
Q13_A|t2	1.00	-1.555	0.215	0.063
Q13_A|t3	1.00	-0.584	0.298	0.042
Q13_A|t4	1.00	0.570	0.555	0.044
Q14_A|t1	1.00	-1.783	0.201	0.073
Q14_A|t2	0.99	-0.865	0.229	0.045
Q14_A|t3	0.80	0.165	0.255	0.040
Q14_A|t4	1.00	1.166	0.548	0.053
Q23_A|t1	1.00	-1.188	0.128	0.051
Q23_A|t2	1.00	-0.610	0.231	0.042
Q23_A|t3	1.00	0.342	0.329	0.040
Q24_A|t1	1.00	-1.033	0.156	0.048
Q24_A|t2	1.00	-0.334	0.245	0.040
Q24_A|t3	1.00	0.654	0.410	0.043
Q25_A|t1	1.00	-1.087	0.144	0.049
Q25_A|t2	1.00	-0.431	0.244	0.041
Q25_A|t3	1.00	0.625	0.084	0.040
psycho~~physical	1.00	0.914	0.035	0.013
psycho~~social	1.00	0.750	0.031	0.024
psycho~~environment	1.00	0.663	0.025	0.025
physical~~social	1.00	0.636	0.025	0.027
physical~~environment	1.00	0.631	0.028	0.025
social~~environment	1.00	0.557	0.040	0.031

	Average Param	Average Bias	Coverage	Rel Bias	Std Bias
psycho=~Q6_P	0.659	0.000	0.94	0.000	0.012
psycho=~Q7_P	0.677	-0.001	0.96	-0.002	-0.060
psycho=~Q11_P	0.671	0.005	0.92	0.007	0.215
psycho=~Q19_P	0.869	0.002	0.95	0.003	0.175

psycho=~Q26_P	0.648	0.000	0.94	0.001	0.020
physical=~Q3_F	0.510	0.001	0.89	0.003	0.043
physical=~Q4_F	0.423	-0.003	0.94	-0.007	-0.092
physical=~Q10_F	0.811	0.002	0.94	0.003	0.164
physical=~Q15_F	0.643	0.005	0.93	0.007	0.179
physical=~Q16_F	0.590	0.003	0.94	0.004	0.099
physical=~Q17_F	0.923	0.000	0.97	0.000	-0.033
physical=~Q18_F	0.888	0.000	0.95	0.000	0.026
social=~Q20_S	0.819	0.002	0.95	0.003	0.098
social=~Q21_S	0.586	0.000	0.96	-0.001	-0.018
social=~Q22_S	0.748	0.002	0.95	0.002	0.074
environment=~Q8_A	0.510	-0.006	0.94	-0.011	-0.171
environment=~Q9_A	0.629	0.003	0.92	0.004	0.094
environment=~Q12_A	0.715	0.004	0.96	0.005	0.156
environment=~Q13_A	0.663	-0.001	0.95	-0.001	-0.027
environment=~Q14_A	0.732	-0.002	0.94	-0.003	-0.103
environment=~Q23_A	0.580	-0.001	0.96	-0.003	-0.052
environment=~Q24_A	0.492	-0.003	0.96	-0.007	-0.108
environment=~Q25_A	0.542	-0.001	0.97	-0.002	-0.031
Q3_F~~Q4_F	0.387	0.004	0.99	0.010	0.146
Q6_P|t1	-1.931	-0.002	0.93	0.001	-0.019
Q6_P|t2	-1.319	0.004	0.95	-0.003	0.071
Q6_P|t3	-0.598	0.009	0.96	-0.016	0.244
Q6_P|t4	0.541	0.005	0.94	0.009	0.114
Q7_P|t1	-2.244	-0.023	0.94	0.010	-0.192
Q7_P|t2	-1.238	0.008	0.96	-0.007	0.178
Q7_P|t3	-0.290	0.003	0.99	-0.010	0.093
Q7_P|t4	1.183	-0.001	0.98	-0.001	-0.019
Q11_P|t1	-2.033	-0.007	0.94	0.003	-0.071
Q11_P|t2	-1.296	0.000	0.97	0.000	-0.008
Q11_P|t3	-0.415	0.001	0.93	-0.002	0.016
Q11_P|t4	0.589	0.002	0.97	0.003	0.041
Q19_P|t1	-1.062	0.006	0.93	-0.006	0.125
Q19_P|t2	-0.405	0.000	0.98	0.000	0.003
Q19_P|t3	0.829	0.001	0.94	0.002	0.027
Q26_P|t1	-1.846	-0.003	0.93	0.002	-0.036
Q26_P|t2	-1.378	0.001	0.94	-0.001	0.015
Q26_P|t3	-0.796	0.006	0.91	-0.007	0.126
Q26_P|t4	0.723	-0.001	0.95	-0.001	-0.015
Q3_F|t1	-2.528	-0.023	0.99	0.009	-0.148
Q3_F|t2	-1.471	0.001	0.94	-0.001	0.019
Q3_F|t3	-0.819	0.001	0.90	-0.001	0.025
Q3_F|t4	0.116	-0.008	0.93	-0.071	-0.210
Q4_F|t1	-2.188	-0.012	0.97	0.006	-0.121
Q4_F|t2	-1.302	-0.001	0.97	0.001	-0.021
Q4_F|t3	-0.754	0.001	0.91	-0.001	0.010
Q4_F|t4	0.167	0.000	1.00	0.001	0.005
Q10_F|t1	-2.344	-0.015	0.95	0.007	-0.119
Q10_F|t2	-1.285	0.002	0.93	-0.002	0.038

Q10_F|t3	-0.112	0.003	0.96	-0.031	0.097
Q10_F|t4	0.924	0.000	0.94	0.000	-0.003
Q15_F|t1	-2.528	-0.013	0.93	0.005	-0.069
Q15_F|t2	-1.820	0.019	0.92	-0.010	0.183
Q15_F|t3	-1.248	0.027	0.91	-0.021	0.173
Q15_F|t4	-0.167	-0.015	0.90	0.089	-0.162
Q16_F|t1	-0.738	0.014	0.93	-0.019	0.151
Q16_F|t2	-0.116	0.030	0.94	-0.261	0.206
Q16_F|t3	0.917	-0.031	0.93	-0.034	-0.107
Q17_F|t1	-1.070	0.017	0.91	-0.016	0.181
Q17_F|t2	-0.476	0.034	0.91	-0.071	0.176
Q17_F|t3	0.906	-0.034	0.92	-0.038	-0.120
Q18_F|t1	-1.041	0.012	0.92	-0.011	0.125
Q18_F|t2	-0.479	0.026	0.92	-0.054	0.152
Q18_F|t3	0.793	-0.037	0.91	-0.046	-0.131
Q20_S|t1	-1.178	0.011	0.92	-0.010	0.081
Q20_S|t2	-0.248	0.022	0.94	-0.090	0.130
Q20_S|t3	0.993	-0.038	0.91	-0.039	-0.151
Q21_S|t1	-0.783	0.011	0.96	-0.015	0.107
Q21_S|t2	-0.035	0.014	0.92	-0.410	0.100
Q21_S|t3	0.985	-0.047	0.94	-0.048	-0.155
Q22_S|t1	-1.150	0.015	0.95	-0.013	0.105
Q22_S|t2	-0.153	0.027	0.96	-0.178	0.160
Q22_S|t3	1.070	-0.058	0.91	-0.054	-0.158
Q8_A|t1	-1.416	0.012	0.95	-0.009	0.091
Q8_A|t2	-0.492	0.019	0.93	-0.039	0.135
Q8_A|t3	0.558	0.029	0.91	0.053	0.173
Q8_A|t4	1.728	-0.075	0.97	-0.043	-0.145
Q9_A|t1	-1.859	0.010	0.92	-0.005	0.069
Q9_A|t2	-1.096	0.019	0.96	-0.018	0.136
Q9_A|t3	-0.141	0.032	0.93	-0.231	0.150
Q9_A|t4	1.372	-0.051	0.94	-0.037	-0.115
Q12_A|t1	-1.648	0.014	0.88	-0.008	0.119
Q12_A|t2	-0.947	0.018	0.89	-0.019	0.114
Q12_A|t3	0.080	0.018	0.94	0.219	0.161
Q12_A|t4	0.826	-0.068	0.93	-0.083	-0.138
Q13_A|t1	-2.668	0.045	0.88	-0.017	0.149
Q13_A|t2	-1.603	0.048	0.90	-0.030	0.223
Q13_A|t3	-0.653	0.069	0.91	-0.105	0.230
Q13_A|t4	0.692	-0.122	0.90	-0.177	-0.220
Q14_A|t1	-1.820	0.037	0.91	-0.020	0.185
Q14_A|t2	-0.913	0.048	0.89	-0.053	0.210
Q14_A|t3	0.109	0.056	0.90	0.516	0.220
Q14_A|t4	1.291	-0.125	0.92	-0.097	-0.228
Q23_A|t1	-1.212	0.024	0.92	-0.020	0.189
Q23_A|t2	-0.665	0.054	0.88	-0.082	0.235
Q23_A|t3	0.410	-0.067	0.92	-0.164	-0.204
Q24_A|t1	-1.062	0.029	0.88	-0.027	0.185
Q24_A|t2	-0.386	0.053	0.91	-0.136	0.215

Q24_A|t3	0.748	-0.093	0.92	-0.125	-0.228
Q25_A|t1	-1.118	0.031	0.90	-0.027	0.212
Q25_A|t2	-0.487	0.056	0.90	-0.114	0.227
Q25_A|t3	0.606	0.019	0.87	0.032	0.229
psycho~~physical	0.922	-0.008	0.93	-0.009	-0.234
psycho~~social	0.755	-0.005	0.94	-0.007	-0.161
psycho~~environment	0.659	0.004	0.91	0.007	0.177
physical~~social	0.638	-0.002	0.98	-0.003	-0.078
physical~~environment	0.630	0.001	0.95	0.001	0.025
social~~environment	0.550	0.007	0.88	0.013	0.178

	Rel SE Bias	Not Cover Below	Not Cover Above
psycho=~Q6_P	-0.057	0.04	0.02
psycho=~Q7_P	0.055	0.03	0.01
psycho=~Q11_P	-0.065	0.07	0.01
psycho=~Q19_P	-0.006	0.03	0.02
psycho=~Q26_P	-0.038	0.05	0.01
physical=~Q3_F	-0.144	0.05	0.06
physical=~Q4_F	-0.059	0.03	0.03
physical=~Q10_F	0.025	0.06	0.00
physical=~Q15_F	-0.019	0.07	0.00
physical=~Q16_F	-0.047	0.05	0.01
physical=~Q17_F	0.017	0.01	0.02
physical=~Q18_F	-0.093	0.05	0.00
social=~Q20_S	-0.037	0.03	0.02
social=~Q21_S	0.097	0.03	0.01
social=~Q22_S	0.038	0.03	0.02
environment=~Q8_A	-0.092	0.04	0.02
environment=~Q9_A	-0.039	0.05	0.03
environment=~Q12_A	0.006	0.02	0.02
environment=~Q13_A	0.038	0.05	0.00
environment=~Q14_A	-0.023	0.03	0.03
environment=~Q23_A	0.042	0.02	0.02
environment=~Q24_A	0.008	0.01	0.03
environment=~Q25_A	-0.020	0.01	0.02
Q3_F~~Q4_F	0.086	0.01	0.00
Q6_P|t1	-0.028	0.03	0.04
Q6_P|t2	-0.050	0.03	0.02
Q6_P|t3	0.077	0.04	0.00
Q6_P|t4	-0.054	0.04	0.02
Q7_P|t1	-0.070	0.02	0.04
Q7_P|t2	0.095	0.03	0.01
Q7_P|t3	0.207	0.01	0.00
Q7_P|t4	0.040	0.00	0.02
Q11_P|t1	-0.069	0.03	0.03
Q11_P|t2	-0.010	0.02	0.01
Q11_P|t3	-0.101	0.04	0.03
Q11_P|t4	0.003	0.01	0.02
Q19_P|t1	-0.077	0.06	0.01
Q19_P|t2	0.037	0.01	0.01

Q19_P|t3	-0.066	0.02	0.04
Q26_P|t1	-0.083	0.03	0.04
Q26_P|t2	-0.027	0.03	0.03
Q26_P|t3	-0.056	0.06	0.03
Q26_P|t4	-0.014	0.01	0.04
Q3_F|t1	-0.052	0.01	0.00
Q3_F|t2	-0.072	0.03	0.03
Q3_F|t3	-0.096	0.06	0.04
Q3_F|t4	-0.013	0.03	0.04
Q4_F|t1	0.018	0.02	0.01
Q4_F|t2	0.009	0.02	0.01
Q4_F|t3	-0.144	0.05	0.04
Q4_F|t4	0.006	0.00	0.00
Q10_F|t1	-0.064	0.03	0.02
Q10_F|t2	-0.090	0.03	0.04
Q10_F|t3	0.076	0.02	0.02
Q10_F|t4	-0.121	0.03	0.03
Q15_F|t1	-0.220	0.07	0.00
Q15_F|t2	-0.283	0.05	0.03
Q15_F|t3	-0.663	0.05	0.04
Q15_F|t4	-0.575	0.03	0.07
Q16_F|t1	-0.549	0.06	0.01
Q16_F|t2	-0.736	0.05	0.01
Q16_F|t3	-0.842	0.03	0.04
Q17_F|t1	-0.487	0.07	0.02
Q17_F|t2	-0.788	0.07	0.02
Q17_F|t3	-0.842	0.05	0.03
Q18_F|t1	-0.496	0.06	0.02
Q18_F|t2	-0.763	0.06	0.02
Q18_F|t3	-0.844	0.05	0.04
Q20_S|t1	-0.637	0.04	0.04
Q20_S|t2	-0.772	0.03	0.03
Q20_S|t3	-0.817	0.03	0.06
Q21_S|t1	-0.594	0.03	0.01
Q21_S|t2	-0.728	0.04	0.04
Q21_S|t3	-0.848	0.02	0.04
Q22_S|t1	-0.658	0.03	0.02
Q22_S|t2	-0.771	0.03	0.01
Q22_S|t3	-0.868	0.02	0.07
Q8_A|t1	-0.582	0.03	0.02
Q8_A|t2	-0.713	0.04	0.03
Q8_A|t3	-0.755	0.06	0.03
Q8_A|t4	-0.865	0.01	0.02
Q9_A|t1	-0.445	0.04	0.04
Q9_A|t2	-0.660	0.04	0.00
Q9_A|t3	-0.818	0.06	0.01
Q9_A|t4	-0.873	0.02	0.04
Q12_A|t1	-0.441	0.08	0.04
Q12_A|t2	-0.703	0.07	0.04

Q12_A|t3	-0.644	0.04	0.02
Q12_A|t4	-0.906	0.04	0.03
Q13_A|t1	-0.436	0.12	0.00
Q13_A|t2	-0.708	0.09	0.01
Q13_A|t3	-0.860	0.08	0.01
Q13_A|t4	-0.921	0.04	0.06
Q14_A|t1	-0.636	0.07	0.02
Q14_A|t2	-0.804	0.07	0.04
Q14_A|t3	-0.845	0.07	0.03
Q14_A|t4	-0.903	0.02	0.06
Q23_A|t1	-0.603	0.07	0.01
Q23_A|t2	-0.819	0.10	0.02
Q23_A|t3	-0.877	0.02	0.06
Q24_A|t1	-0.695	0.07	0.05
Q24_A|t2	-0.836	0.09	0.00
Q24_A|t3	-0.894	0.01	0.07
Q25_A|t1	-0.663	0.08	0.02
Q25_A|t2	-0.834	0.08	0.02
Q25_A|t3	-0.523	0.09	0.04
psycho~~physical	-0.646	0.02	0.05
psycho~~social	-0.224	0.00	0.06
psycho~~environment	0.003	0.08	0.01
physical~~social	0.069	0.01	0.01
physical~~environment	-0.111	0.02	0.03
social~~environment	-0.224	0.10	0.02

	Average CI Width	SD CI Width
psycho=~Q6_P	0.086	0.004
psycho=~Q7_P	0.084	0.004
psycho=~Q11_P	0.083	0.004
psycho=~Q19_P	0.051	0.003
psycho=~Q26_P	0.090	0.004
physical=~Q3_F	0.112	0.005
physical=~Q4_F	0.122	0.004
physical=~Q10_F	0.058	0.003
physical=~Q15_F	0.099	0.005
physical=~Q16_F	0.097	0.004
physical=~Q17_F	0.039	0.003
physical=~Q18_F	0.045	0.003
social=~Q20_S	0.088	0.005
social=~Q21_S	0.118	0.005
social=~Q22_S	0.094	0.005
environment=~Q8_A	0.119	0.004
environment=~Q9_A	0.105	0.005
environment=~Q12_A	0.090	0.004
environment=~Q13_A	0.103	0.005
environment=~Q14_A	0.088	0.005
environment=~Q23_A	0.115	0.005
environment=~Q24_A	0.125	0.005
environment=~Q25_A	0.119	0.005

Q3_F~~Q4_F	0.112	0.004
Q6_P|t1	0.318	0.022
Q6_P|t2	0.211	0.006
Q6_P|t3	0.162	0.001
Q6_P|t4	0.160	0.001
Q7_P|t1	0.431	0.056
Q7_P|t2	0.202	0.005
Q7_P|t3	0.154	0.001
Q7_P|t4	0.198	0.004
Q11_P|t1	0.348	0.029
Q11_P|t2	0.209	0.006
Q11_P|t3	0.157	0.001
Q11_P|t4	0.162	0.001
Q19_P|t1	0.187	0.004
Q19_P|t2	0.157	0.001
Q19_P|t3	0.173	0.003
Q26_P|t1	0.297	0.020
Q26_P|t2	0.218	0.007
Q26_P|t3	0.171	0.002
Q26_P|t4	0.167	0.002
Q3_F|t1	0.590	0.126
Q3_F|t2	0.230	0.009
Q3_F|t3	0.172	0.003
Q3_F|t4	0.152	0.000
Q4_F|t1	0.403	0.040
Q4_F|t2	0.210	0.006
Q4_F|t3	0.169	0.002
Q4_F|t4	0.153	0.000
Q10_F|t1	0.474	0.072
Q10_F|t2	0.208	0.006
Q10_F|t3	0.152	0.000
Q10_F|t4	0.178	0.003
Q15_F|t1	0.587	0.131
Q15_F|t2	0.287	0.020
Q15_F|t3	0.202	0.009
Q15_F|t4	0.153	0.002
Q16_F|t1	0.168	0.003
Q16_F|t2	0.153	0.003
Q16_F|t3	0.178	0.003
Q17_F|t1	0.187	0.006
Q17_F|t2	0.158	0.002
Q17_F|t3	0.178	0.004
Q18_F|t1	0.185	0.005
Q18_F|t2	0.159	0.002
Q18_F|t3	0.171	0.004
Q20_S|t1	0.197	0.008
Q20_S|t2	0.154	0.004
Q20_S|t3	0.182	0.004
Q21_S|t1	0.170	0.003

Q21_S|t2	0.153	0.004
Q21_S|t3	0.182	0.004
Q22_S|t1	0.194	0.007
Q22_S|t2	0.153	0.005
Q22_S|t3	0.189	0.008
Q8_A|t1	0.222	0.011
Q8_A|t2	0.159	0.001
Q8_A|t3	0.163	0.016
Q8_A|t4	0.272	0.014
Q9_A|t1	0.299	0.026
Q9_A|t2	0.190	0.006
Q9_A|t3	0.154	0.009
Q9_A|t4	0.220	0.009
Q12_A|t1	0.256	0.016
Q12_A|t2	0.179	0.005
Q12_A|t3	0.153	0.003
Q12_A|t4	0.182	0.071
Q13_A|t1	0.668	0.190
Q13_A|t2	0.245	0.021
Q13_A|t3	0.164	0.002
Q13_A|t4	0.172	0.028
Q14_A|t1	0.287	0.030
Q14_A|t2	0.176	0.006
Q14_A|t3	0.155	0.012
Q14_A|t4	0.208	0.006
Q23_A|t1	0.199	0.009
Q23_A|t2	0.164	0.003
Q23_A|t3	0.158	0.007
Q24_A|t1	0.186	0.008
Q24_A|t2	0.157	0.002
Q24_A|t3	0.170	0.006
Q25_A|t1	0.190	0.008
Q25_A|t2	0.159	0.002
Q25_A|t3	0.157	0.026
psycho~~physical	0.049	0.010
psycho~~social	0.094	0.005
psycho~~environment	0.099	0.005
physical~~social	0.106	0.005
physical~~environment	0.098	0.007
social~~environment	0.121	0.010

Source: Article Notebook

Interpreting Parameter Summary

Look for:

- **Power > .80**: Adequate power to detect non-zero parameters
- **Coverage 0.91–0.98**: Proper confidence interval performance

- **Rel Bias** $< |\mathbf{0.10}|$: Estimates are unbiased
- **Rel SE Bias** $< |\mathbf{0.10}|$: Standard errors are accurate

All parameters of interest outside these ranges deserve attention. Thresholds are generally not parameters of interest and can be ignored for interpretation purposes.

0.11 Summary and Conclusions

This tutorial demonstrated a complete CFA workflow for ordinal data using the WHOQOL-BREF instrument:

1. **Data preparation**: Loaded data directly from Mendeley Data repository
2. **Population model**: Defined a teaching example based on Lin & Yao (2022)
3. **Model specification**: Tested four competing models:
   - Original 4-factor model (all items)
   - Bifactor model with general QOL factor
   - Second-order hierarchical model
   - Revised 4-factor model (excluding problematic item Q5)
4. **Model evaluation**: Used DWLS estimation appropriate for ordinal data, examining overall fit, local fit, reliability, and path diagrams
5. **Model comparison**: Systematically compared all models on key fit indices
6. **Power analysis**: Assessed statistical power via Monte Carlo simulation

! Key Findings and Recommendations

1. **Item Q5 is problematic** across multiple models and should be excluded
2. **The revised 4-factor correlated model** (without Q5) is recommended as the best balance of fit, interpretability, and statistical properties
3. **All four domains** (Psychological, Physical, Social, Environment) show adequate reliability
4. **Sample size (N = 1,047)** provides adequate power for parameter estimation
5. **DWLS/WLSMV estimation** is essential for proper analysis of ordinal Likert data

0.11.1 Pedagogical Takeaways

This tutorial illustrated several important principles for conducting "typical CFA" in applied settings:

- Always use appropriate estimators for ordinal data (DWLS/WLSMV, not ML)
- Examine both overall and local fit—overall indices can mask specific problems
- Be cautious with modification indices; only make theoretically justified changes
- Consider multiple competing models rather than accepting the first specification
- Document and report problematic items transparently
- Use simulation-based power analysis when feasible
- Bifactor models often fit better but require strong assumptions and careful interpretation

For further methodological guidance, see Rogers (2022) and Rogers (2024).

References

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